



上海科技大学
ShanghaiTech University



YesAI 可信与通用
智能实验室

模型和数据联合驱动的深度学习



主页

石野

2024-11



立志成才 报国裕民

模型和数据联合驱动



上海科技大学
ShanghaiTech University

Charging



Robotics



Quantum



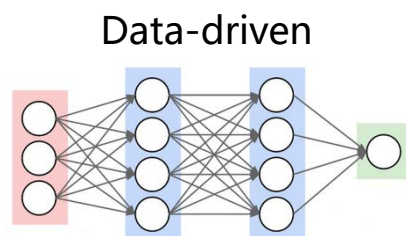
Smart Grid



Autonomous Driving



Health-care



Model-driven

$$\begin{aligned} ||W||_1 &\leq A \\ f_1(W_{ij}) &\leq B \\ f_2(W_{ij}) &= f_3(W_{ij}) \\ &\dots \end{aligned}$$

Models

- Physics/Knowledge/Prior
- Optimization Models
- Foundation Models

Outline



1. Controllable diffusion model for reliable generation

- Conditional Diffusion with Spherical Gaussian Constraint
- Gaze-guided Hand-object Interaction Synthesis
- Diffusion-based Reinforcement Learning

2. Fusing physical information into data-driven models

- Faster Implicit Differentiation
- More accurate 3D Motion Capture
- Reliable Decision-making via Safe RL

3. Federated Learning with Foundation Models

- Personalized Federated Learning with Transformer
- Prompt Tuning Boost Federated Learning
- Theoretical Analysis for prompt-based Federated Learning



Backgrounds: Diffusion Models



上海科技大学
ShanghaiTech University

Diffusion Models: Diffusion models are a class of likelihood-based generative models that outperform other generative models(e.g. GAN, VAE) in various tasks.

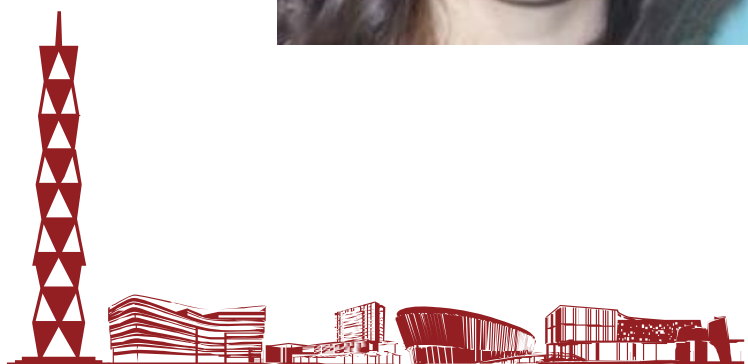
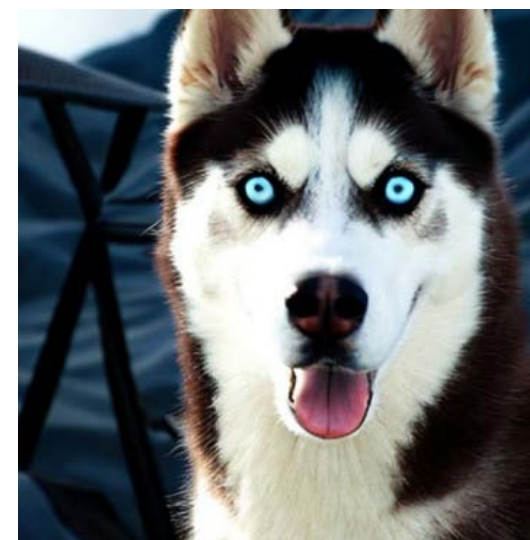
DDPM



Stable Diffusion



DiT



立志成才 报国裕民

Backgrounds: Diffusion Models

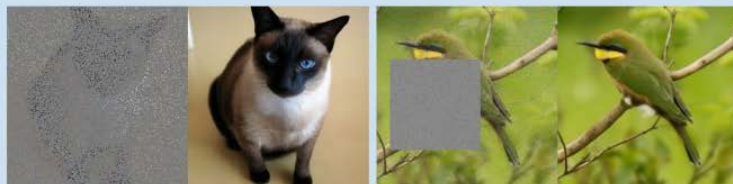


上海科技大学
ShanghaiTech University

Diffusion Models: Diffusion models are a class of likelihood-based generative models that outperform other generative models(e.g. GAN, VAE) in various tasks.

Linear

(a) Inpainting



(c) Gaussian deblur



(b) Super-resolution



(d) Motion deblur



Non-linear

(e) Phase retrieval



(f) Non-uniform deblur



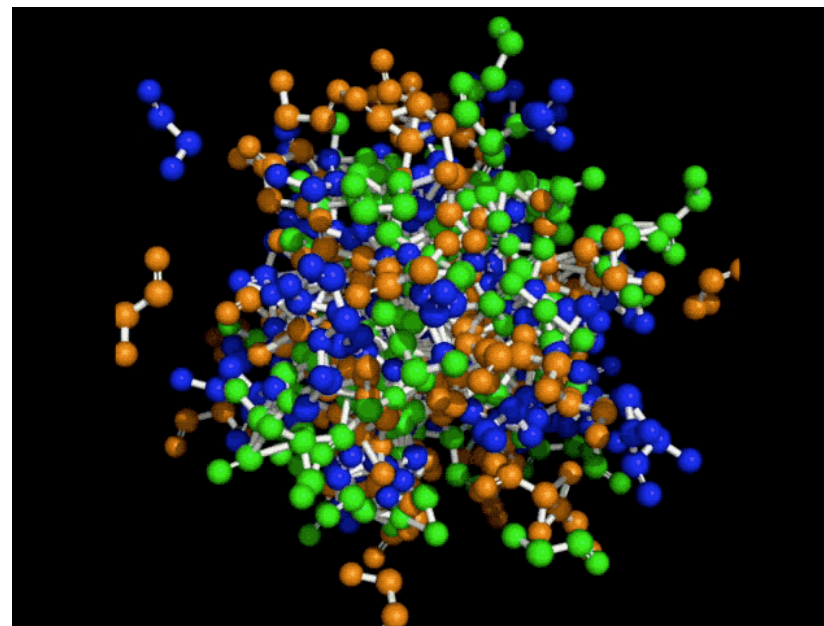
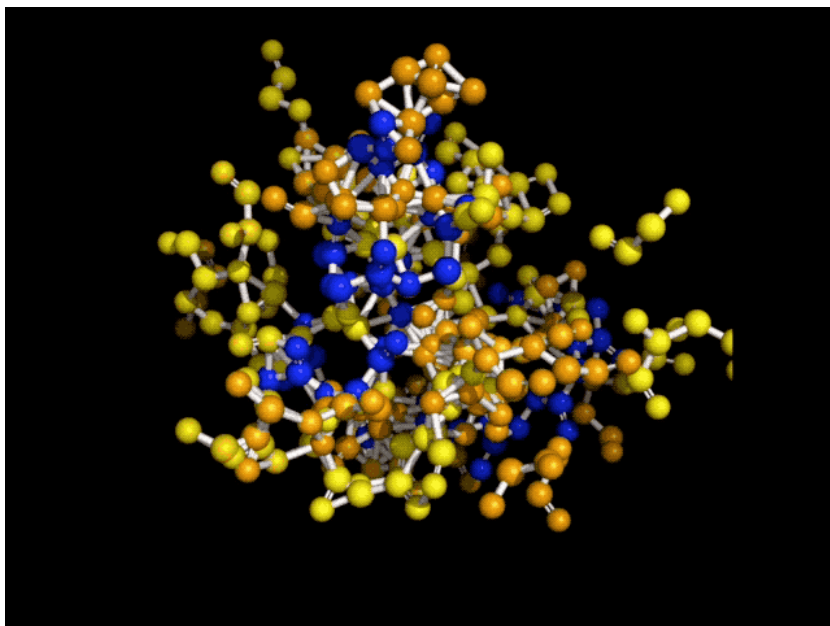
立志成才 报国裕民

Backgrounds: Diffusion Models



上海科技大学
ShanghaiTech University

Diffusion Models: Diffusion models are a class of likelihood-based generative models that outperform other generative models(e.g. GAN, VAE) in various tasks.



立志成才 报国裕民

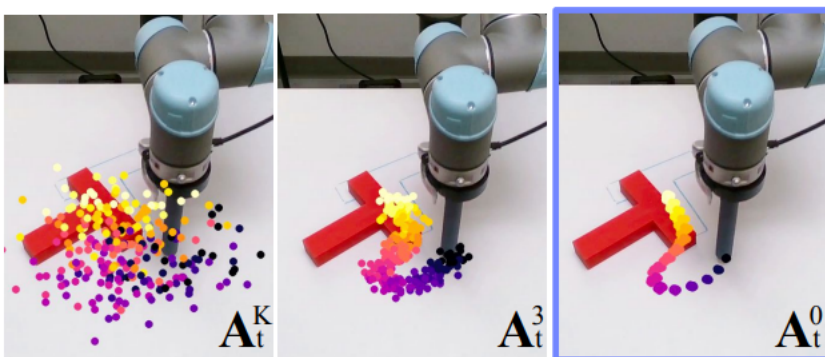
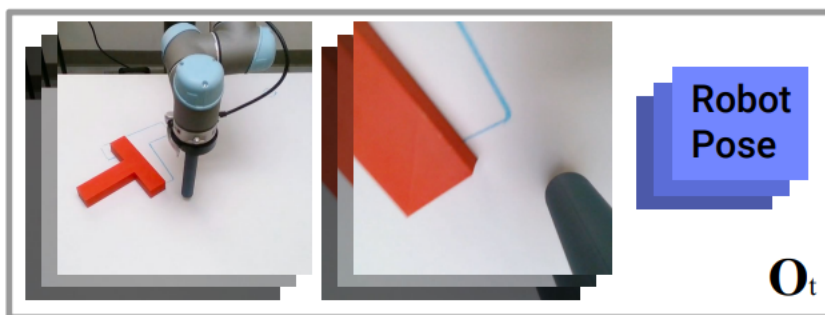
Backgrounds: Diffusion Models



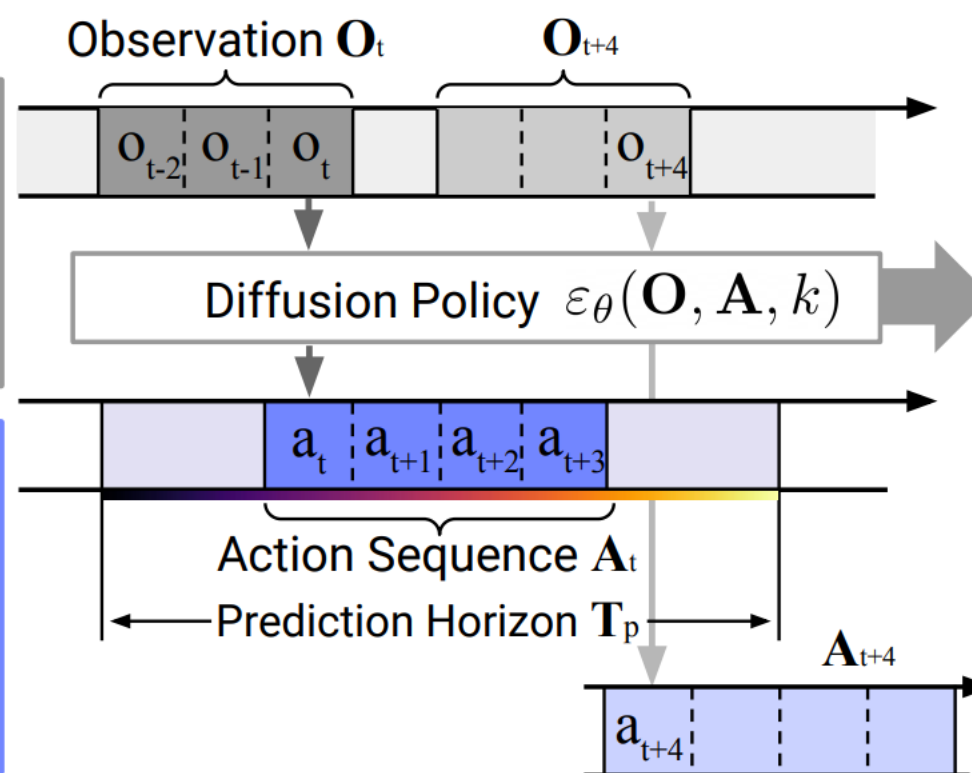
上海科技大学
ShanghaiTech University

Diffusion Models: Diffusion models are a class of likelihood-based generative models that outperform other generative models(e.g. GAN, VAE) in various tasks.

Input: Image Observation Sequence



Output: Action Sequence



a) Diffusion Policy General Formulation

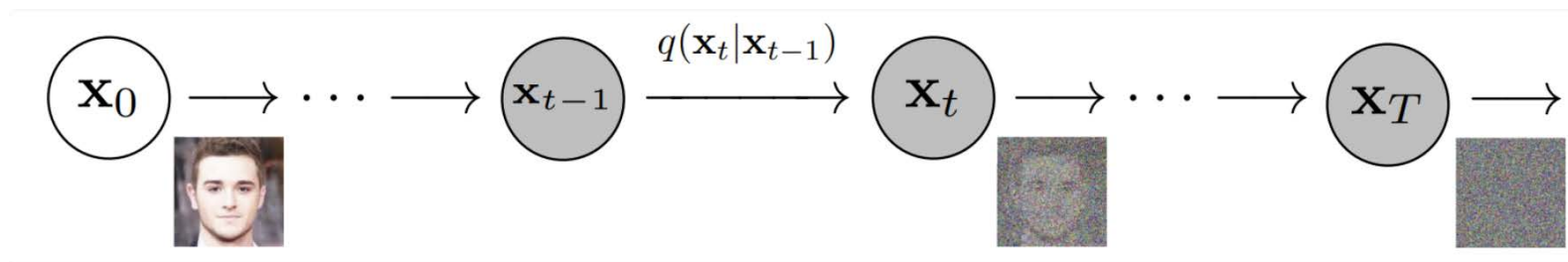
立志成才 报国裕民

Backgrounds: Diffusion Models



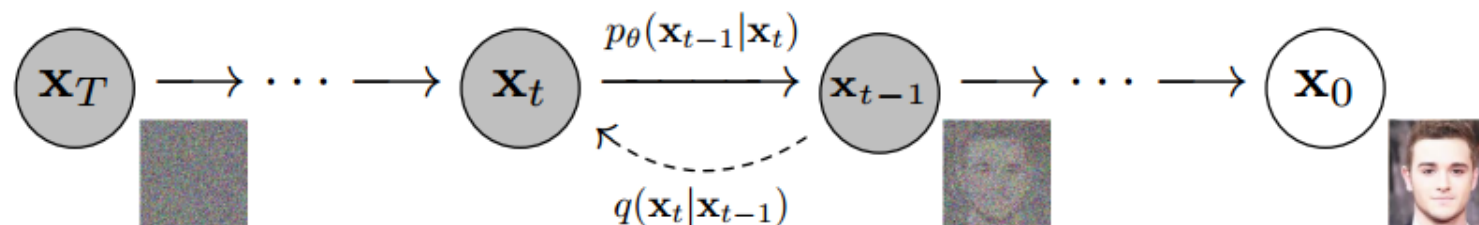
上海科技大学
ShanghaiTech University

Forward process: $q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_0, (1 - \alpha_t)I)$.



Reverse process:
$$\mathbf{x}_{t-1} = \underbrace{\sqrt{\alpha_{t-1}} \left(\frac{\mathbf{x}_t - \sqrt{1 - \alpha_t} \epsilon_{\theta}^{(t)}(\mathbf{x}_t)}{\sqrt{\alpha_t}} \right)}_{\text{"predicted } \mathbf{x}_0"} + \underbrace{\sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \epsilon_{\theta}^{(t)}(\mathbf{x}_t)}_{\text{"direction pointing to } \mathbf{x}_t"} + \underbrace{\sigma_t \epsilon_t}_{\text{random noise}}$$

Loss function: $L_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon_t} [\|\epsilon_t - \epsilon_{\theta}(x_t, t)\|^2]$.



立志成才 报国裕民

Forward Process

A nice property of the above process is that we can sample $\mathbf{x}_t$ at any arbitrary time step t in a closed form using reparameterization trick. Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$:

$$\begin{aligned}\mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} && \text{;where } \boldsymbol{\epsilon}_{t-1}, \boldsymbol{\epsilon}_{t-2}, \dots \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\boldsymbol{\epsilon}}_{t-2} && \text{;where } \bar{\boldsymbol{\epsilon}}_{t-2} \text{ merges two Gaussians (*)} \\ &= \dots \\ &= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon} \\ q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})\end{aligned}$$

(*) Recall that when we merge two Gaussians with different variance, $\mathcal{N}(\mathbf{0}, \sigma_1^2 \mathbf{I})$ and $\mathcal{N}(\mathbf{0}, \sigma_2^2 \mathbf{I})$, the new distribution is $\mathcal{N}(\mathbf{0}, (\sigma_1^2 + \sigma_2^2) \mathbf{I})$. Here the merged standard deviation is $\sqrt{(1 - \alpha_t) + \alpha_t(1 - \alpha_{t-1})} = \sqrt{1 - \alpha_t \alpha_{t-1}}$.

Usually, we can afford a larger update step when the sample gets noisier, so $\beta_1 < \beta_2 < \dots < \beta_T$ and therefore $\bar{\alpha}_1 > \dots > \bar{\alpha}_T$.

Connection with stochastic gradient Langevin dynamics

Langevin dynamics is a concept from physics, developed for statistically modeling molecular systems. Combined with stochastic gradient descent, *stochastic gradient Langevin dynamics* (Welling & Teh 2011) can produce samples from a probability density $p(\mathbf{x})$ using only the gradients $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ in a Markov chain of updates:

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \frac{\delta}{2} \nabla_{\mathbf{x}} \log p(\mathbf{x}_{t-1}) + \sqrt{\delta} \boldsymbol{\epsilon}_t, \quad \text{where } \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

where δ is the step size. When $T \rightarrow \infty, \delta \rightarrow 0$, $\mathbf{x}_T$ equals to the true probability density $p(\mathbf{x})$.

Compared to standard SGD, stochastic gradient Langevin dynamics injects Gaussian noise into the parameter updates to avoid collapses into local minima.

Reverse Process



上海科技大学

Using Bayes' rule, we have:

$$\begin{aligned}q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) &= q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)} \\ &\propto \exp \left(-\frac{1}{2} \left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_{t-1})^2}{\beta_t} + \frac{(\mathbf{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0)^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)^2}{1 - \bar{\alpha}_t} \right) \right) \\ &= \exp \left(-\frac{1}{2} \left(\frac{\mathbf{x}_t^2 - 2\sqrt{\alpha_t} \mathbf{x}_t \mathbf{x}_{t-1} + \alpha_t \mathbf{x}_{t-1}^2}{\beta_t} + \frac{\mathbf{x}_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0 \mathbf{x}_{t-1} + \bar{\alpha}_{t-1} \mathbf{x}_0^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)^2}{1 - \bar{\alpha}_t} \right) \right) \\ &= \exp \left(-\frac{1}{2} \left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) \mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} \mathbf{x}_0 \right) \mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0) \right) \right)\end{aligned}$$

where $C(\mathbf{x}_t, \mathbf{x}_0)$ is some function not involving $\mathbf{x}_{t-1}$ and details are omitted. Following the standard Gaussian density function, the mean and variance can be parameterized as follows (recall that $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$):

$$\begin{aligned}\tilde{\beta}_t &= 1 / \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) = 1 / \left(\frac{\alpha_t - \bar{\alpha}_t + \beta_t}{\beta_t(1 - \bar{\alpha}_{t-1})} \right) = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \\ \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) &= \left(\frac{\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} \mathbf{x}_0 \right) / \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) \\ &= \left(\frac{\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} \mathbf{x}_0 \right) \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \\ &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0\end{aligned}$$

Thanks to the nice property, we can represent $\mathbf{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t)$ and plug it into the above equation and obtain:

$$\begin{aligned}\tilde{\mu}_t &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t) \\ &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_t \right)\end{aligned}$$

<https://lilianweng.github.io/posts/2021-07-11-diffusion-models/#nice>

立志成才 报国裕民

It is also straightforward to get the same result using Jensen's inequality. Say we want to minimize the cross entropy as the learning objective,

$$\begin{aligned}
L_{CE} &= -\mathbb{E}_{q(\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0) \\
&= -\mathbb{E}_{q(\mathbf{x}_0)} \log \left(\int p_{\theta}(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \right) \\
&= -\mathbb{E}_{q(\mathbf{x}_0)} \log \left(\int q(\mathbf{x}_{1:T}|\mathbf{x}_0) \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} d\mathbf{x}_{1:T} \right) \\
&= -\mathbb{E}_{q(\mathbf{x}_0)} \log \left(\mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right) \\
&\leq -\mathbb{E}_{q(\mathbf{x}_{0:T})} \log \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \\
&= \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{0:T})} \right] = L_{VLB}
\end{aligned}$$

To convert each term in the equation to be analytically computable, the objective can be further rewritten to be a combination of several KL-divergence and entropy terms (See the detailed step-by-step process in Appendix B in [Sohl-Dickstein et al., 2015](#)):

$$\begin{aligned}
L_{VLB} &= \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{0:T})} \right] \\
&= \mathbb{E}_q \left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_{\theta}(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\
&= \mathbb{E}_q \left[-\log p_{\theta}(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\
&= \mathbb{E}_q \left[-\log p_{\theta}(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
&= \mathbb{E}_q \left[-\log p_{\theta}(\mathbf{x}_T) + \sum_{t=2}^T \log \left(\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)} \cdot \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} \right) + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
&= \mathbb{E}_q \left[-\log p_{\theta}(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
&= \mathbb{E}_q \left[-\log p_{\theta}(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
&= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_{\theta}(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) \right] \\
&= \mathbb{E}_q \left[\underbrace{D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_T))}_{L_T} + \sum_{t=2}^T \underbrace{D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) \right]_{L_0}
\end{aligned}$$

Parameterization of L_t for Training Loss

Recall that we need to learn a neural network to approximate the conditioned probability distributions in the reverse diffusion process, $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$. We would like to train $\boldsymbol{\mu}_{\theta}$ to predict $\tilde{\boldsymbol{\mu}}_t = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_t \right)$. Because $\mathbf{x}_t$ is available as input at training time, we can reparameterize the Gaussian noise term instead to make it predict $\boldsymbol{\epsilon}_t$ from the input $\mathbf{x}_t$ at time step t :

$$\boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right)$$

$$\text{Thus } \mathbf{x}_{t-1} = \mathcal{N}(\mathbf{x}_{t-1}; \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$

The loss term L_t is parameterized to minimize the difference from $\tilde{\boldsymbol{\mu}}$:

$$\begin{aligned}
L_t &= \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{1}{2 \|\boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t)\|_2^2} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t)\|^2 \right] \\
&= \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{1}{2 \|\boldsymbol{\Sigma}_{\theta}\|_2^2} \left\| \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_t \right) - \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) \right\|^2 \right] \\
&= \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t) \|\boldsymbol{\Sigma}_{\theta}\|_2^2} \|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t)\|^2 \right] \\
&= \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t) \|\boldsymbol{\Sigma}_{\theta}\|_2^2} \|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t} \boldsymbol{\epsilon}_t, t)\|^2 \right]
\end{aligned}$$

Simplification

Empirically, [Ho et al. \(2020\)](#) found that training the diffusion model works better with a simplified objective that ignores the weighting term:

$$\begin{aligned}
L_t^{\text{simple}} &= \mathbb{E}_{t \sim [1, T], \mathbf{x}_0, \boldsymbol{\epsilon}_t} \left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t)\|^2 \right] \\
&= \mathbb{E}_{t \sim [1, T], \mathbf{x}_0, \boldsymbol{\epsilon}_t} \left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t} \boldsymbol{\epsilon}_t, t)\|^2 \right]
\end{aligned}$$

The final simple objective is:

$$L_{\text{simple}} = L_t^{\text{simple}} + C$$

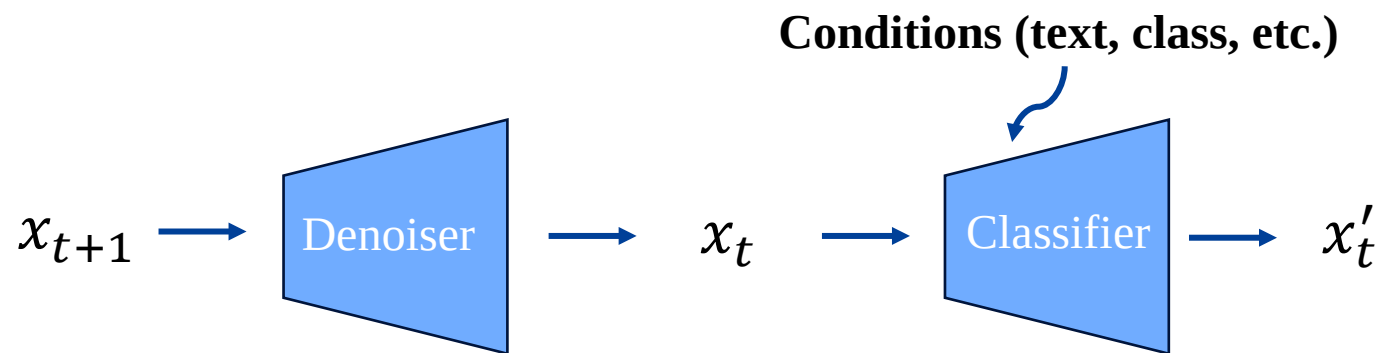
Training-free Conditional Diffusion Models(CDMs)



上海科技大学
ShanghaiTech University

- **Training-required CDMs:** neither need to train a conditional diffusion denoiser nor noise-dependent classifier.

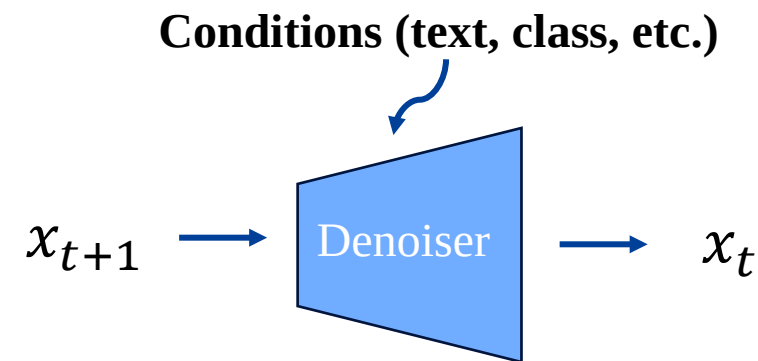
Classifier Guidance:



$$\nabla_{x_t} \log p_t(x_t | y) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_t(y | x_t).$$

OpenAI 2021

Classifier-free guidance:



$$\tilde{\epsilon}_{\theta}(z_{\lambda}, c) = (1 + w) \epsilon_{\theta}(z_{\lambda}, c) - w \epsilon_{\theta}(z_{\lambda})$$

Google 2022

Training-free Conditional Diffusion Models (CDMs)

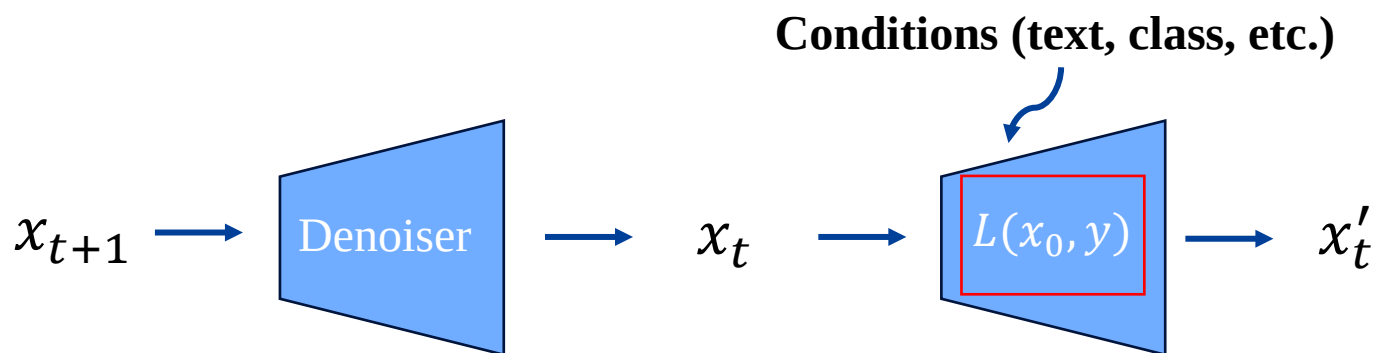


上海科技大学
ShanghaiTech University

- **DPS**: Replace the $p_t(y | x_t)$ in classifier guidance to likelihood of a loss function.

$\min L(x_0, y)$ ——— **Off-the-shelf** loss function to achieve conditional generation

$s.t. x_0 \in M$ ——— Stay on the data manifold



$$\nabla_{x_t} \log p_t(x_t | y) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_t(y | x_t).$$

[ICLR 2023 spotlight] Diffusion Posterior Sampling for General Noisy Inverse Problems (DPS)

立志成才 报国强民

Training-free Conditional Diffusion Models(CDMs)



上海科技大学
ShanghaiTech University

Guided Diffusion: $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t).$

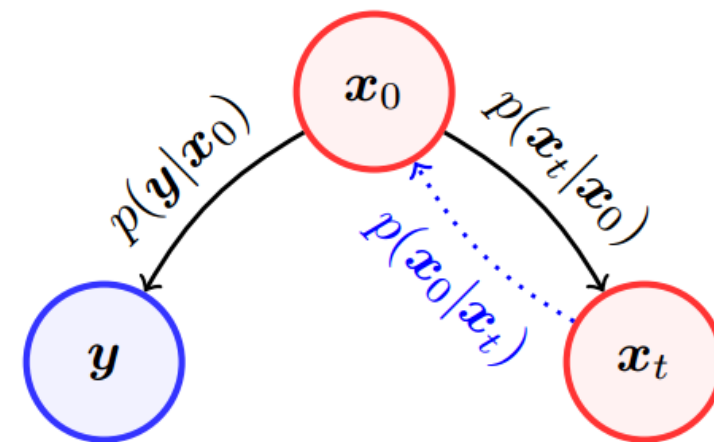
Tweedie's formula: estimate expectation of x_0 given x_t .

$$\hat{x}_0(x_t) \approx \mathbb{E}[x_0|x_t] = (x_t + b_t^2 \nabla_{x_t} \log p(x_t))/a_t,$$

$$x_t \sim \mathcal{N}(a_t x_0, b_t^2 I)$$

DPS guidance: calculate gradient w.r.t. x_t through x_0 .

$$x_{t-1} = \underbrace{DDIM(x_t, \epsilon_\theta(x_t, t))}_{\text{sampling step}} - \underbrace{\gamma \nabla_{x_t} L(\hat{x}_0(x_t), y)}_{\text{correction step}}.$$



[ICLR 2023 spotlight] Diffusion Posterior Sampling for General Noisy Inverse Problems (DPS)

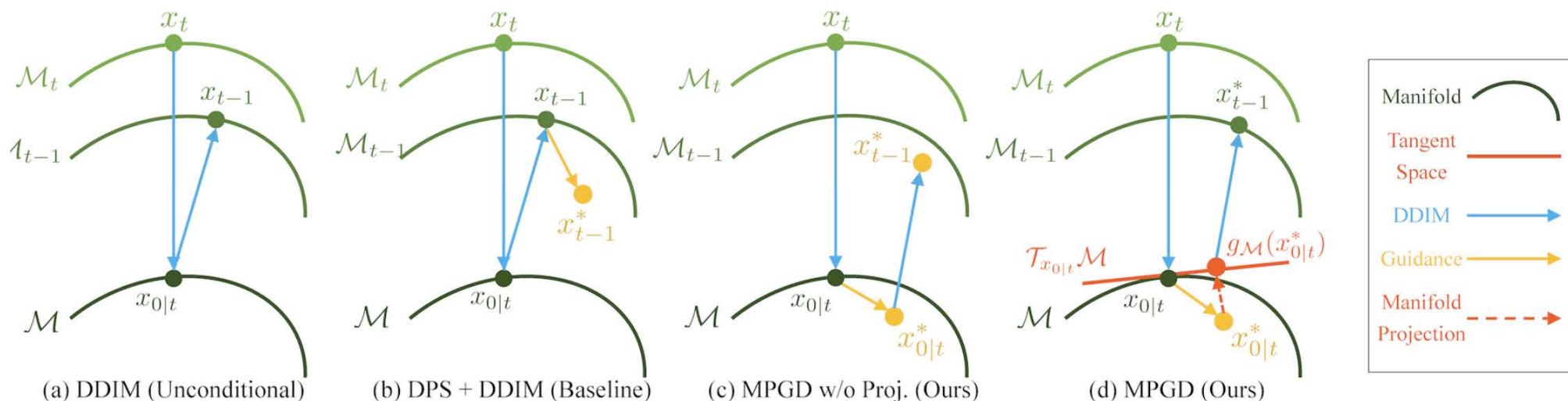
立志成才 报國裕民

Manifold Deviation Problems



上海科技大学
ShanghaiTech University

1. Linear manifold assumption: Many Previous works are based on the linear manifold assumption. However, **the linear manifold assumption is quite strong**, which will inevitably introduce errors.



Credit to MPGD

立志成才 报国裕民

Manifold Deviation Problems



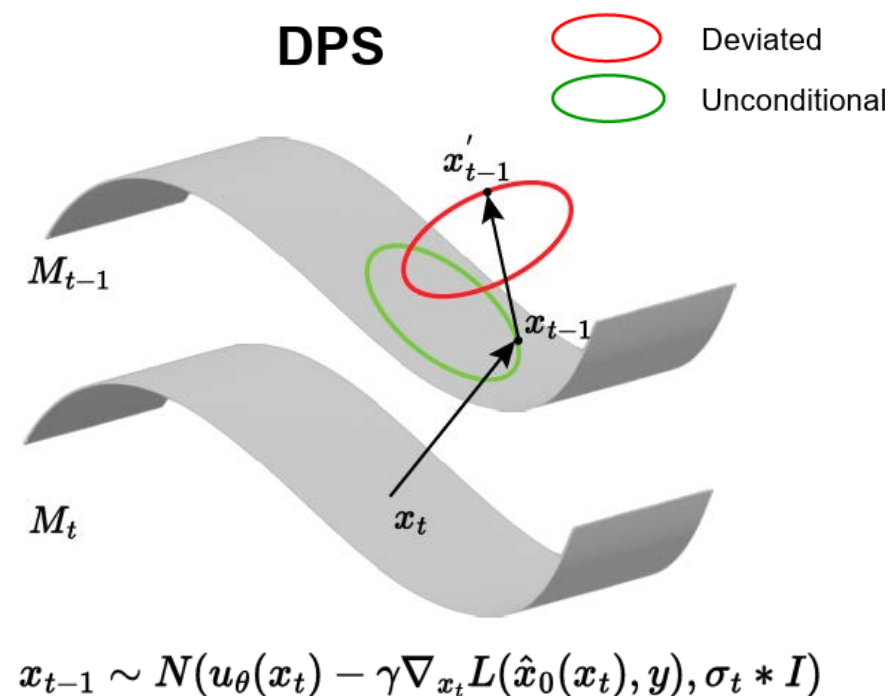
上海科技大学
ShanghaiTech University

2. Jensen gap: There exists Jensen gap J between $E_{x_0 \sim p(x_0|x_t)} [L(x_0, y)]$ and $L(\hat{x}_0(x_t), y)$. Although DPS provides an upper bound for the Jensen gap, there also exists a lower bound of the Jensen gap, leading to serious manifold deviation problem.

$$\mathcal{J} \geq \frac{1}{2}\beta \sum_{i=1}^n \lambda_i.$$

where $\Sigma = P^T \Lambda P$ via spectral decomposition and $\lambda_1, \dots, \lambda_n$ are positive diagonal elements in Λ .

$$\begin{aligned} p(y|x_t) &= \int p(y|x_0, x_t) p(x_0|x_t) dx_0 \\ &= \int p(y|x_0) p(x_0|x_t) dx_0, \end{aligned}$$



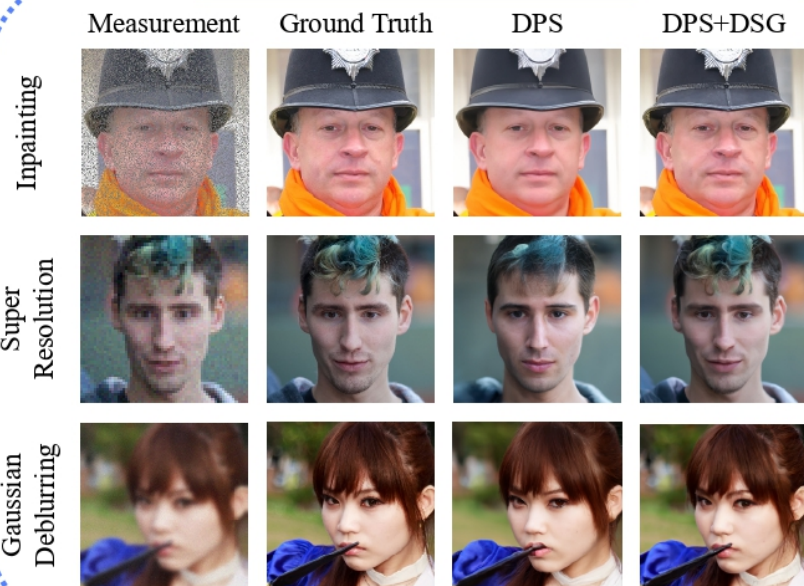
立志成才 报国裕民

Overview of our DSG

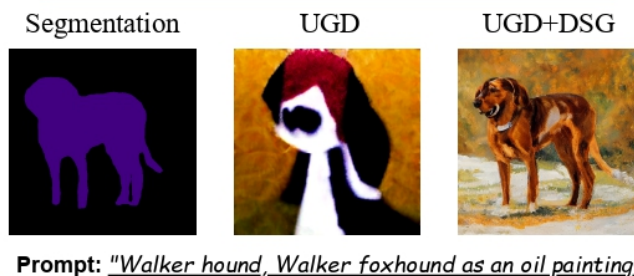


上海科技大学
ShanghaiTech University

Linear Inverse Problem



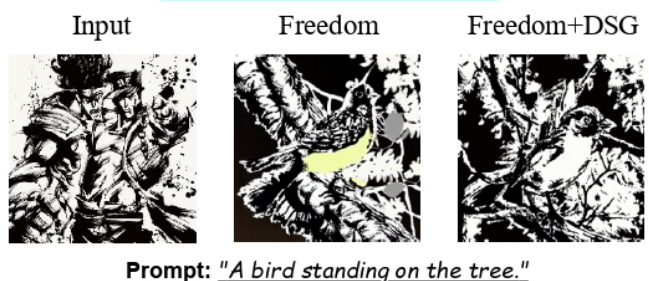
Segmentation-Text Guidance



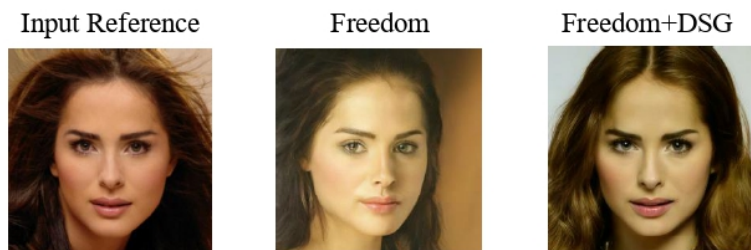
Style Guidance



Text-Style Guidance



Face ID Guidance



Plug-and-Play!

Training free!

Can be used for
Various tasks!

Overview



Few lines of code and significantly improve the performance:

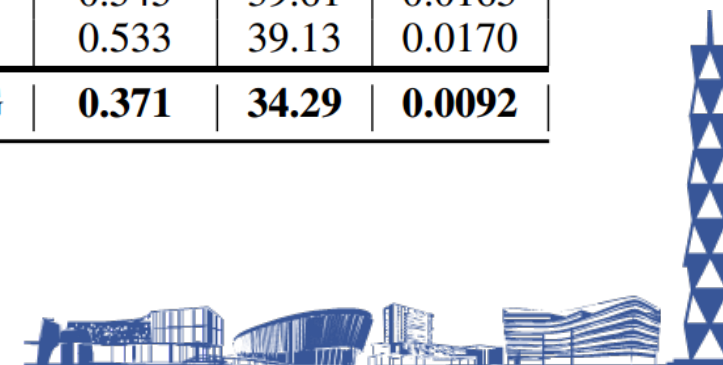
```
diff=measure-self.forward(x_0_hat)
dist=norm(diff)
grad=autograd.grad(
    outputs=distance, inputs=x_t)[0]
# DPS: x_t=x_t_mean+sigma_t*eps_t-
#       grad*self.gamma
# Ours:
norm=norm(grad, dim=[1, 2, 3])
x_t=x_t_mean-sqrt(n)*sigma_t*grad/norm
```

Table 1. Quantitative evaluation of the FFHQ dataset in Linear Inverse Problem

Methods	Inpainting		Super resolution		Gaussian deblurring	
	LPIPS↓	FID↓	LPIPS↓	FID↓	LPIPS↓	FID↓
DDRM (Kawar et al., 2022)	0.665	114.9	0.339	59.57	0.427	63.02
MCG (Chung et al., 2022)	0.414	39.19	0.637	144.5	0.550	95.04
PnP-ADMM (Chan et al., 2016)	0.677	114.7	0.433	97.27	0.519	100.6
Score-SDE (Song et al., 2020b)	0.659	127.1	0.701	170.7	0.667	120.3
(ILVR (Choi et al., 2021))						
DPS (Chung et al., 2023)	0.212	21.19	0.214	39.35	0.257	44.05
LGD (Song et al., 2023)	0.159	28.21	0.231	34.44	0.229	32.57
DPS+DSG(Ours)	0.115	15.77	0.211	30.30	0.208	28.22

Table 3. The quantitative result in CelebA-HQ test set for FaceID guidance

Method	FaceID	FID	KID
Freedom	0.545	39.61	0.0165
LGD	0.533	39.13	0.0170
Freedom+DSG	0.371	34.29	0.0092



Diffusion with Spherical Gaussian Constraint (DSG)



Why not start from the pre-existing unconditional intermediate manifold M_t and subsequently find the point closest to the conditional sampling?

$$\arg \min_{x'} - [\nabla_{x_t} L(\hat{x}_0(x_t), y)]^T (x' - x_t) \quad \text{“gradient descent direction”}$$

$$\text{s.t. } x' \in CI_{1-\delta} \quad \text{“stay on unconditional manifold } M_t \text{”}$$

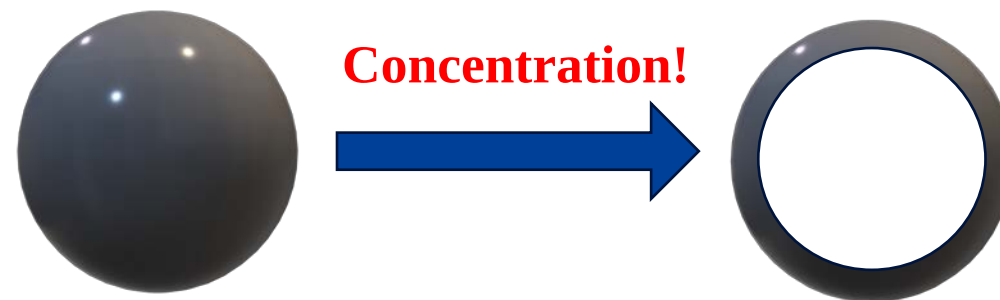
Proposition 4.3. (Concentration of high-dimensional isotropy Gaussian distribution). For any n -dimensional isotropy Gaussian distribution $x \sim \mathcal{N}(\mu, \sigma^2 I)$:

$$P(\|x - \mu\|_2^2 \geq x_{lower} = n\sigma^2 + 2n\sigma^2(\sqrt{\epsilon} + \epsilon)) \leq e^{-n\epsilon},$$

$$P(\|x - \mu\|_2^2 \leq x_{upper} = n\sigma^2 - 2n\sigma^2\sqrt{\epsilon}) \leq e^{-n\epsilon},$$

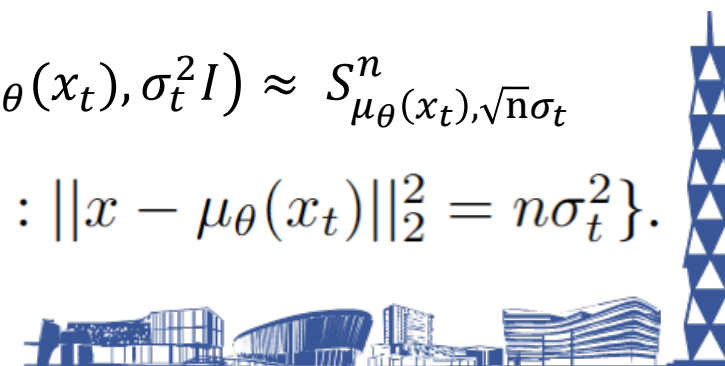
$$\arg \min_{x'} - [\nabla_{x_t} L(\hat{x}_0(x_t), y)]^T (x' - x_t)$$

$$\text{s.t. } x' \in S_{\mu_\theta(x_t), \sqrt{n}\sigma_t}^n$$



$$\text{Spherical Gaussian: } N(\mu_\theta(x_t), \sigma_t^2 I) \approx S_{\mu_\theta(x_t), \sqrt{n}\sigma_t}^n$$

$$S_{\mu_\theta(x_t), \sqrt{n}\sigma_t}^n = \{x \in \mathbb{R}^n : \|x - \mu_\theta(x_t)\|_2^2 = n\sigma_t^2\}.$$



Diffusion with Spherical Gaussian Constraint(DSG)



上海科技大学
ShanghaiTech University

$$\arg \min_{x'} [\nabla_{x_t} L(\hat{x}_0(x_t), y)]^T (x' - x_t)$$

$$\text{s.t. } x' \in S_{\mu_\theta(x_t), \sqrt{n}\sigma_t}^n$$



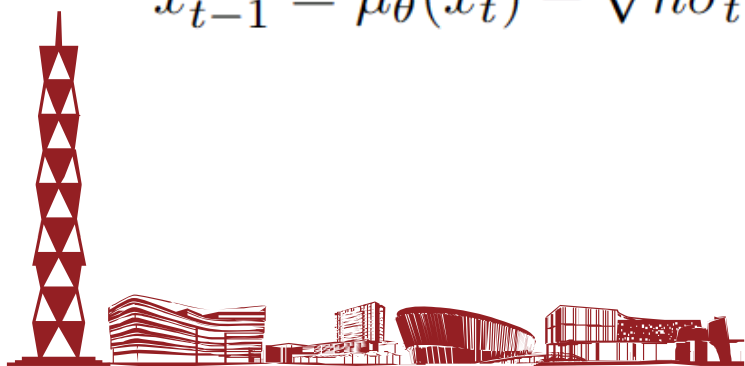
Closed-form Solution:

$$x_{t-1}^* = \mu_\theta(x_t) - \sqrt{n}\sigma_t \frac{\nabla_{x_t} L(\hat{x}_0(x_t), y)}{\|\nabla_{x_t} L(\hat{x}_0(x_t), y)\|_2}.$$

Plug-in with **no extra computational overhead.**

```
diff=measure-self.forward(x_0_hat)
dist=norm(diff)
grad=autograd.grad(
    outputs=distance, inputs=x_t)[0]
# DPS: x_t=x_t_mean+sigma_t*eps_t-
#       grad*self.gamma
# Ours:
norm=norm(grad, dim=[1, 2, 3])
x_t=x_t_mean-sqrt(n)*sigma_t*grad/norm
```

Adaptive and larger step sizes: The step size of DSG is $\sqrt{n}\sigma_t$, which decreases fast in the early stage and then coverage in the final stage. The largest step size in time T is 400x larger than DPS.

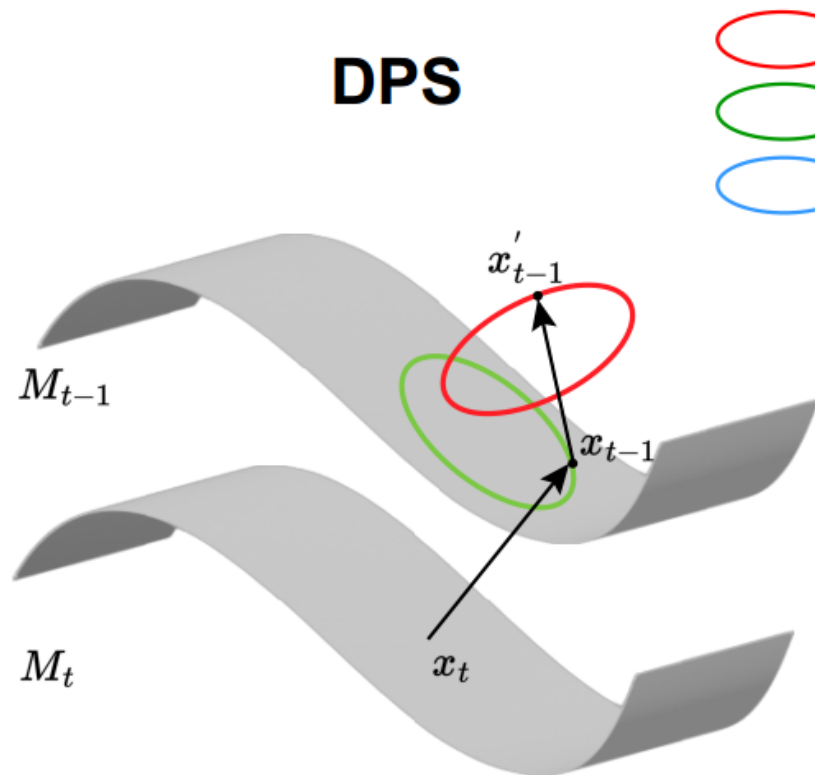


立志成才 报国裕民

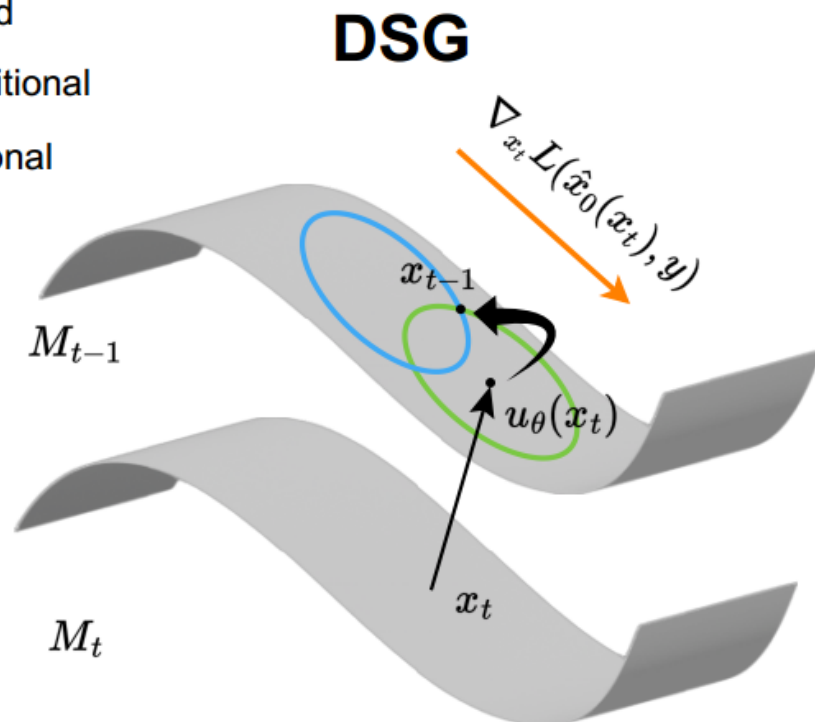
Diffusion with Spherical Gaussian Constraint(DSG)



上海科技大学
ShanghaiTech University



- Deviated
- Unconditional
- Conditional



$$x_{t-1} \sim N(u_{\theta}(x_t) - \gamma \nabla_{x_t} L(\hat{x}_0(x_t), y), \sigma_t * I)$$

$$x_{t-1} = u_{\theta}(x_t) - \sqrt{n} \sigma_t \frac{\nabla_{x_t} L(\hat{x}_0(x_t), y)}{\|\nabla_{x_t} L(\hat{x}_0(x_t), y)\|_2}$$

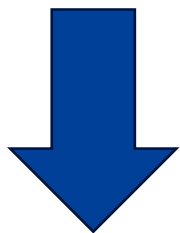
Diffusion with Spherical Gaussian Constraint(DSG)



上海科技大学
ShanghaiTech University

Enhancing diversity like Classifier-free manner:

$$x_{t-1}^* = \mu_\theta(x_t) - \sqrt{n}\sigma_t \frac{\nabla_{x_t} L(\hat{x}_0(x_t), y)}{\|\nabla_{x_t} L(\hat{x}_0(x_t), y)\|_2}.$$



$$d^{sample} = \sigma_t \epsilon_t$$

$$d^* = -\sqrt{n}\sigma_t \cdot \frac{\nabla_{x_t} L(\hat{x}_0(x_t), y)}{\|\nabla_{x_t} L(\hat{x}_0(x_t), y)\|}$$

$$d_m = d^{sample} + g_r(d^* - d^{sample}).$$

$$x_{t-1} = \mu_\theta(x_t) + r \frac{d_m}{\|d_m\|},$$

Algorithm 1 Diffusion with Spherical Gaussian constraint

Input: pure noise $x_T \sim N(0, I)$, guidance interval i , guidance rate g_r

for $t = T$ **to** 1 **do**

$\epsilon_t \sim N(0, I)$

$\hat{x}_0(x_t) = (x_t - \sqrt{1 - \alpha_t} \epsilon_\theta(x_t, t)) / \sqrt{\alpha_t}$

$\mu_\theta(x_t) = \sqrt{\alpha_{t-1}} \hat{x}_0(x_t) + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \epsilon_\theta(x_t, t)$

// Guidance with Spherical Gaussian Constraint

$d^* = -\sqrt{n}\sigma_t \cdot \frac{\nabla_{x_t} L(\hat{x}_0(x_t), y)}{\|\nabla_{x_t} L(\hat{x}_0(x_t), y)\|}$

$d^{sample} = \sigma_t \epsilon_t$

$d_m = d^{sample} + g_r(d^* - d^{sample})$

$x_{t-1} = \mu_\theta(x_t) + r \frac{d_m}{\|d_m\|}$

end for

Return x_0

Experiment (Linear Case)



上海科技大学
ShanghaiTech University

- Tasks: 1. Noisy linear inverse problem in image restoration(IR):



Formally, the general form of the *forward* model³ can be stated as

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \mathbf{n}, \quad \mathbf{y}, \mathbf{n} \in \mathbb{R}^n, \mathbf{x}_0 \in \mathbb{R}^d, \quad (6)$$

where $\mathcal{A}(\cdot) : \mathbb{R}^d \mapsto \mathbb{R}^n$ is the forward measurement operator and $\mathbf{n}$ is the measurement noise. In the case of white Gaussian noise, $\mathbf{n} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$. In explicit form, $p(\mathbf{y}|\mathbf{x}_0) \sim \mathcal{N}(\mathbf{y}|\mathcal{A}(\mathbf{x}_0), \sigma^2 \mathbf{I})$. However, there does not exist explicit dependency between $\mathbf{y}$ and $\mathbf{x}_t$, as can be seen in the probabilistic graph from Fig. 2, where the blue dotted line remains unknown.

Super-resolution. The forward model for super-resolution is defined as

$$\mathbf{y} \sim \mathcal{N}(\mathbf{y}|\mathbf{L}^f \mathbf{x}, \sigma^2 \mathbf{I}), \quad (\text{Gaussian}) \quad (50)$$

$$\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{L}^f \mathbf{x}; \lambda), \quad (\text{Poisson}) \quad (51)$$

where $\mathbf{L}^f \in \mathbb{R}^{n \times d}$ represents the bicubic downsampling block Hankel matrix with the factor f , and $\mathcal{P}$ denotes the Poisson distribution with the parameter λ .

Inpainting. For both box-type and random-type inpainting, the forward model reads

$$\mathbf{y} \sim \mathcal{N}(\mathbf{y}|\mathbf{P}\mathbf{x}, \sigma^2 \mathbf{I}), \quad (\text{Gaussian}) \quad (52)$$

$$\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{P}\mathbf{x}; \lambda), \quad (\text{Poisson}) \quad (53)$$

where $\mathbf{P} \in \{0, 1\}^{n \times d}$ is the masking matrix that consists of elementary unit vectors.

Linear Deblurring. For both Gaussian and motion deblurring, the measurement model is given as

$$\mathbf{y} \sim \mathcal{N}(\mathbf{y}|\mathbf{C}^\psi \mathbf{x}, \sigma^2 \mathbf{I}), \quad (\text{Gaussian}) \quad (54)$$

$$\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{C}^\psi \mathbf{x}; \lambda), \quad (\text{Poisson}) \quad (55)$$

where $\mathbf{C}^\psi \in \mathbb{R}^{n \times d}$ is the block Hankel matrix that effectively induces convolution with the given blur kernel ψ .

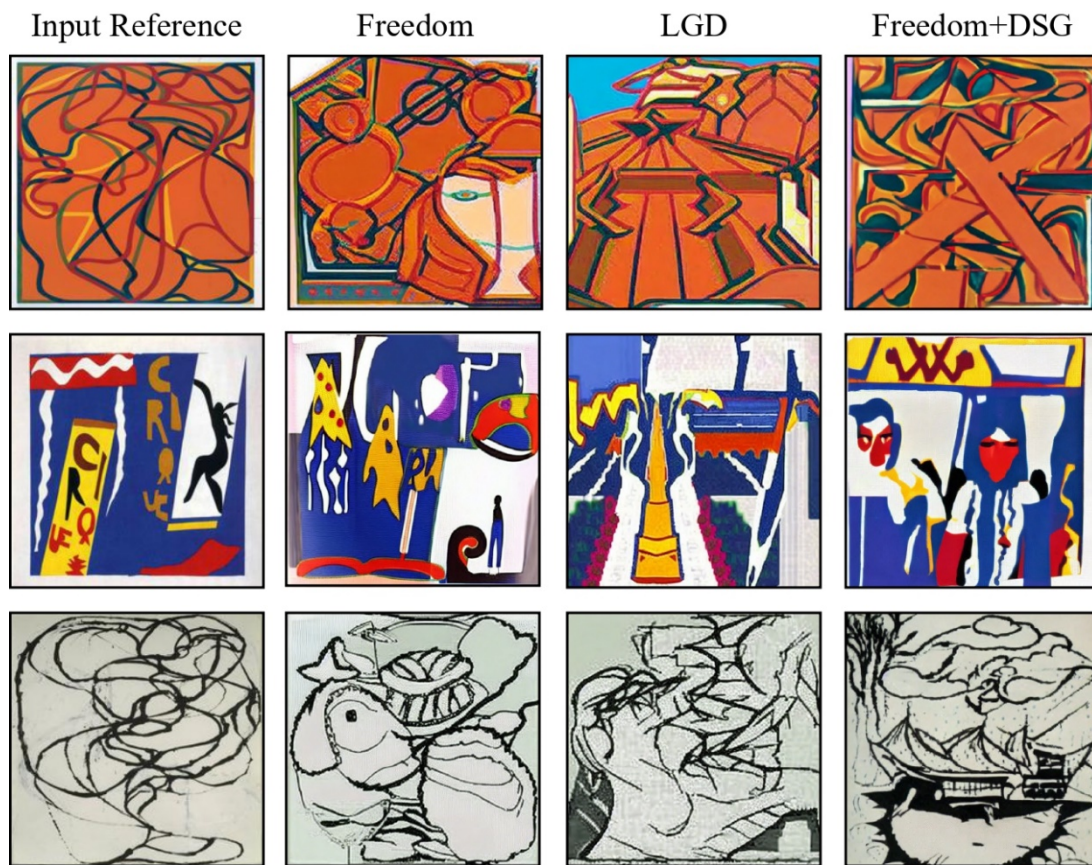
立志成才 报国裕民

Experiment (Non-Linear Case)

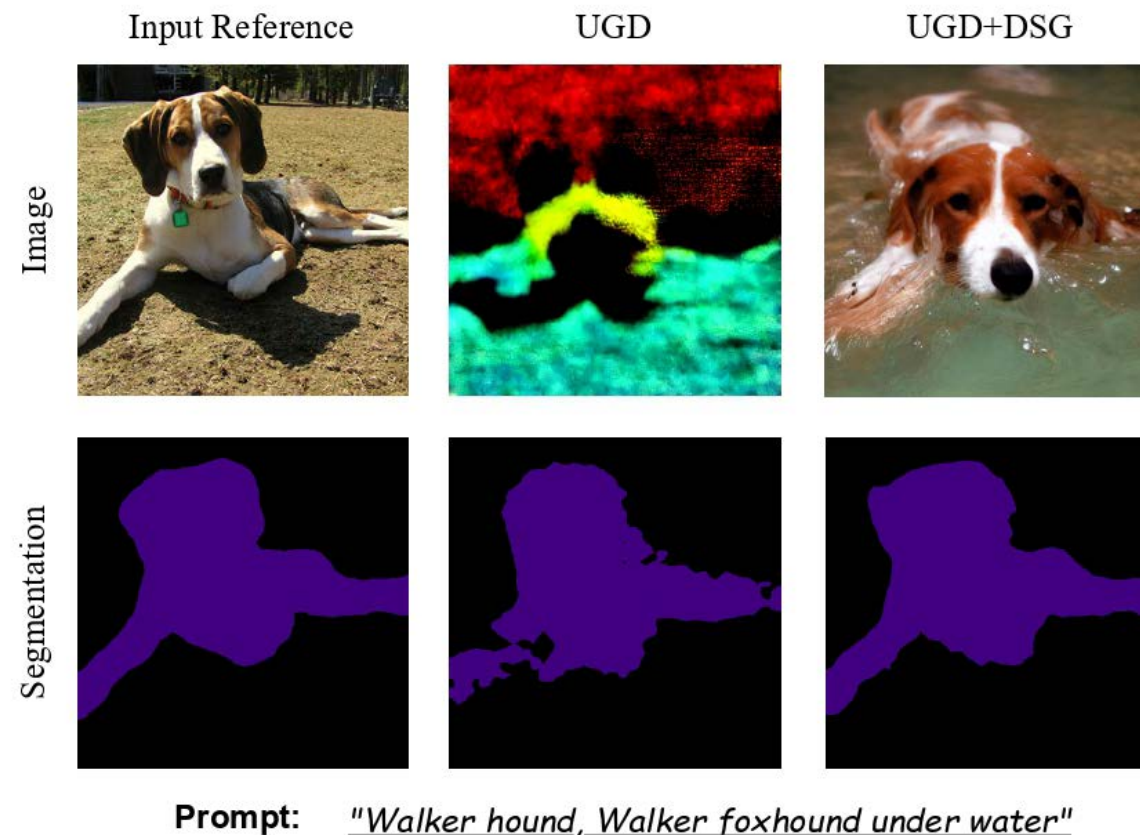


上海科技大学
ShanghaiTech University

• Tasks: 2. Style Guidance:



3. Text-Segmentation Guidance:



Prompt: "Walker hound, Walker foxhound under water"

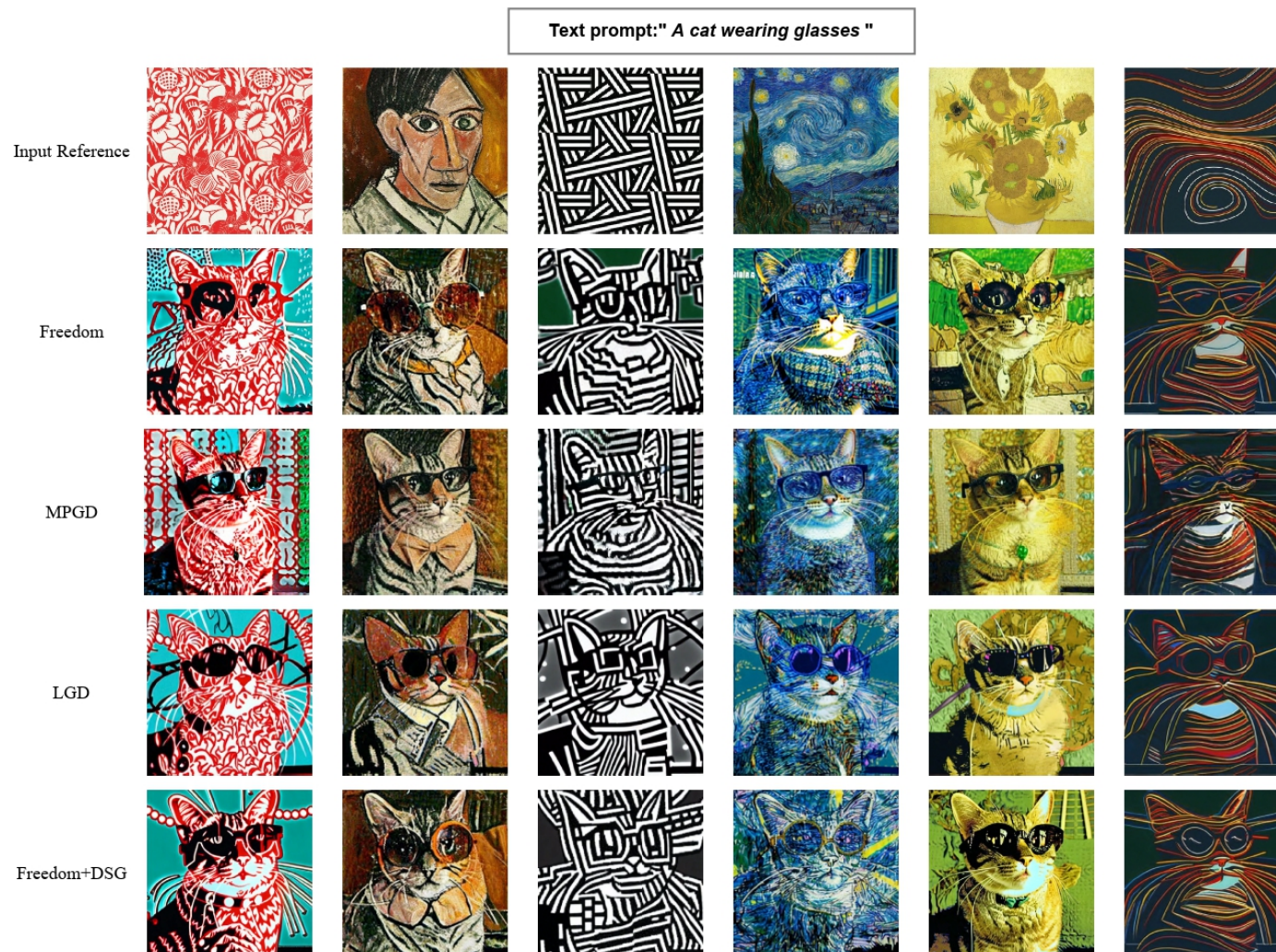
立志成才 报国裕民

Experiment (Non-Linear Case)



上海科技大学
ShanghaiTech University

- Tasks: 4. Text-Style Guidance:



立志成才 报国裕民

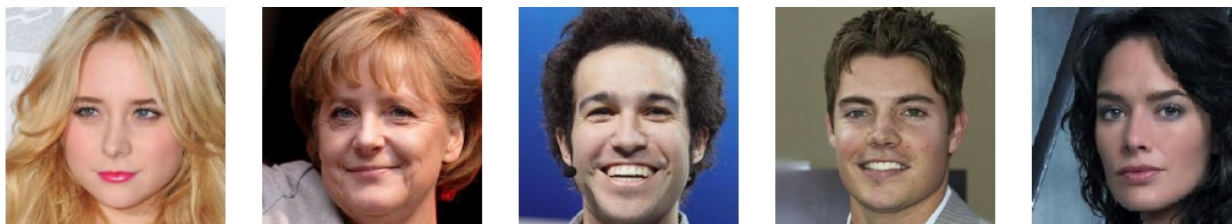
Experiment (Non-Linear Case)



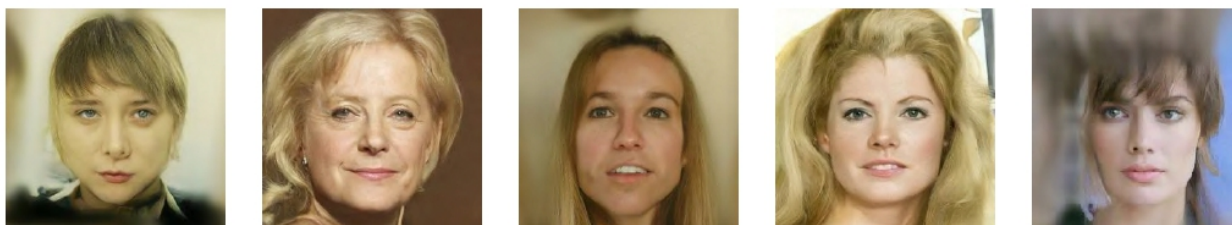
上海科技大学
ShanghaiTech University

- Tasks: 5. FaceID Guidance:

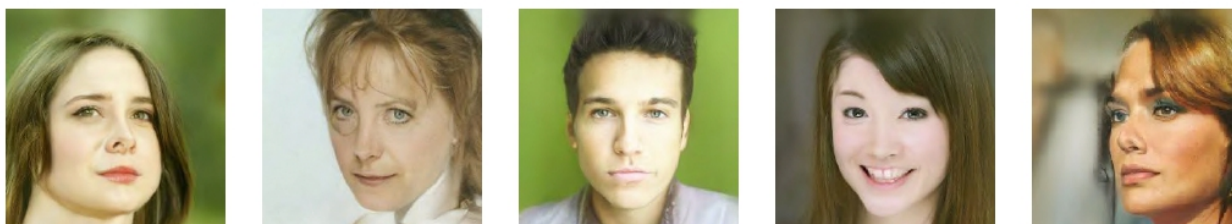
Input Reference



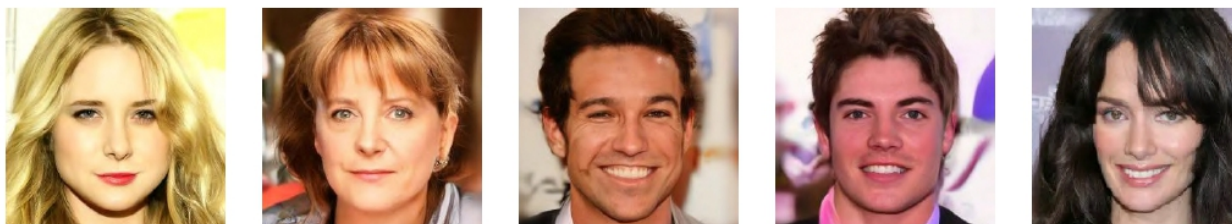
Freedom



LGD



Freedom+DSG



立志成才 报国裕民

Experiment (Linear Case)



SOTA performance in various tasks:

Table 1. Quantitative evaluation of the FFHQ dataset in Linear Inverse Problem

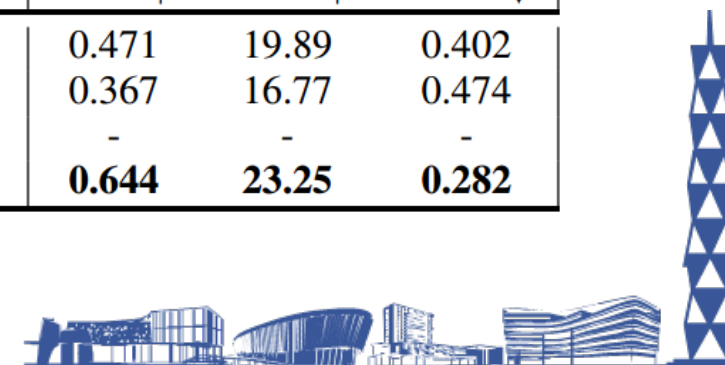
Methods	Inpainting		Super resolution		Gaussian deblurring	
	LPIPS↓	FID↓	LPIPS↓	FID↓	LPIPS↓	FID↓
DDRM (Kawar et al., 2022)	0.665	114.9	0.339	59.57	0.427	63.02
MCG (Chung et al., 2022)	0.414	39.19	0.637	144.5	0.550	95.04
PnP-ADMM (Chan et al., 2016)	0.677	114.7	0.433	97.27	0.519	100.6
Score-SDE (Song et al., 2020b)						
(ILVR (Choi et al., 2021))	0.659	127.1	0.701	170.7	0.667	120.3
DPS (Chung et al., 2023)	0.212	21.19	0.214	39.35	0.257	44.05
LGD (Song et al., 2023)	0.159	28.21	0.231	34.44	0.229	32.57
DPS+DSG(Ours)	0.115	15.77	0.211	30.30	0.208	28.22

Table 3. The quantitative result in CelebA-HQ test set for FaceID guidance

Method	FaceID	FID	KID
Freedom	0.545	39.61	0.0165
LGD	0.533	39.13	0.0170
Freedom+DSG	0.371	34.29	0.0092

Table 5. Quantitative results in Linear Inverse Problem in ImageNet 256*256

Methods	Inpainting			Super resolution			Gaussian deblurring		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
DPS (Chung et al., 2023)	0.828	28.57	0.189	0.635	23.73	0.317	0.471	19.89	0.402
LGD (Song et al., 2023)	0.825	28.11	0.191	0.633	23.13	0.318	0.367	16.77	0.474
DDNM (Wang et al., 2022)	0.875	28.76	0.125	-	-	-	-	-	-
DPS+DSG(Ours)	0.879	29.20	0.116	0.672	23.86	0.282	0.644	23.25	0.282



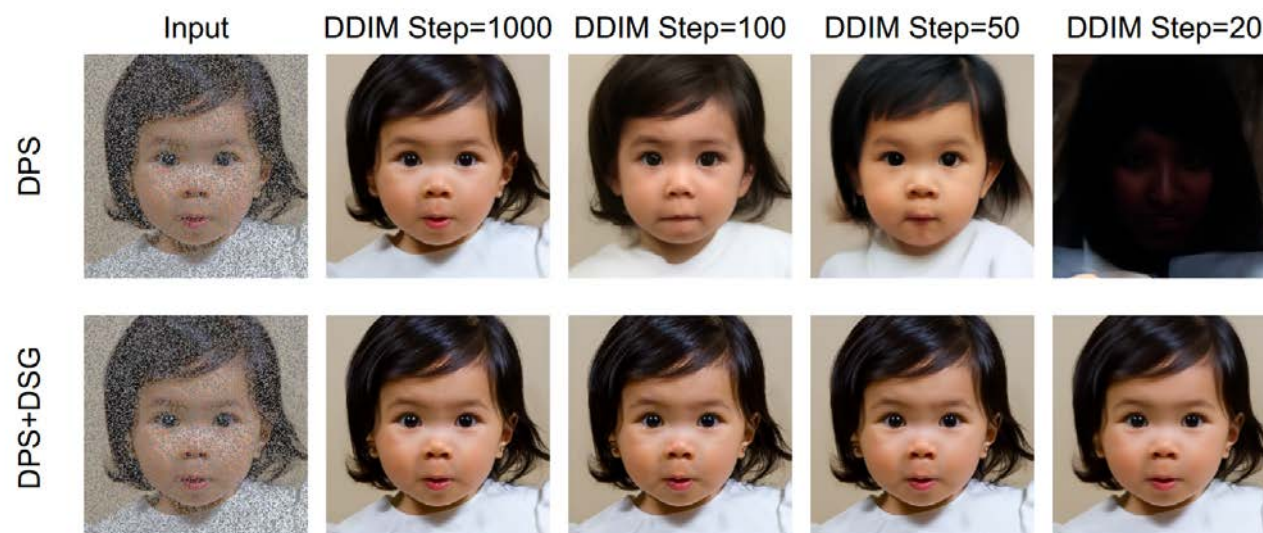
Experiment (Ablation Study)



More robust with smaller DDIM steps:

Table 2. Quantitative evaluation with different DDIM step in Linear Inverse Problem

Methods	Inpainting			Super resolution			Gaussian deblurring			DDIM steps
	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	
DPS (Chung et al., 2023)	0.638	22.40	0.348	0.535	19.70	0.396	0.614	22.31	0.337	100
DPS+DSG(Ours)	0.899	30.95	0.132	0.784	26.82	0.227	0.762	26.16	0.239	100
DPS (Chung et al., 2023)	0.529	18.80	0.424	0.460	17.24	0.461	0.492	17.84	0.440	50
DPS+DSG(Ours)	0.886	30.67	0.176	0.790	26.96	0.231	0.749	25.76	0.264	50
DPS (Chung et al., 2023)	0.360	12.44	0.566	0.347	12.63	0.566	0.323	10.94	0.609	20
DPS+DSG(Ours)	0.812	27.53	0.267	0.771	26.68	0.272	0.771	26.68	0.272	20



Outline



1. Controllable diffusion model for reliable generation

- Conditional Diffusion with Spherical Gaussian Constraint
- **Gaze-guided Hand-object Interaction Synthesis**
- Diffusion-based Reinforcement Learning

2. Fusing physical information into data-driven models

- Faster Implicit Differentiation
- More accurate 3D Motion Capture
- Reliable Decision-making via Safe RL

3. Federated Learning with Foundation Models

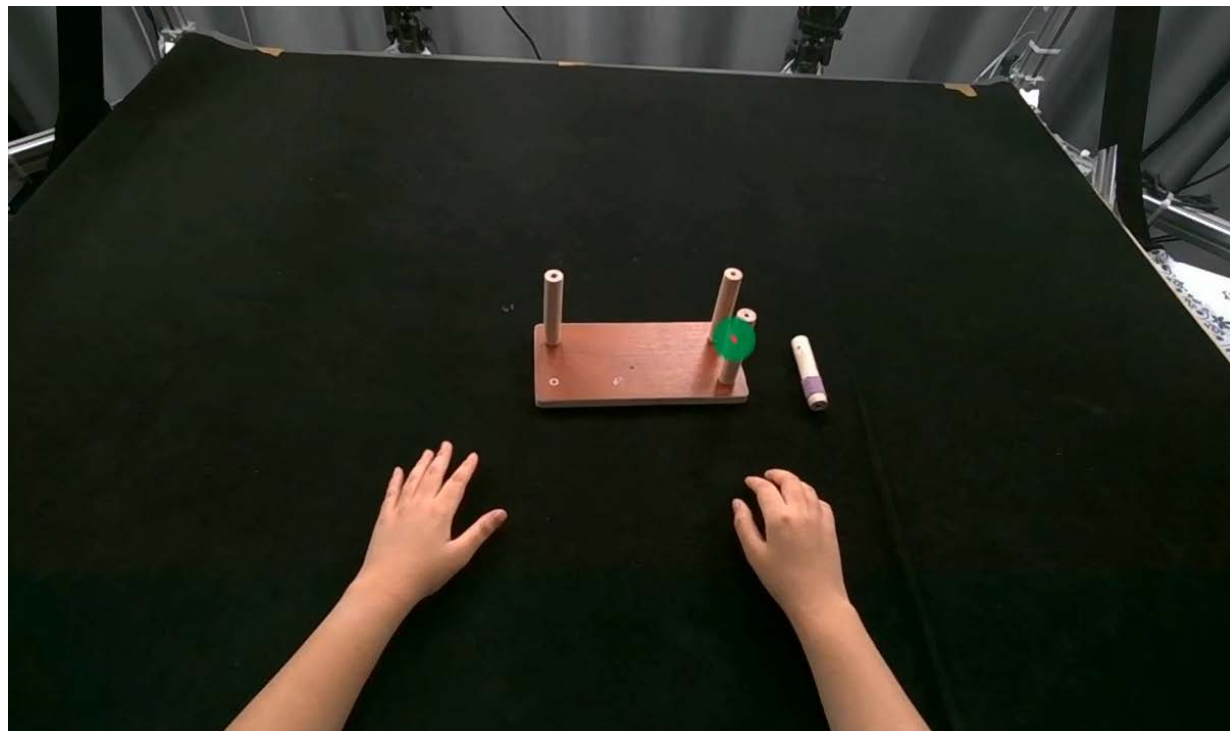
- Personalized Federated Learning with Transformer
- Prompt Tuning Boost Federated Learning
- Theoretical Analysis for prompt-based Federated Learning



Gaze-guided Hand-object Interaction Synthesis



Gaze serves as a significant behavioral signal, rich with information on intent.



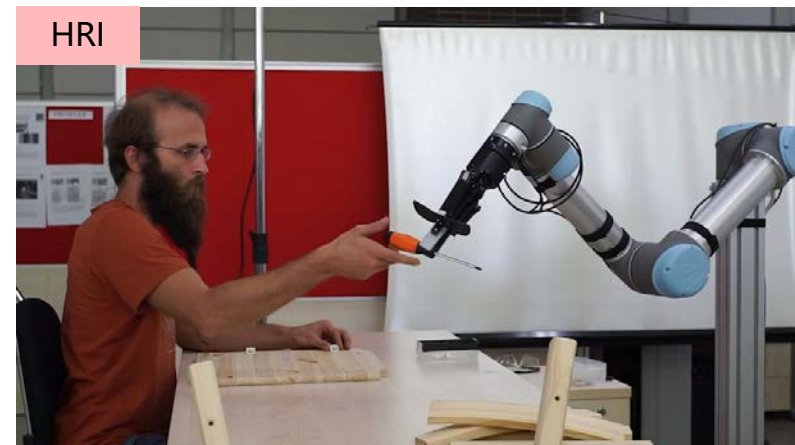
A Novel Task: Gaze-guided Hand-object Interaction Synthesis.

Application about Gaze

VR



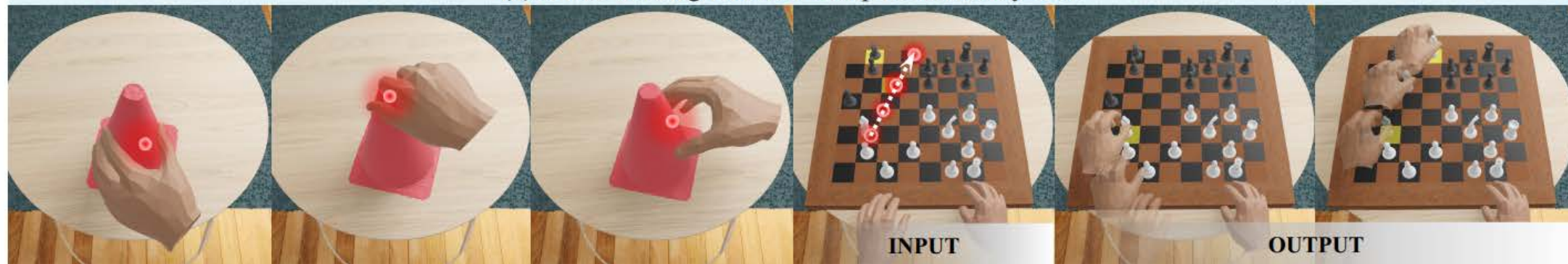
HRI



Gaze-guided Hand-object Interaction Synthesis



(a) Gazes in Long-Horizon Complex Assembly Task

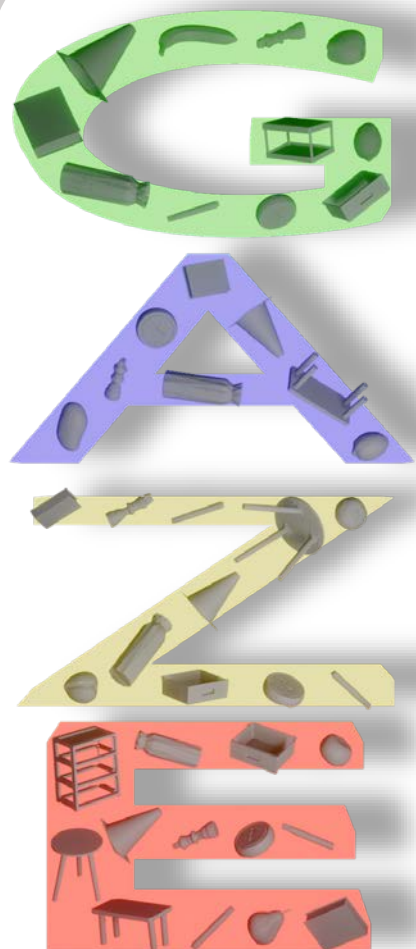


(b) Different Gazes indicate Different Interaction Types

(c) Task Overview



Gaze-guided Hand-object Interaction Synthesis

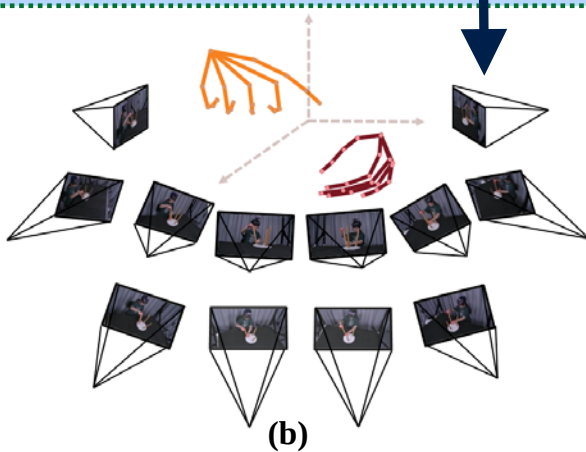


ZCAM Coordinate System

📷 ×12



Per-view
Keypoint
Extraction



Fit MANO

Time



(e)

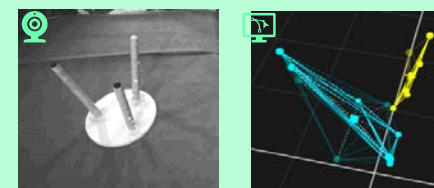
Ego-view Coordinate System

Gaze
Tracking



📍 Gaze Point (c)

Optitrack & Object
Coordinate System



ICP

Feature
Matching



(d)

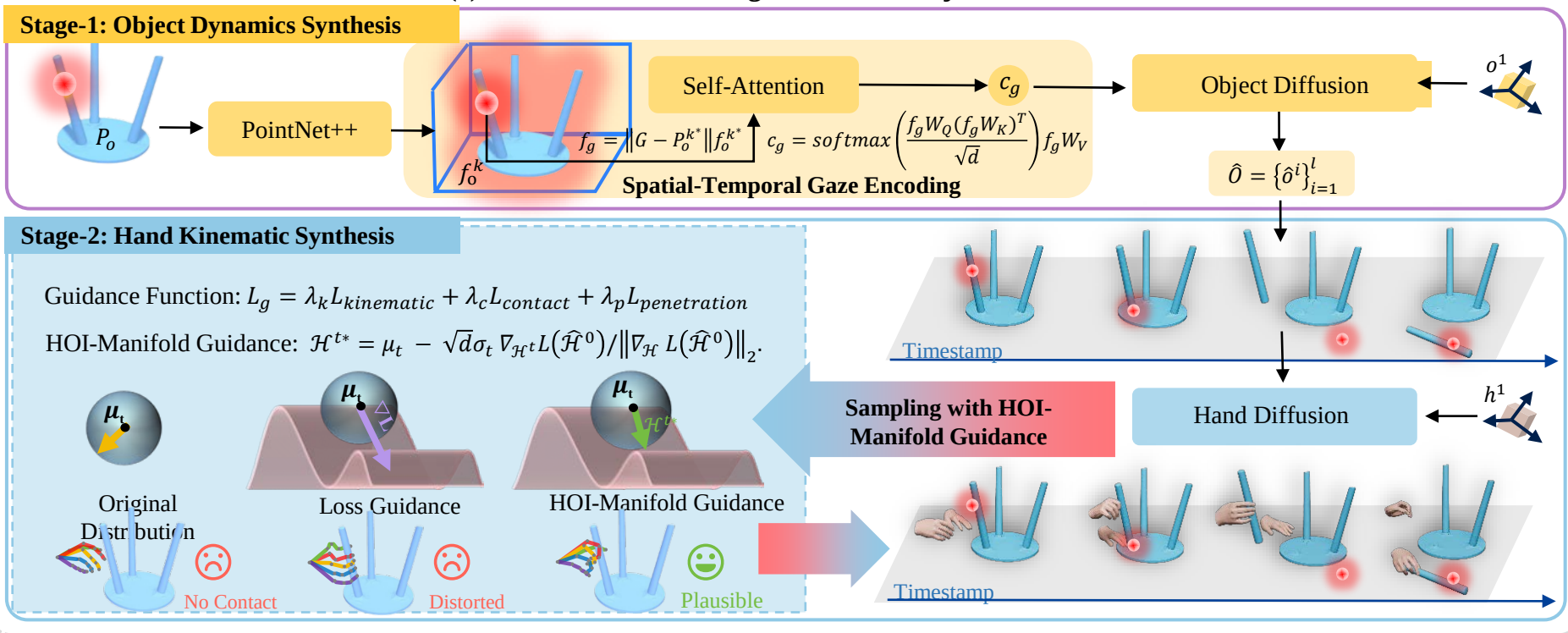
Ego Calibration



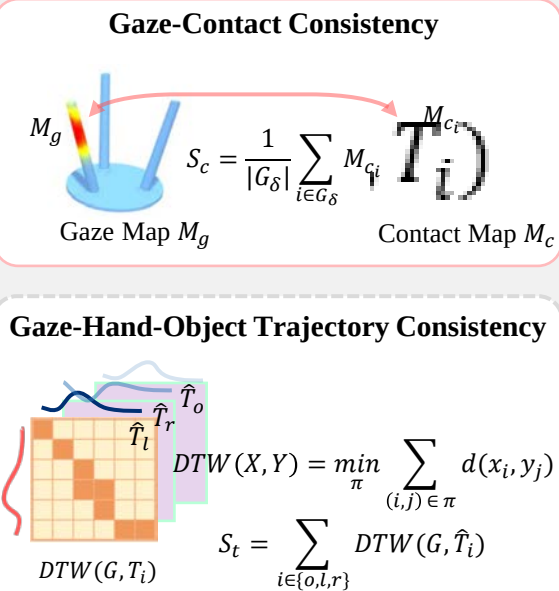
Gaze-guided Hand-object Interaction Synthesis



(a) GHO-Diffusion: Gaze-guided Hand-Object Diffusion



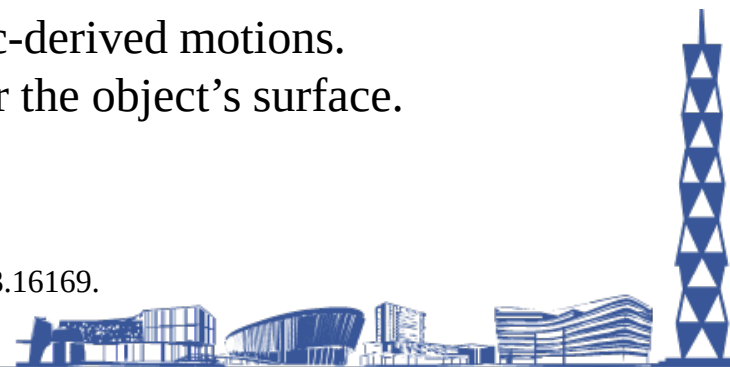
(b) Motion Selection via Gaze-Interaction Consistency Score



Hand-Object Kinematic Guidance: ensure consistency between generated and kinematic-derived motions.

Hand-Object Contact Guidance: minimize the distance between hand joints that are near the object's surface.

Hand-Object Penetration Guidance: avoid penetration of hand-object interaction.

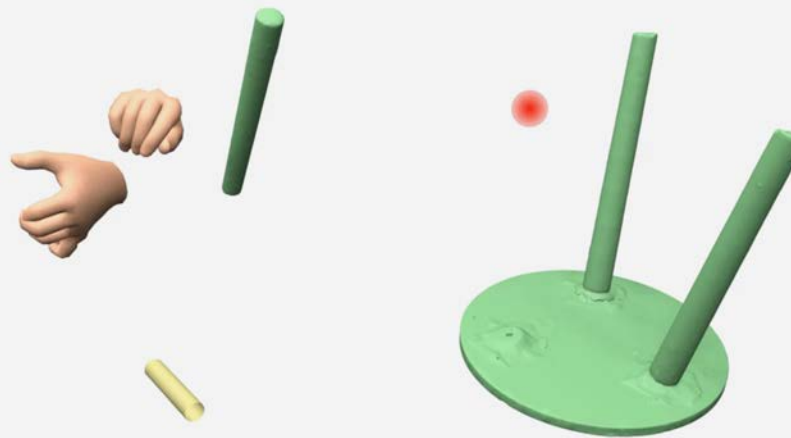


Results Visualization on **Seen** Objects

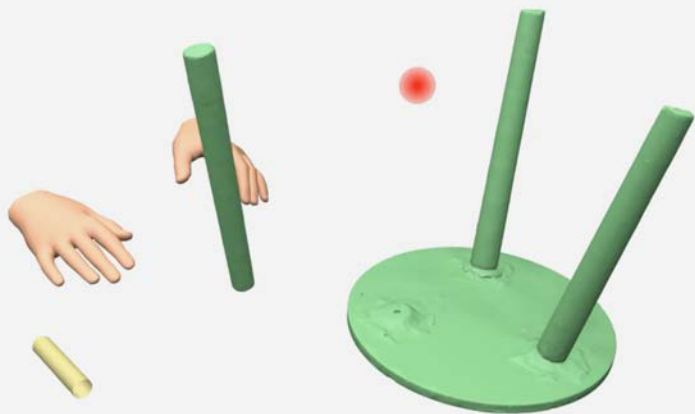
MDM*



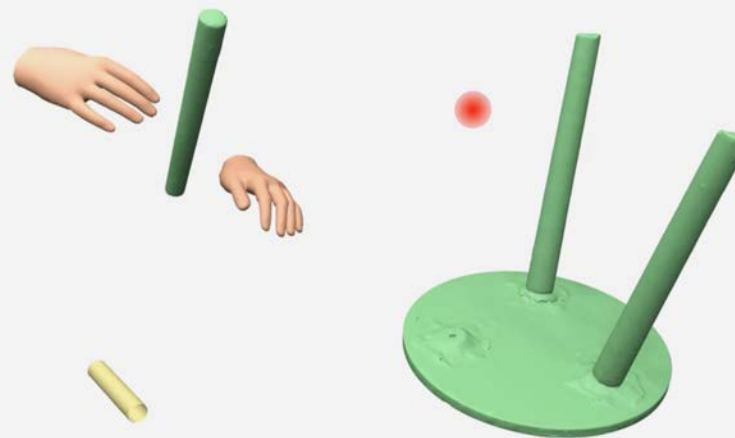
OMOMO*



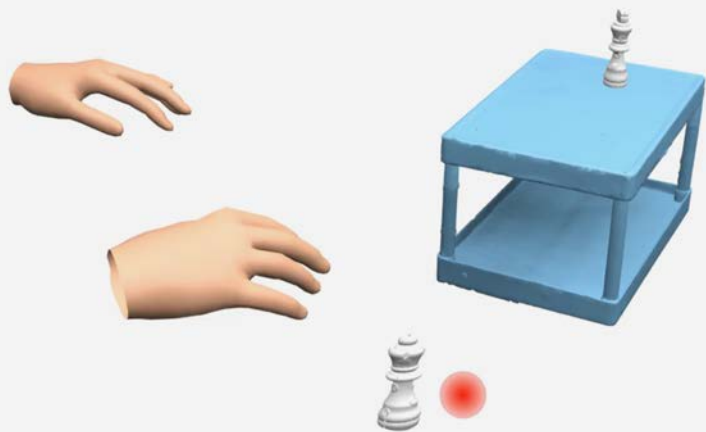
Text2HOI*



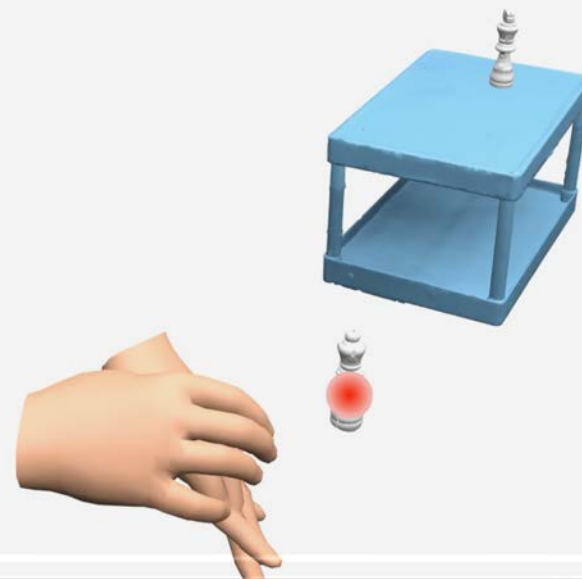
Ours



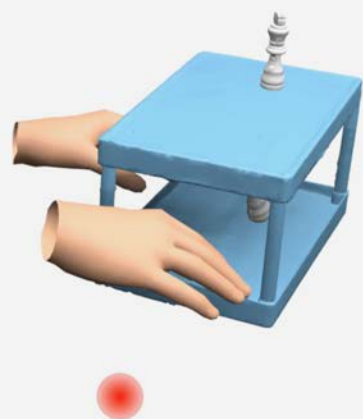
MDM*



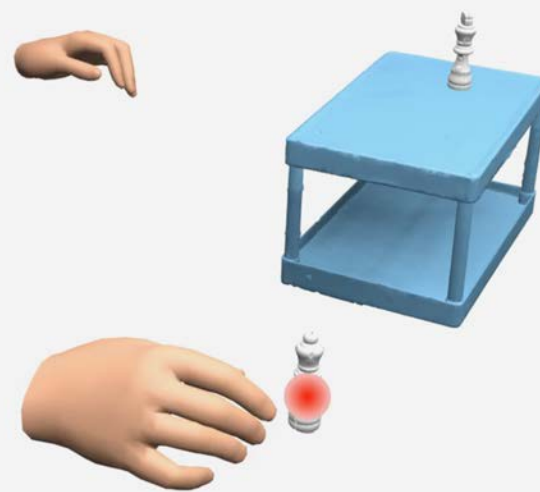
OMOMO*



Text2HOI*

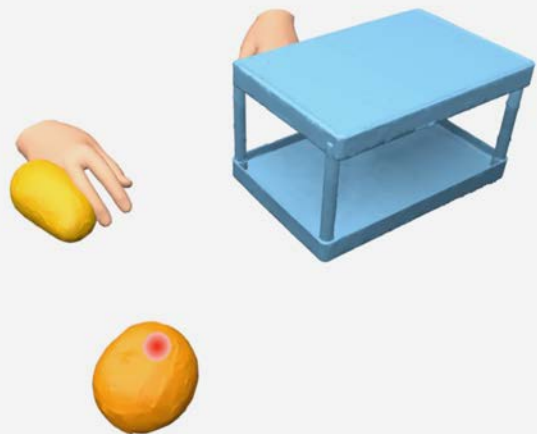


Ours

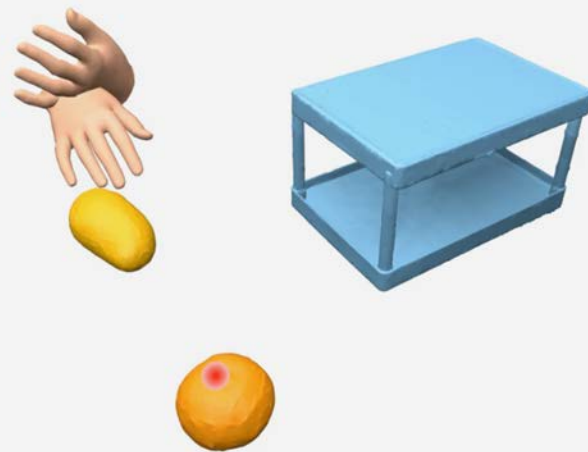


Results Visualization on **Unseen** Objects

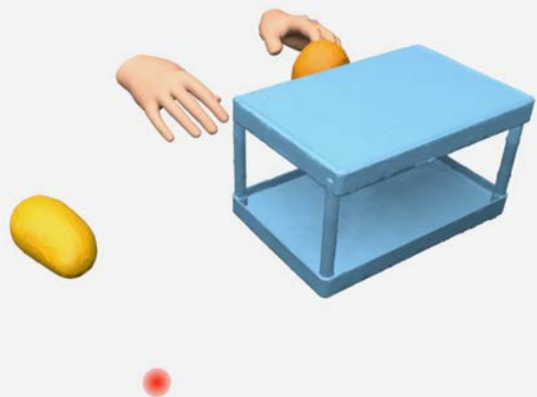
MDM*



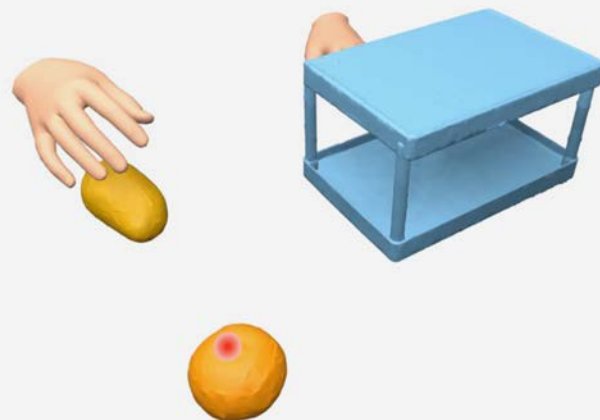
OMOMO*



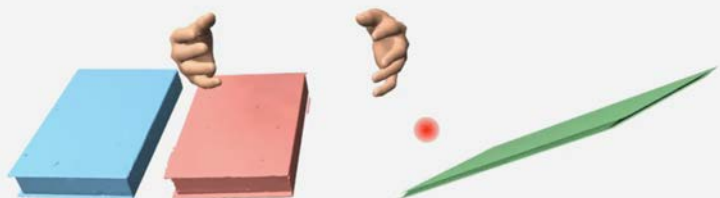
Text2HOI*



Ours



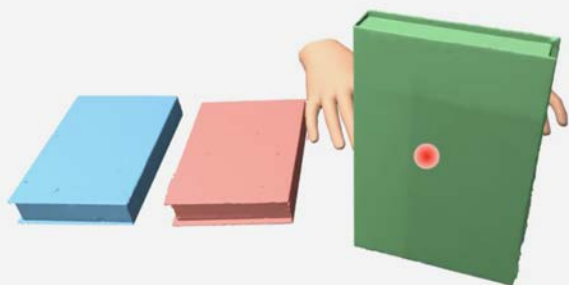
MDM*



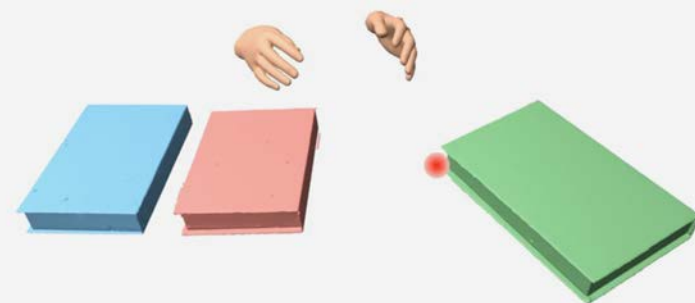
OMOMO*



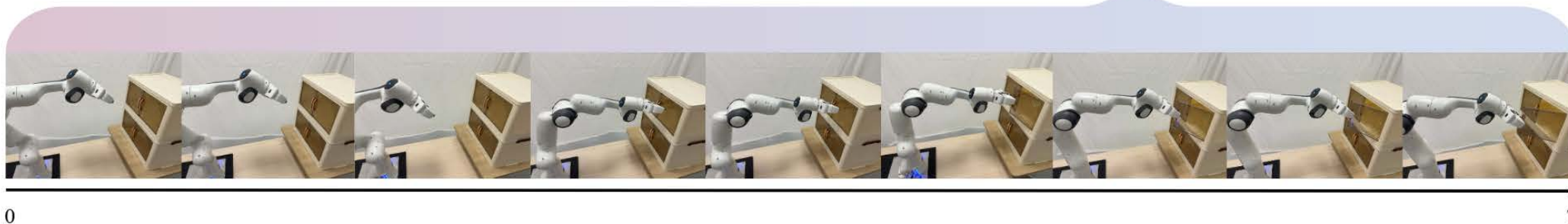
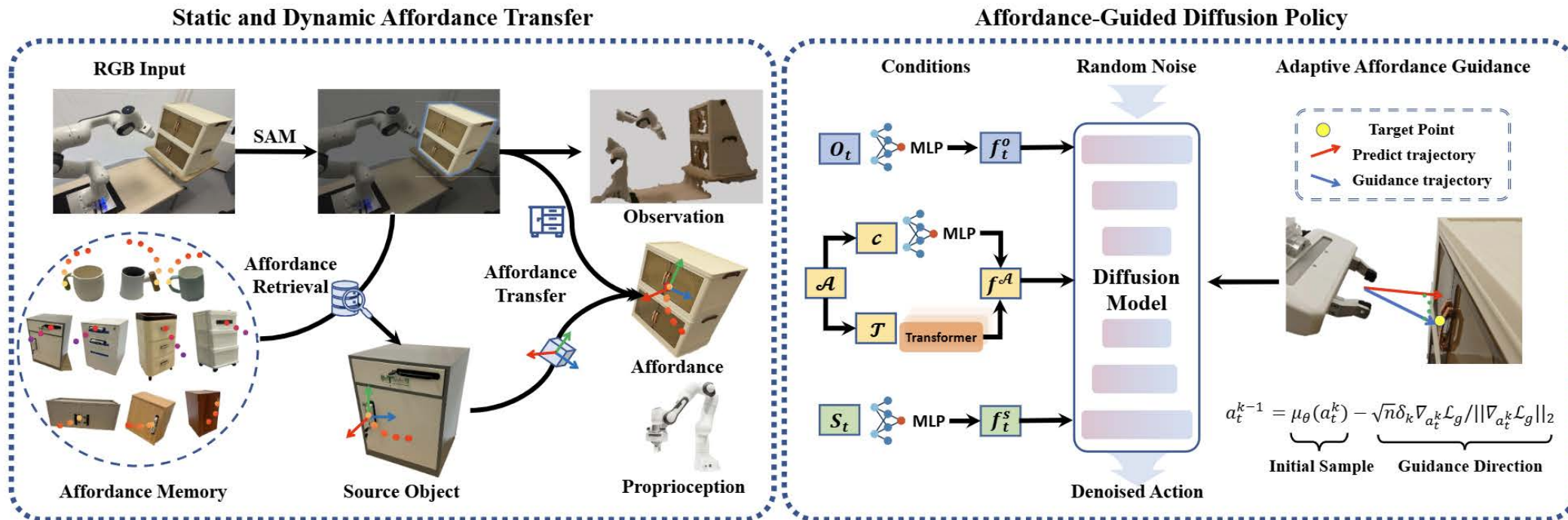
Text2HOI*



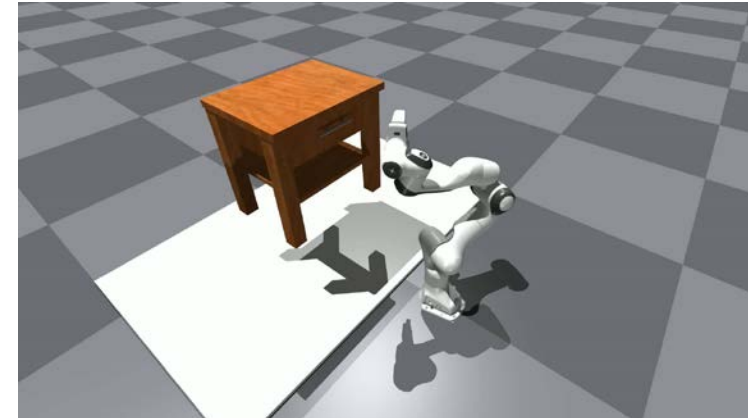
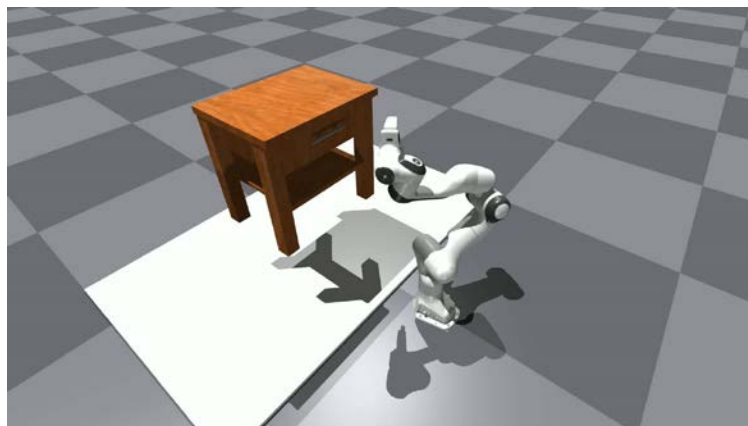
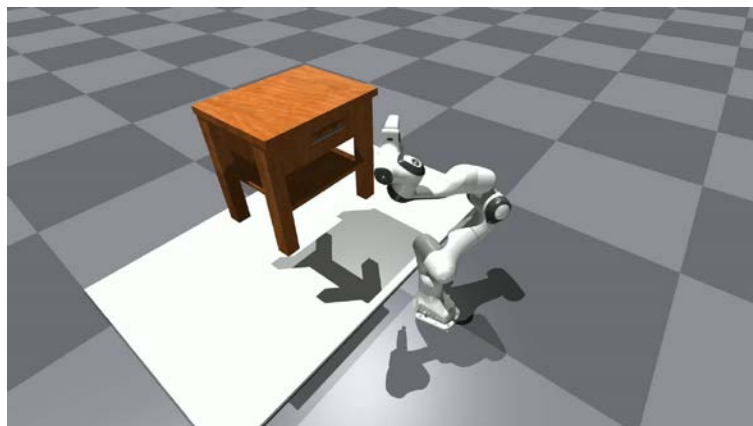
Ours



Generalizable Robot Manipulation via controllable Diffusion Policy



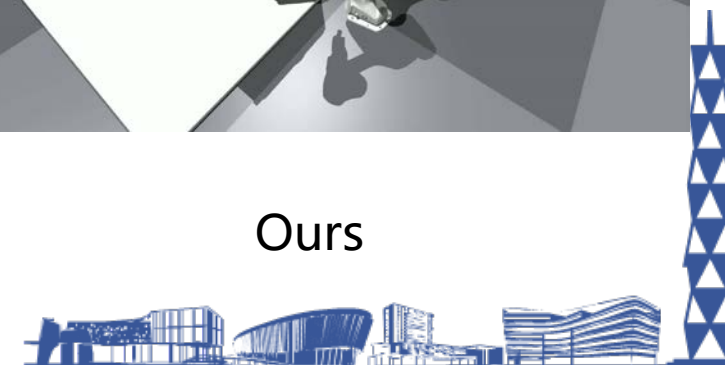
Generalizable Robot Manipulation via controllable Diffusion Policy



Diffusion Policy

3D Diffusion Policy

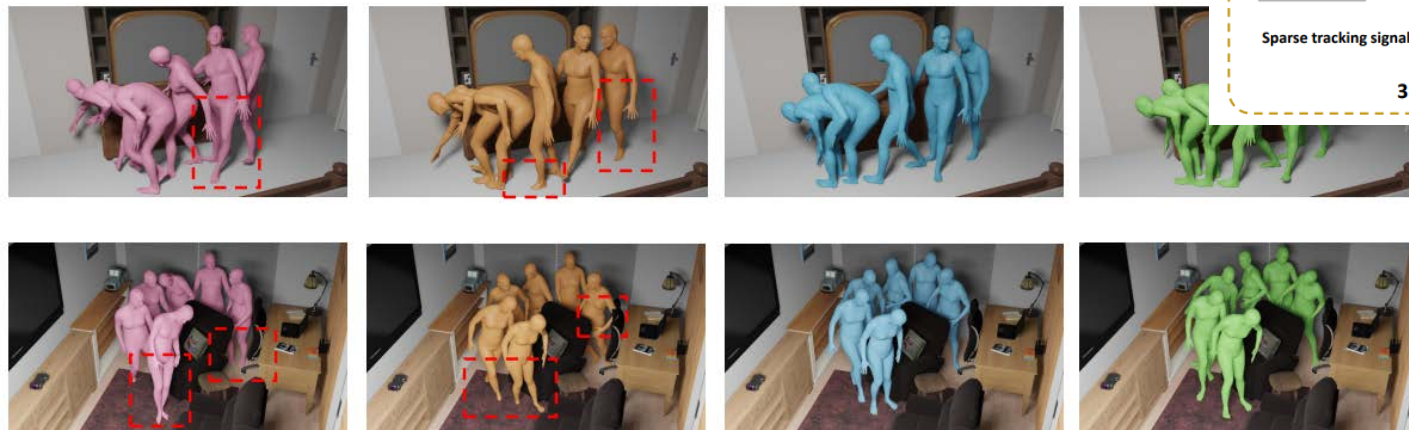
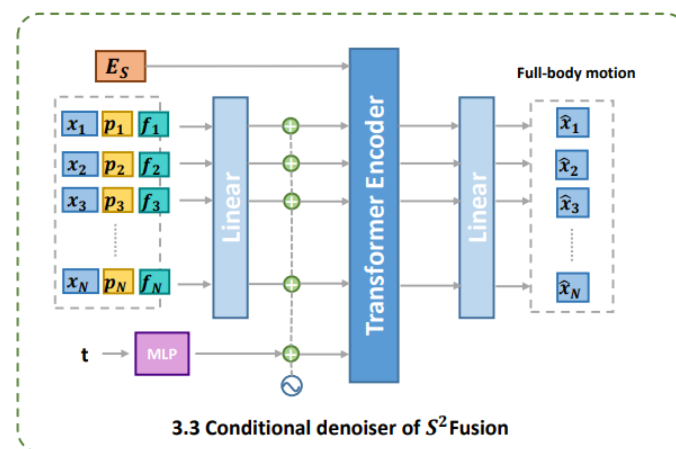
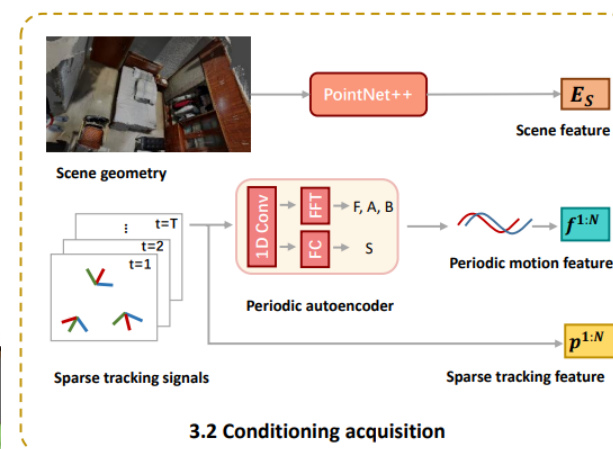
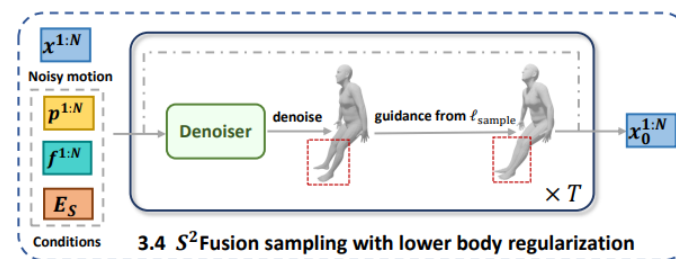
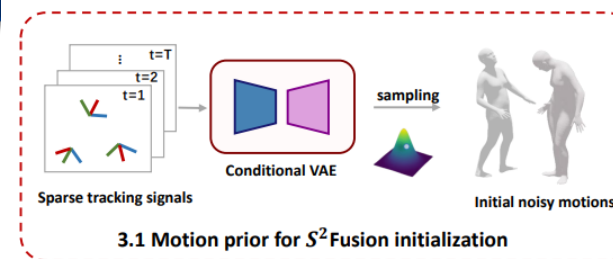
Ours



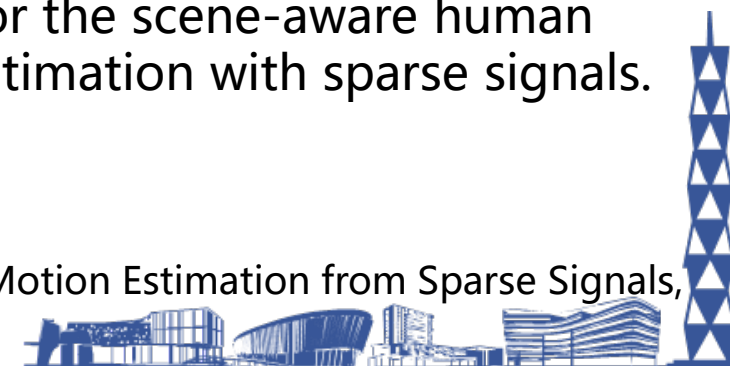
Reliable Generation for Human Pose Estimation via 3 IMUs and Scene



We introduce a new framework that combines scene information and sparse tracking signals for human motion estimation to greatly reduce the inherent ambiguities in estimating full-body motion.



We develop a unified diffusion method, tailored for the scene-aware human motion estimation with sparse signals.



Outline



1. Controllable diffusion model for reliable generation

- Conditional Diffusion with Spherical Gaussian Constraint
- Gaze-guided Hand-object Interaction Synthesis
- **Diffusion-based Reinforcement Learning**

2. Fusing physical information into data-driven models

- Faster Implicit Differentiation
- More accurate 3D Motion Capture
- Reliable Decision-making via Safe RL

3. Federated Learning with Foundation Models

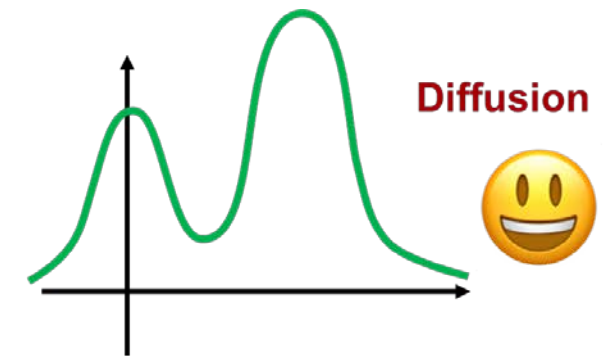
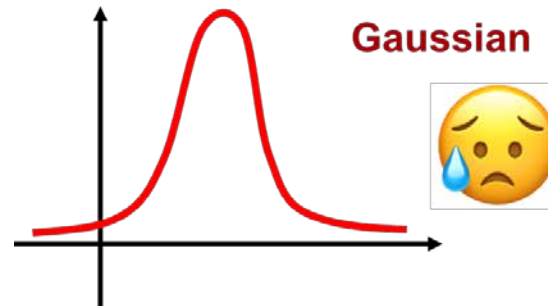
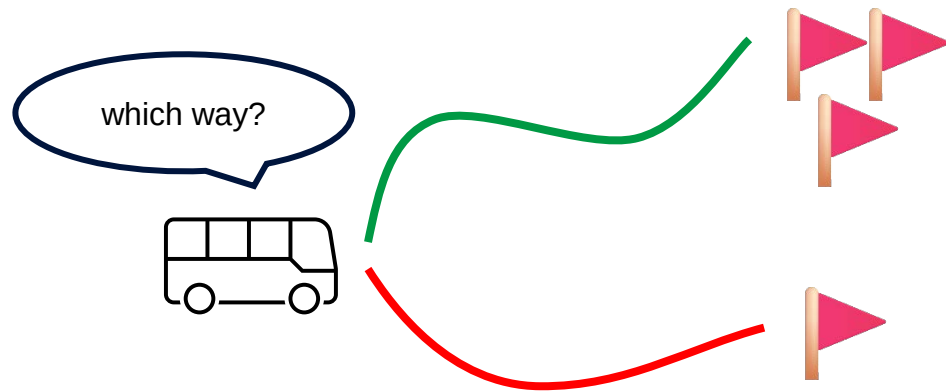
- Personalized Federated Learning with Transformer
- Prompt Tuning Boost Federated Learning
- Theoretical Analysis for prompt-based Federated Learning



Background: Diffusion in Online RL



1. **Exploration capability** of Gaussian policy or deterministic policy is limited
2. **Expressiveness and multimodality** of diffusion avoid policy falling into the local optimality



Background: Diffusion in Online RL



Directions of applying diffusion in Online RL:

1. Use the variational loss of diffusion to do policy improvement

$$\mathbb{E}_{t \sim [1, T], \mathbf{x}_0, \epsilon_t} \left[\left\| \epsilon_t - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, t \right) \right\|^2 \right].$$

2. Apply deterministic policy loss to train the diffusion model (like Diffusion-QL [1])

$$\pi = \arg \min_{\pi_\theta} \mathcal{L}(\theta) = \mathcal{L}_d(\theta) + \mathcal{L}_q(\theta) = \mathcal{L}_d(\theta) - \boxed{\alpha \cdot \mathbb{E}_{\mathbf{s} \sim \mathcal{D}, \mathbf{a}^0 \sim \pi_\theta} \left[Q_\phi \left(\mathbf{s}, \mathbf{a}^0 \right) \right]}$$

[1] Wang Z, Hunt J J, Zhou M. Diffusion policies as an expressive policy class for offline reinforcement learning[J]. arXiv preprint arXiv:2208.06193, 2022..

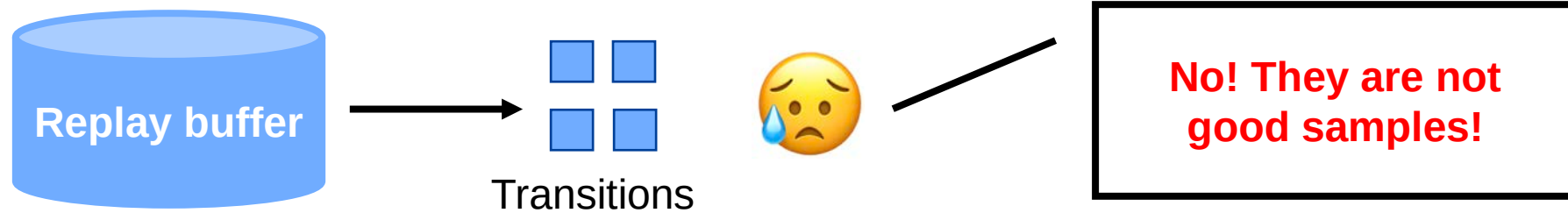


Background: Diffusion in Online RL

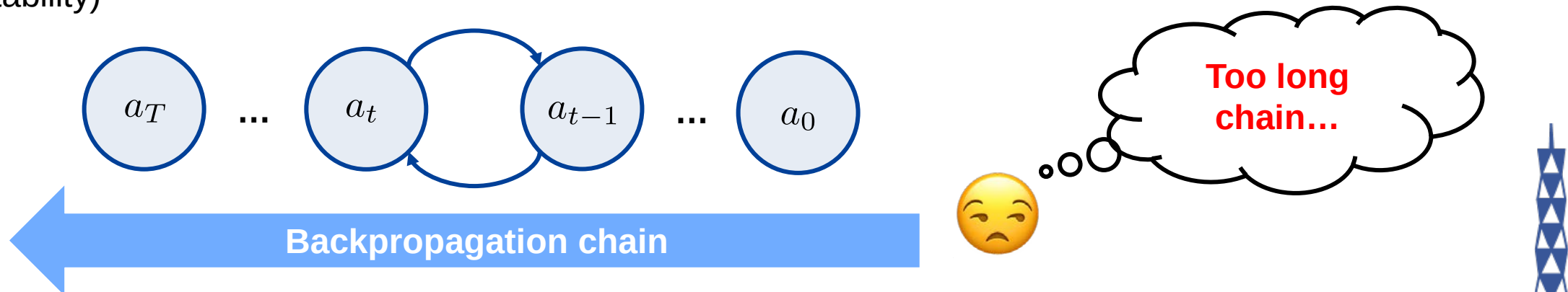


Difficulties:

Direction 1: Unavailability of **good** actions (not like imitation learning or offline RL)



Direction 2: Too long backpropagation chain (leads to high training cost and training instability)



Solution: Q-weighted Variational Policy Optimization



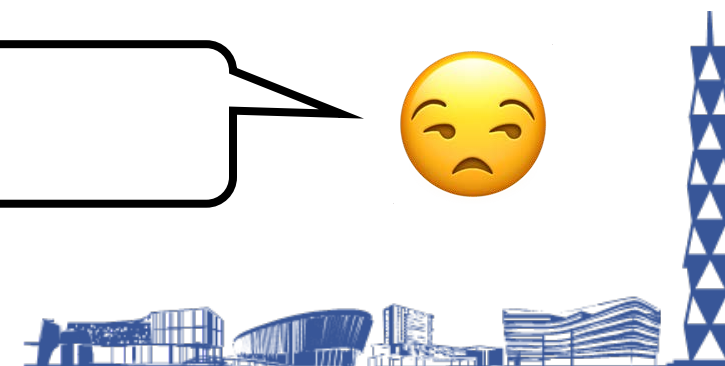
Can we use the variational loss to train diffusion policy?

Distinguish bad and good samples according to their **Q-value**?



But samples from replay buffer are still not good enough!

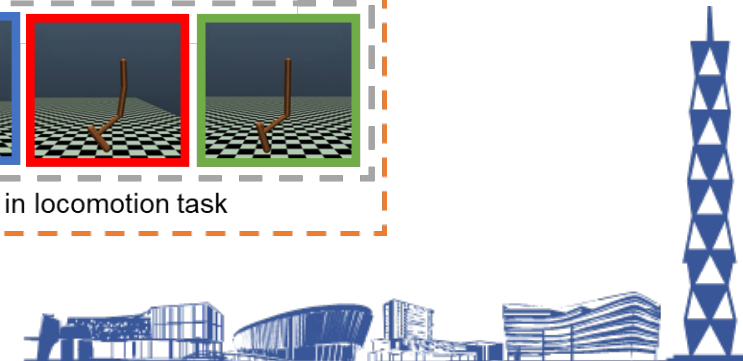
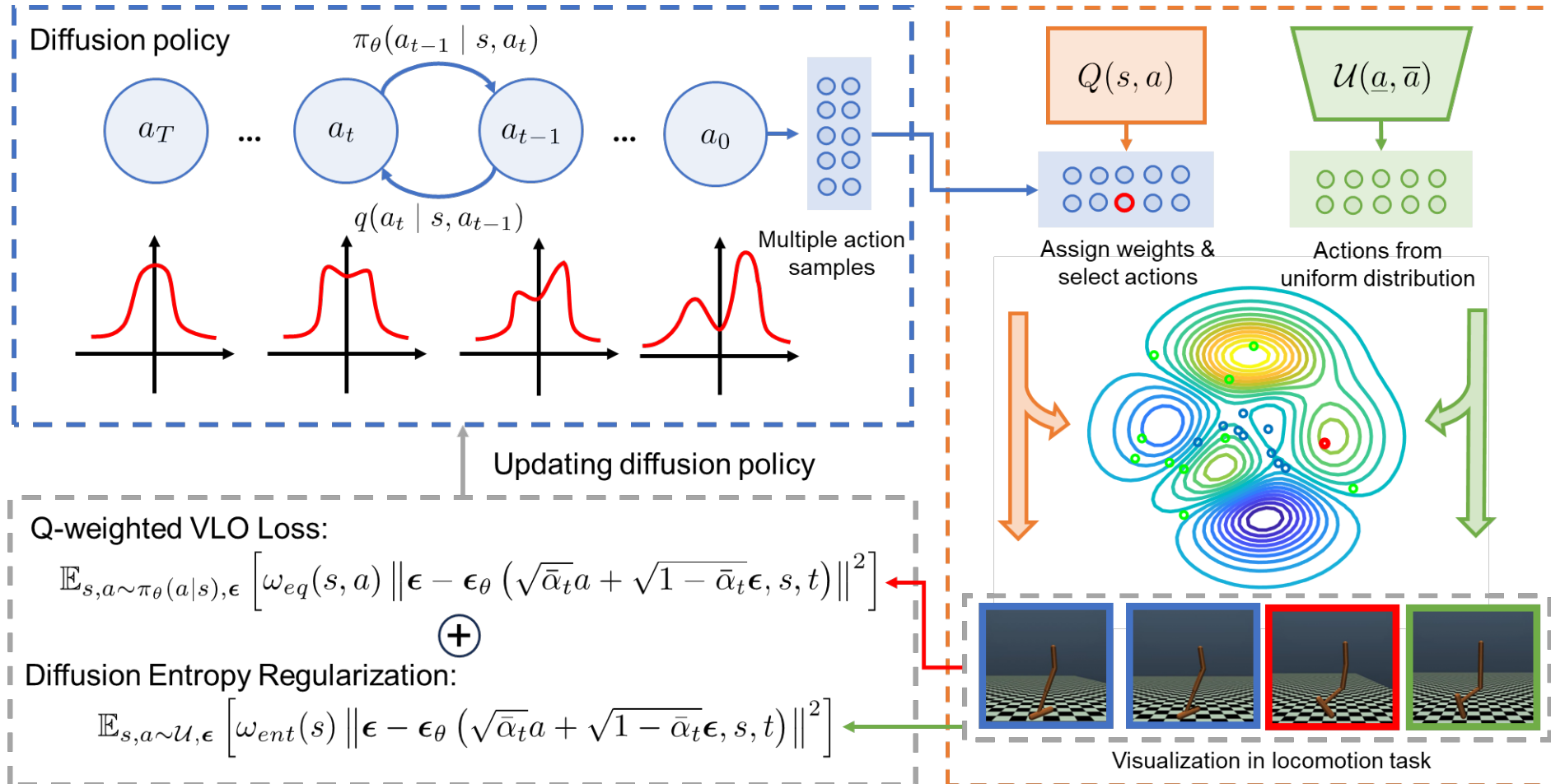
What if use samples from **current policy**?



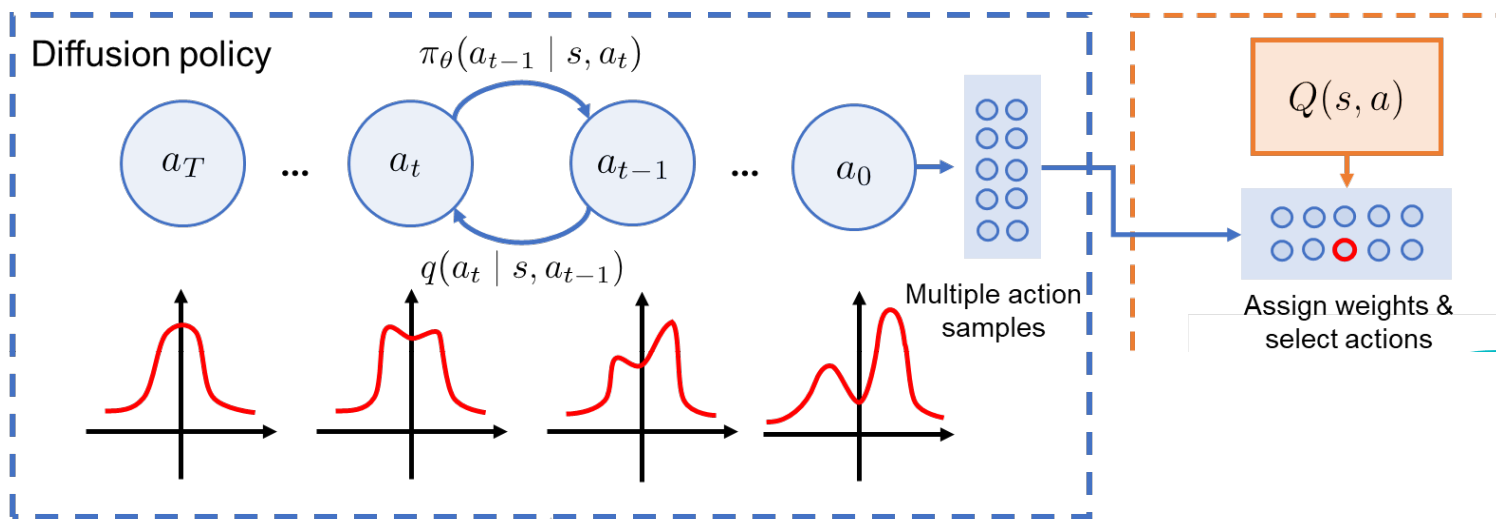
Solution: Q-weighted Variational Policy Optimization



Motivated by these two ideas, we propose Q-weighted variational policy optimization.



Q-weighted Variational Loss



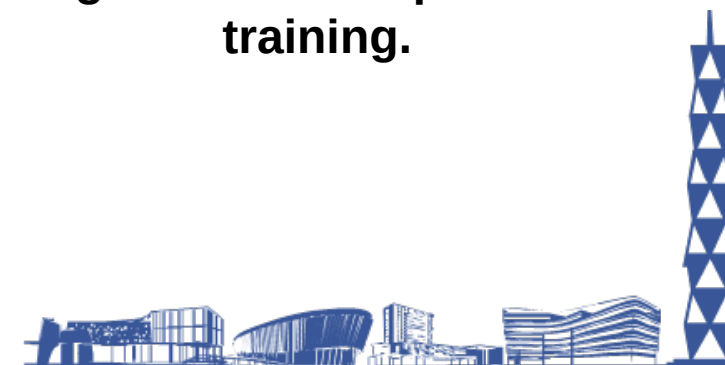
Q-weighted VLO Loss (tight lower bound of RL policy objective):

$$\mathbb{E}_{s, a \sim \pi_{\theta}(a|s), \epsilon} \left[\omega_{eq}(s, a) \left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\bar{\alpha}_t} a + \sqrt{1 - \bar{\alpha}_t} \epsilon, s, t \right) \right\|^2 \right]$$

Since the weight must be nonnegative, we define the weight as

$$\omega_{eq}(s, a) \triangleq \omega_{qadv}(s, a) = \begin{cases} A(s, a), & A(s, a) \geq 0 \\ 0, & A(s, a) < 0 \end{cases}$$

Only choose **the best** generated sample for training.



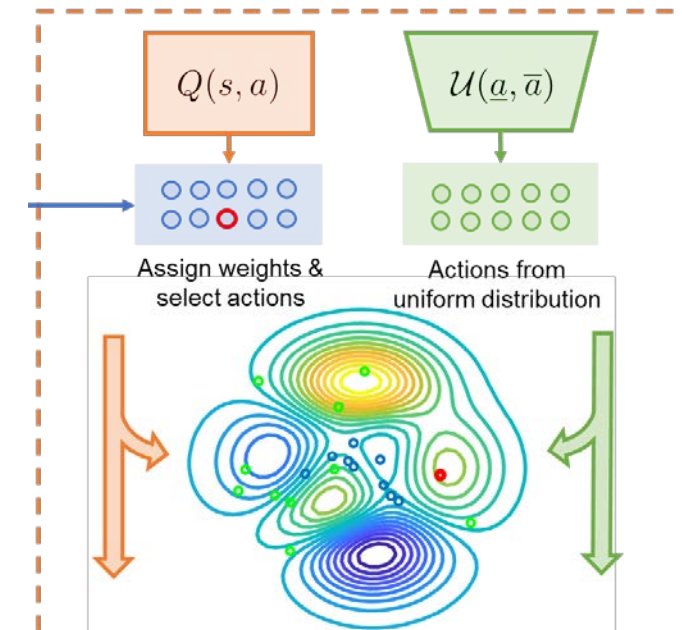
Diffusion Entropy Regularization



We also add Diffusion Entropy Regularization term in objective for enhancing exploration capability:

$$\mathbb{E}_{s,a \sim \mathcal{U}, \epsilon} \left[\omega_{ent}(s) \left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\bar{\alpha}_t} a + \sqrt{1 - \bar{\alpha}_t} \epsilon, s, t \right) \right\|^2 \right]$$

- Maximizing entropy can be viewed as **minimizing the distance** between current policy and the **uniform distribution**.
- Hence, we use samples from the uniform distribution to train diffusion policy for maximizing the diffusion entropy.



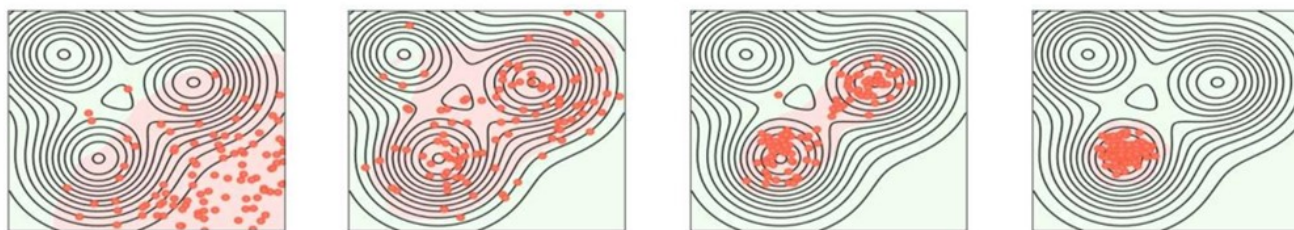
Comparison w/ & w/o Entropy Term



We also add Diffusion Entropy Regularization term in objective for enhancing exploration capability:

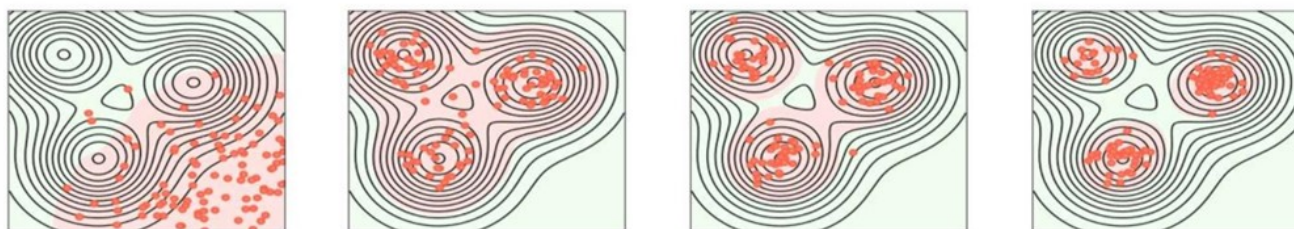
$$\mathbb{E}_{s,a \sim \mathcal{U}, \epsilon} \left[\omega_{ent}(s) \left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\bar{\alpha}_t} a + \sqrt{1 - \bar{\alpha}_t} \epsilon, s, t \right) \right\|^2 \right]$$

Diffusion Policy
w/o entropy term



Training Procedure

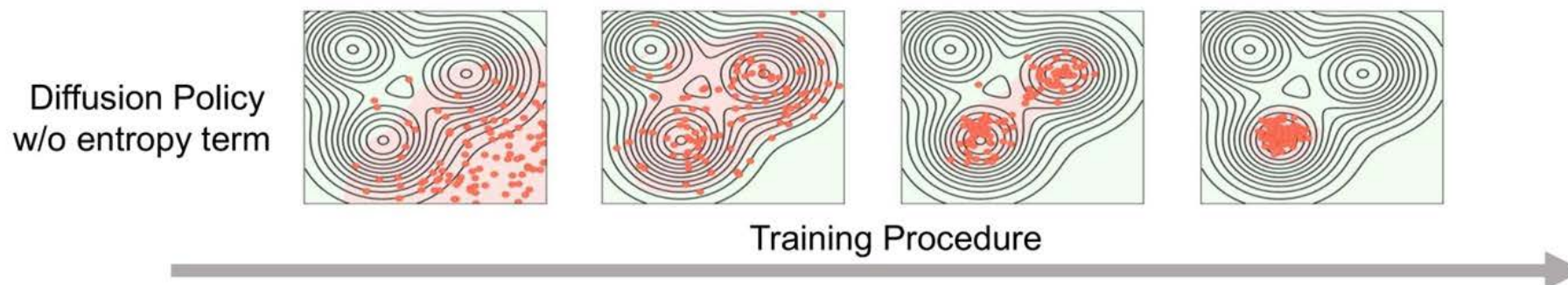
Diffusion Policy
w/ entropy term



Reducing Diffusion Policy Variance via Action Selection

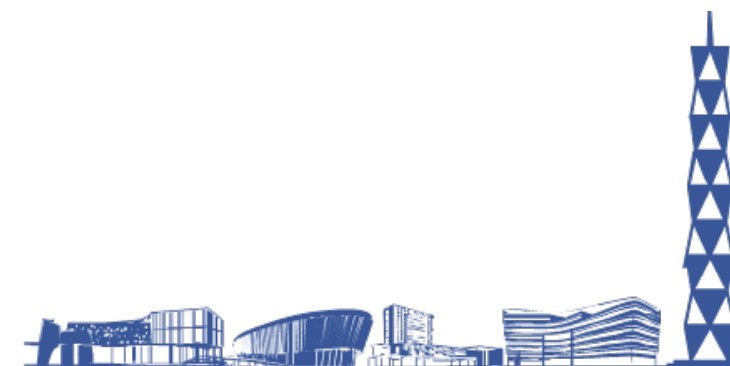


- Diffusion policy has a **large policy variance**
- Reducing this variance can improve **the quality** of collected transitions

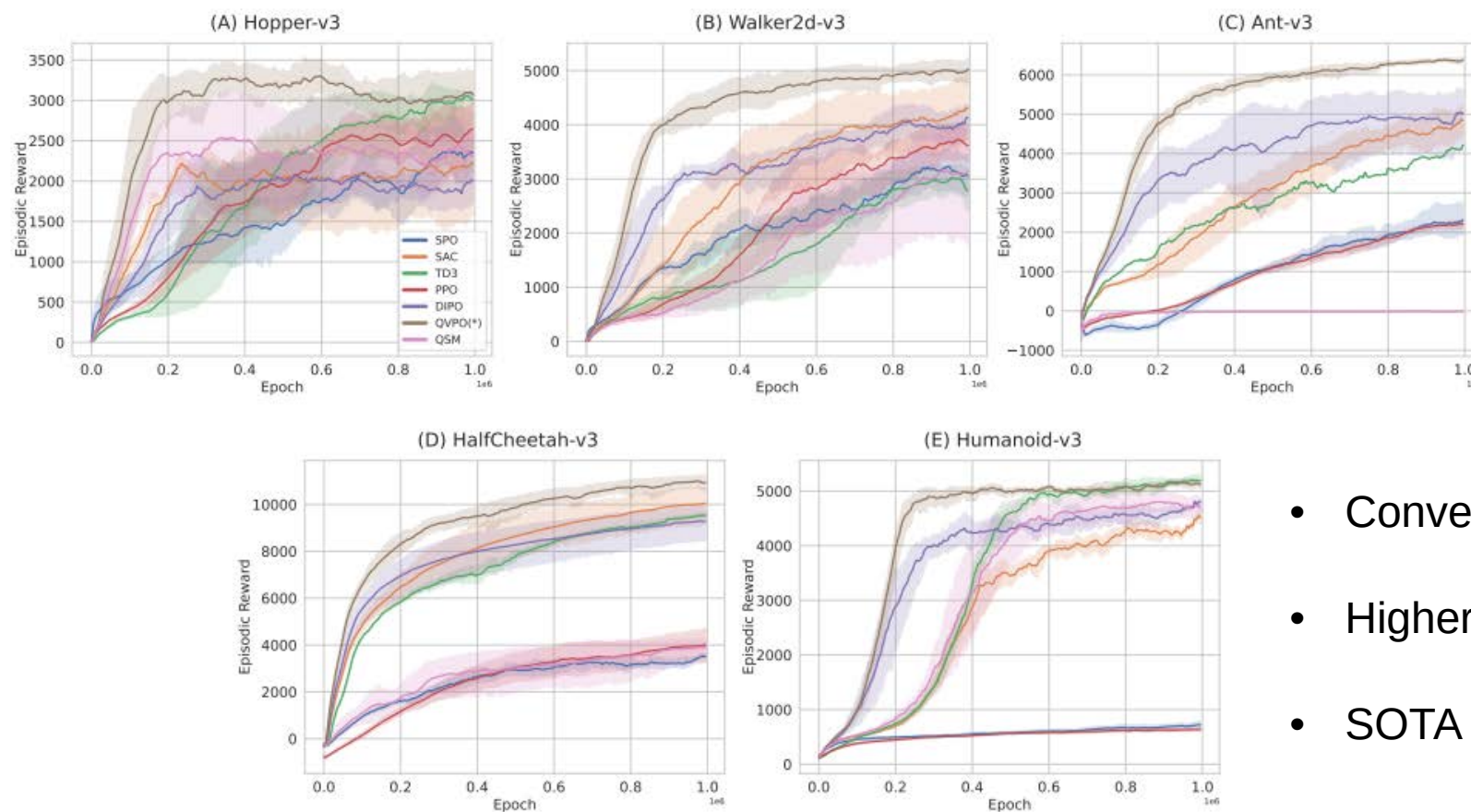


We design **efficient behavior policy** via action selection for sample efficiency:

$$\pi_{\theta}^K(a \mid s) \triangleq \operatorname{argmax}_{a \in \{a_1, \dots, a_K \sim \pi_{\theta}(a|s)\}} Q(s, a).$$



Results



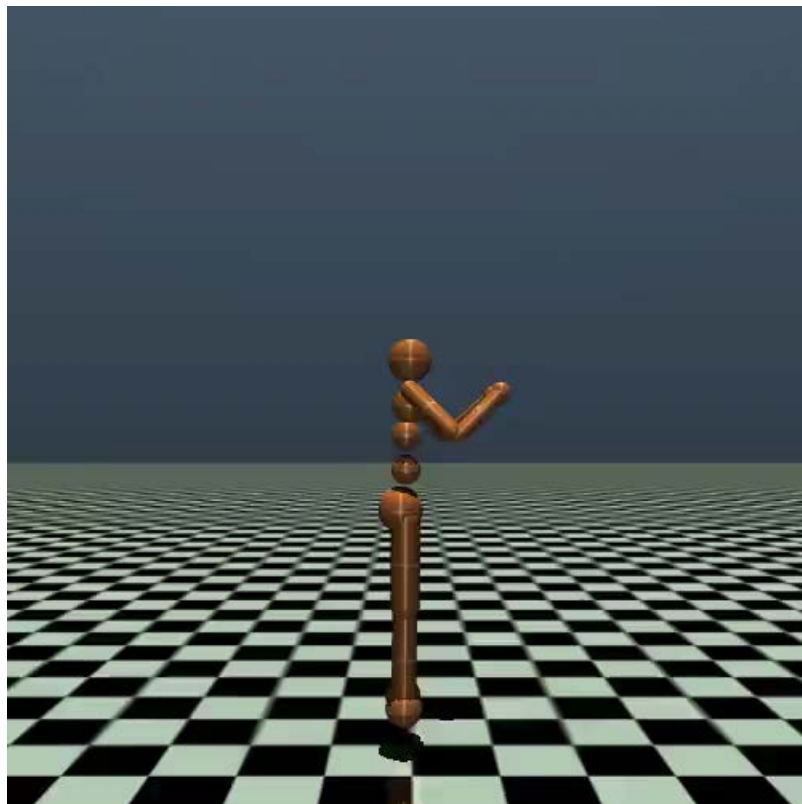
- Converge faster (sample efficiency)
- Higher cumulative reward
- SOTA method in online RL



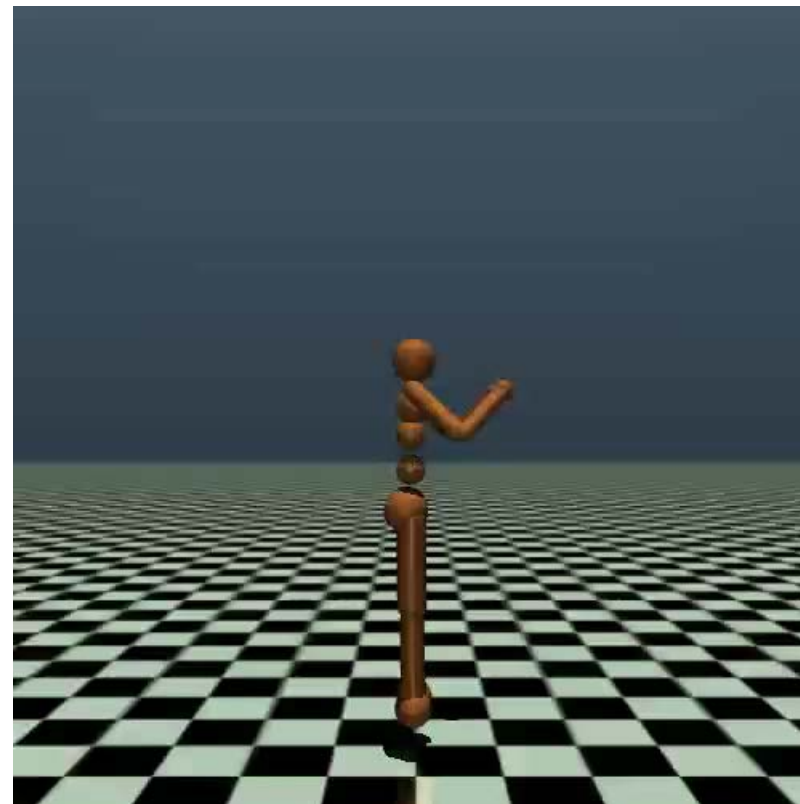
Comparison with SAC in Humanoid-v3



SAC



our QVPO





Ren[d]er every frame On
 Switch camera (#cams = 1) [Tab] (camera ID = 1)
 C[ontact forces Off
 T[r]ansparent Off
 Stop [Space]
 Referenc[e] frames Off
 H[ide Menu
 Cap[t]ure frame
 Toggle geomgroup visibility 0-4

FPS 21
 Solver iterations [2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1]
 Step 1520
 timestep 0.00200



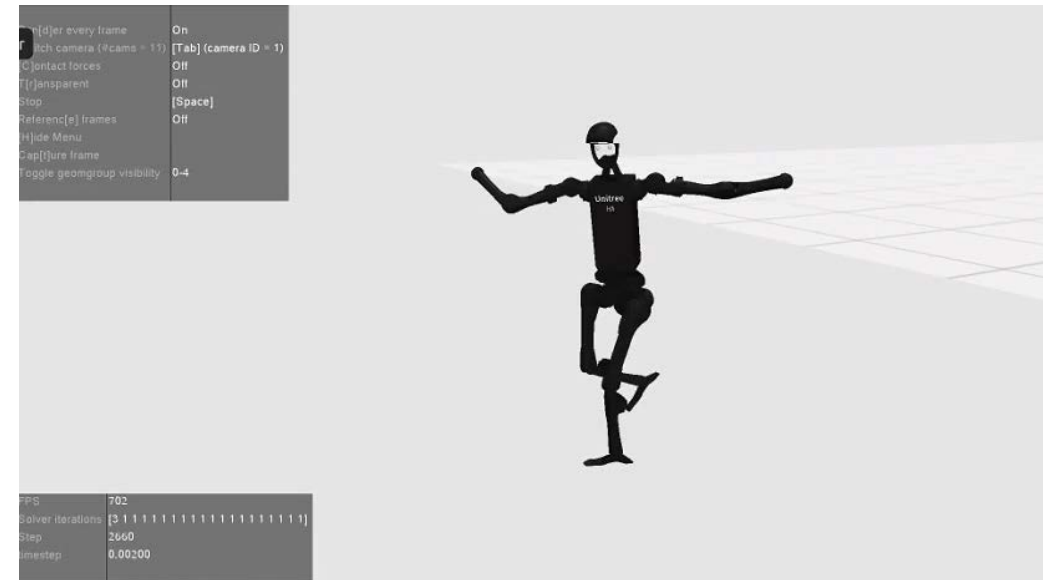
More Tasks in Unitree Humanoid H1



Walk (QVPO)



Run (QVPO)



More Tasks in Unitree Humanoid H1





Fusing physical information into deep neural networks



Outline



1. Controllable diffusion model for reliable generation

- Conditional Diffusion with Spherical Gaussian Constraint
- Gaze-guided Hand-object Interaction Synthesis
- Diffusion-based Reinforcement Learning

2. Fusing physical information into data-driven models

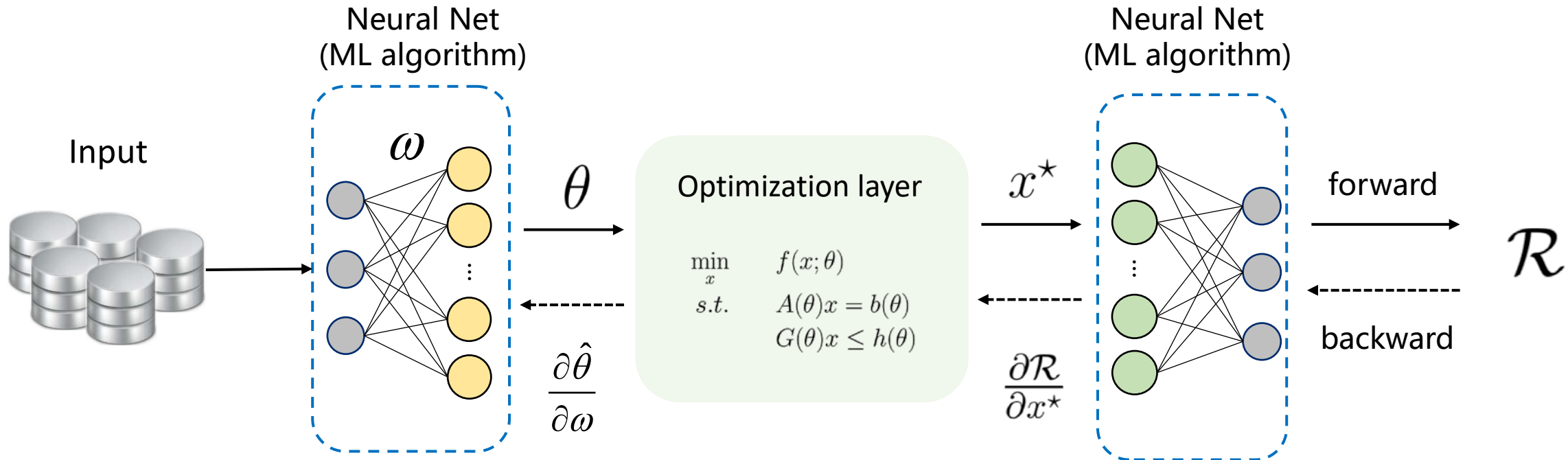
- **Faster Implicit Differentiation**
- More accurate 3D Motion Capture
- Reliable Decision-making via Safe RL

3. Federated Learning with Foundation Models

- Personalized Federated Learning with Transformer
- Prompt Tuning Boost Federated Learning
- Theoretical Analysis for prompt-based Federated Learning



Integrate Optimization into Neural Networks



➤ Definition

A layer in the neural network is defined as an optimization layer if its input is the optimization parameters and its output is the solution to the optimization problem.



Differentiable optimization layers



Optimization layer

$$\begin{array}{ll} \min_x & f(x; \theta) \\ \text{s.t.} & A(\theta)x = b(\theta) \\ & G(\theta)x \leq h(\theta) \end{array}$$

Its optimal is obtained by KKT conditions:

$$\begin{cases} \nabla f(x) + A^T(\theta)\lambda + G^T(\theta)\nu = 0 \\ A(\theta)x - b(\theta) = 0 \\ \mathbf{diag}(\nu)(G(\theta)x - h(\theta)) = 0 \end{cases}$$



How about back-propagation?



Differentiable optimization layers

Let $\mathcal{F}(x, \theta)$ denote a continuously differentiable function with $\mathcal{F}(x^*, \theta) = \mathbf{0}$.
Suppose the Jacobian of $\mathcal{F}(x, \theta)$ is invertible at (x^*, θ) , then the derivative of the solution x^* with respect to θ is

$$\frac{\partial x^*}{\partial \theta} = -[\mathcal{J}_{\mathcal{F};x}]^{-1} \mathcal{J}_{\mathcal{F};\theta}$$

The Jacobian matrix w.r.t the variables is :

$$[\mathcal{J}_{\mathcal{F};x^*} \mid \mathcal{J}_{\mathcal{F};\lambda} \mid \mathcal{J}_{\mathcal{F};\nu}] = \left[\begin{array}{c|c|c} f_{xx}^T(x^*) & A^T(\theta) & B^T(\theta) \\ A(\theta) & 0 & 0 \\ \text{diag}(\nu)G(\theta) & \text{diag}(G(\theta)x - h(\theta)) & 0 \end{array} \right] \quad n + n_c$$

$$[\mathcal{J}_{\mathcal{F};A} \mid \mathcal{J}_{\mathcal{F};b} \mid \mathcal{J}_{\mathcal{F};G} \mid \mathcal{J}_{\mathcal{F};h}] = \left[\begin{array}{c|c|c|c} \nu^T \otimes I & 0 & \lambda^T \otimes I & 0 \\ I \otimes (x^*)^T & -I & 0 & 0 \\ 0 & 0 & \text{diag}(\lambda)I \otimes (x^*)^T & -I \end{array} \right]$$



Too large!



Alternating Differentiation



$$\max_{\lambda, \nu} \min_{x, s \geq 0} L_{\rho}(x, s, \lambda, \nu; \theta)$$

$$= f(x) + \langle \lambda, Ax - b \rangle + \langle \nu, Gx + s - h \rangle + \frac{\rho}{2} (\|Ax - b\|^2 + \|Gx + s - h\|^2),$$

The alternating method is applied to reduce the computational complexity.

➤ Forward pass:

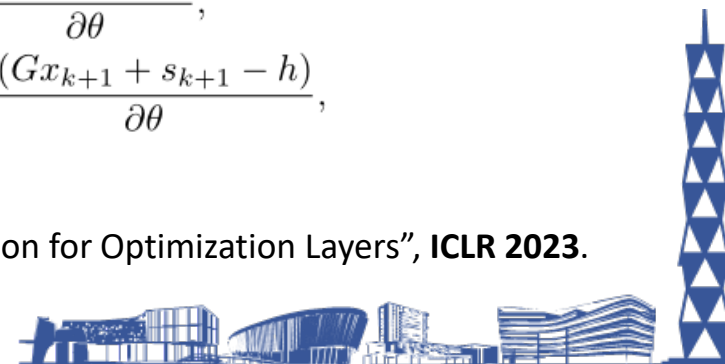
$$\begin{cases} x_{k+1} = \arg \min_x L(x, s_k, \lambda_k, \nu_k), \\ s_{k+1} = \arg \min_{s \geq 0} L(x_{k+1}, s, \lambda_k, \nu_k), \\ \lambda_{k+1} = \lambda_k + \rho(Ax_{k+1} - b), \\ \nu_{k+1} = \nu_k + \rho(Gx_{k+1} + s_{k+1} - h). \end{cases}$$

differentiate

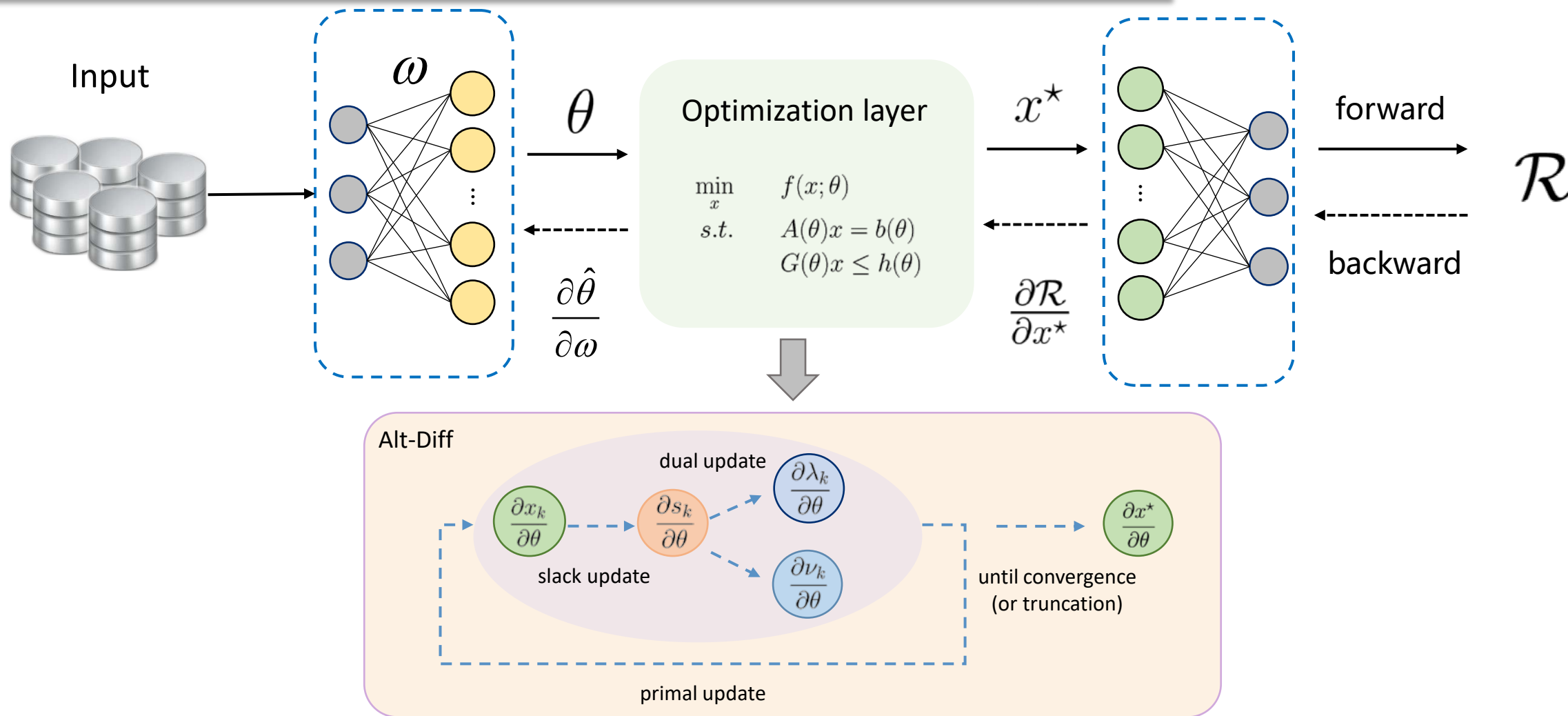


➤ Backward pass:

$$\begin{cases} \frac{\partial x_{k+1}}{\partial \theta} = -(f_{xx}(x_{k+1})^T + \rho A^T A + \rho G^T G)^{-1} L_{x\theta}(x_{k+1}), \\ \frac{\partial s_{k+1}}{\partial \theta} = -\frac{1}{\rho} \mathbf{sgn}(s_{k+1}) \cdot \mathbf{1}^T \odot \left(\frac{\partial \nu_k}{\partial \theta} + \rho \frac{\partial (Gx_{k+1} - h)}{\partial \theta} \right), \\ \frac{\partial \lambda_{k+1}}{\partial \theta} = \frac{\partial \lambda_k}{\partial \theta} + \rho \frac{\partial (Ax_{k+1} - b)}{\partial \theta}, \\ \frac{\partial \nu_{k+1}}{\partial \theta} = \frac{\partial \nu_k}{\partial \theta} + \rho \frac{\partial (Gx_{k+1} + s_{k+1} - h)}{\partial \theta}, \end{cases}$$



Pipeline



Haixiang Sun, **Ye Shi***, Jingya Wang, Hoang Duong Tuan, H. Vincent Poor, Dacheng Tao, "Alternating Differentiation for Optimization Layers", **ICLR 2023**.



Convergence of Alt-Diff



➤ Theorem

Suppose x_k is the truncated solution at the k-th iteration, the error between the gradient $\frac{\partial x_k}{\partial \theta}$ obtained by the truncated Alt-Diff and the real gradient $\frac{\partial x^*}{\partial \theta}$ is bounded as follows:

$$\left\| \frac{\partial x_k}{\partial \theta} - \frac{\partial x^*}{\partial \theta} \right\| \leq C_1 \|x_k - x^*\|$$

where $C_1 = \frac{L_3}{\mu_2} + \frac{\mu_3 L_2}{\mu_2^2}$ is a constant.

➤ Corollary

The error of the gradient w.r.t. θ in loss function R is bounded as follows:

$$\|\nabla \mathcal{R}(\theta; x_k) - \nabla \mathcal{R}(\theta; x^*)\| \leq C_2 \|x_k - x^*\|$$

where $C_2 = L_1 + \frac{\mu_3 L_1 + \mu_1 L_3}{\mu_2} + \frac{\mu_1 \mu_3 L_2}{\mu_2^2}$ is a constant.



Comparison with existing solvers



In sparse quadratic cases:

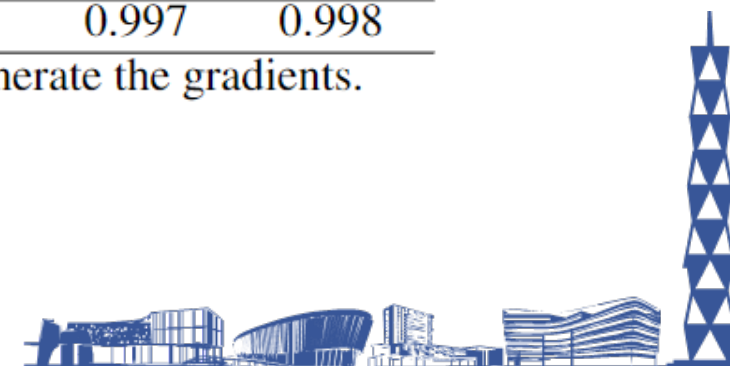
$$\min_x \|x - y\|_2^2$$

$$s.t. \quad 1^T x = 1, \quad 0 \leq x \leq u$$

	Small	Medium	Large	Large+
Num of variable n	3000	5000	10000	20000
Num of constraints n_c	6000	10000	20000	40000
OptNet	18.46	77.87	834.59	-
Forward	16.61	71.45	779.00	-
Backward	1.85	6.42	55.59	-
CvxpyLayer (total)	3.61	18.14	123.79	1024.69
Initialization	3.50	17.79	122.61	1020.41
Canonicalization	0.00	0.00	0.01	0.02
Forward	0.07	0.27	0.44	1.35
Backward	0.03	0.07	0.73	2.91
Alt-Diff (total)	3.16	9.96	58.51	539.19
Inversion	1.21	4.71	34.35	287.83
Forward and backward	1.95	5.25	24.15	251.36
Cos Dist.	0.999	0.998	0.997	0.998

“-” represents that the solver cannot generate the gradients.

Tolerance ϵ	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
OptNet	8.58	9.32	9.48	9.59	10.13
CvxpyLayer	212.34	216.66	212.86	215.75	213.08
Alt-Diff	1.18	3.50	5.75	8.04	10.76



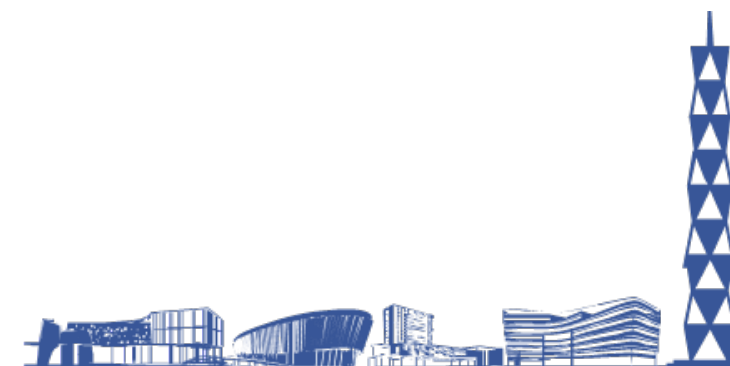
Comparison with existing solvers



In a non-quadratic case:

$$\begin{aligned} \min_x \quad & -y^T x + H(x) \\ \text{s.t.} \quad & Ax = b, \quad Gx \leq h \end{aligned}$$

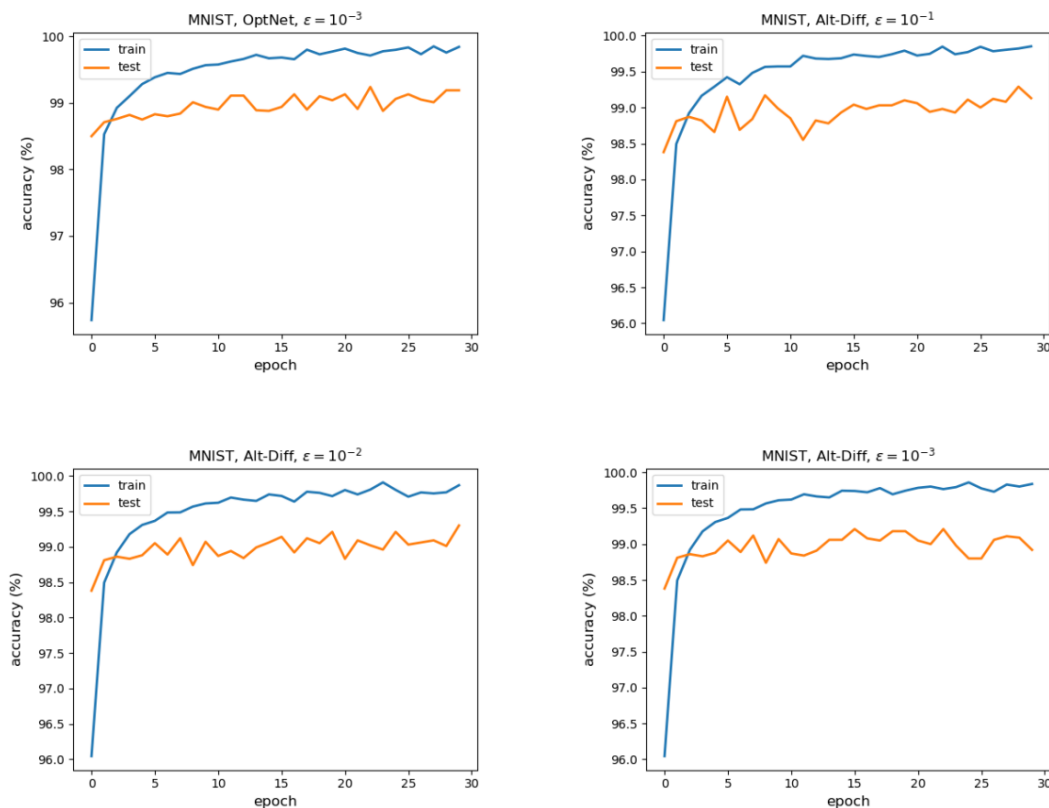
	tiny	small	medium	large
Num of ineq. m	100	300	500	1000
Num of ineq. m	30	100	200	500
Num of eq. p	10	20	20	200
Elements	1.96×10^4	1.76×10^5	5.18×10^5	2.89×10^6
CvxpyLayer, total	0.11	0.97	2.09	9.67
Initialization	0.01	0.04	0.08	0.29
Canonicalization	0.00	0.00	0.01	0.05
Forward	0.07	0.34	0.73	2.76
Backward	0.03	0.59	1.36	6.56
Alt-Diff, total	0.03	0.33	1.10	5.02
Cosine Dist.	0.999	0.996	0.999	0.999



Training neural networks



Image classification



models	Test accuracy (%)	Time per epoch (s)
OptNet	98.95 ± 0.25	307.16 ± 18.21
Alt-Diff	98.97 ± 0.17	167.66 ± 4.40

Also faster running speed!



Similar results



Outline



1. Controllable diffusion model for reliable generation

- Conditional Diffusion with Spherical Gaussian Constraint
- Gaze-guided Hand-object Interaction Synthesis
- Diffusion-based Reinforcement Learning

2. Fusing physical information into data-driven models

- Faster Implicit Differentiation
- **More accurate 3D Motion Capture**
- Reliable Decision-making via Safe RL

3. Federated Learning with Foundation Models

- Personalized Federated Learning with Transformer
- Prompt Tuning Boost Federated Learning
- Theoretical Analysis for prompt-based Federated Learning



Optimization-based Methods for 3D Pose



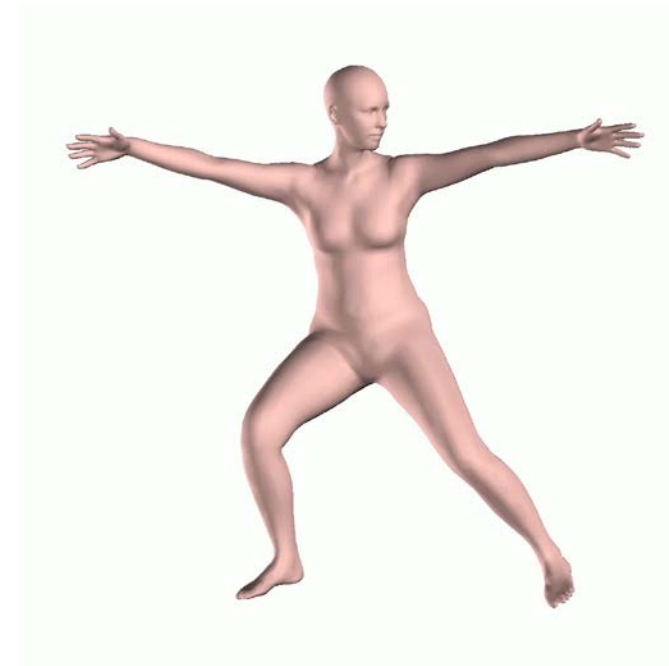
Pros:

- Without training
- 2D joints are easy to annotate/detect
- Well **alignment**



Cons:

- Physical implausibility
- Tedious optimization process
- Low accuracy



Regression-based Methods for 3D Pose

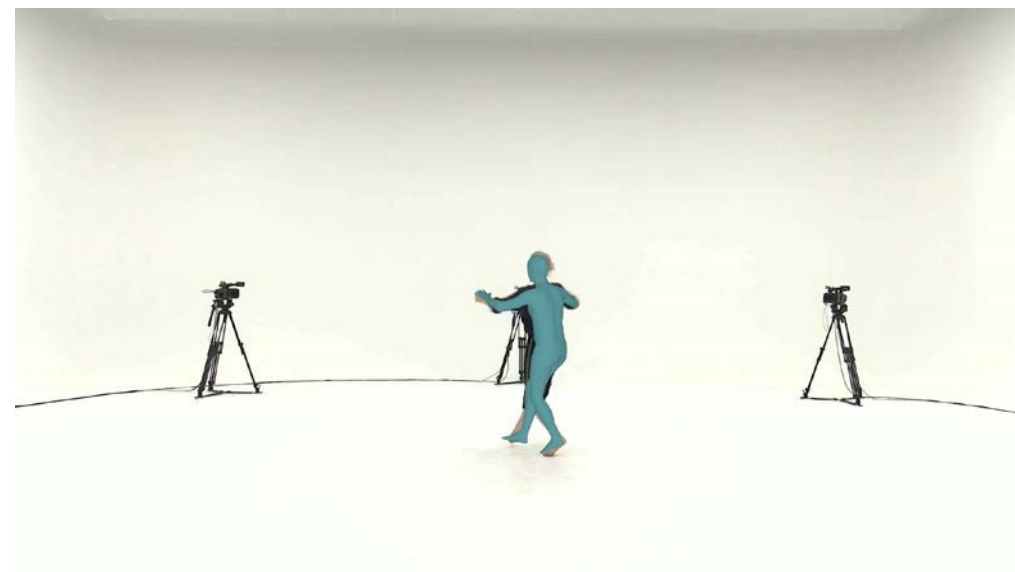


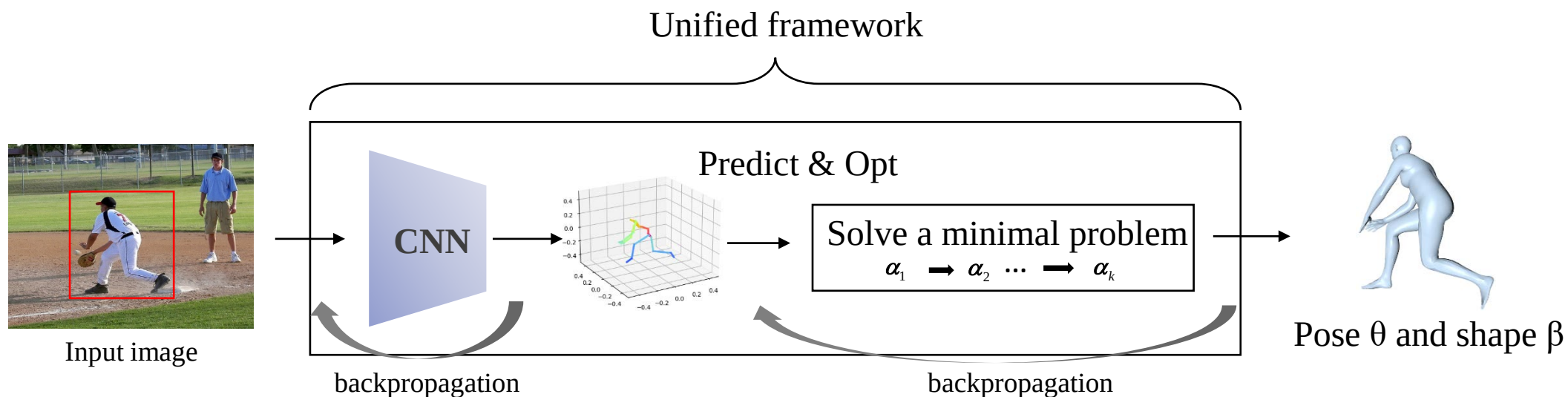
Pros:

- Physical plausibility
- Runtime efficiency
- Promising results

Cons:

- Limited training data in the wild
- Hard to label 3D pose and shape
- Noticeable **misalignment** due to articulated pose



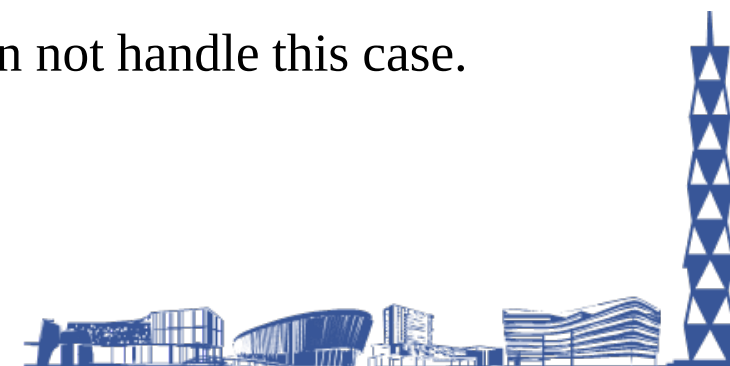


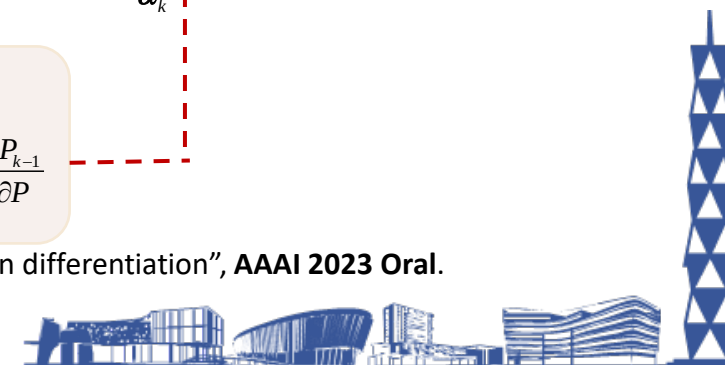
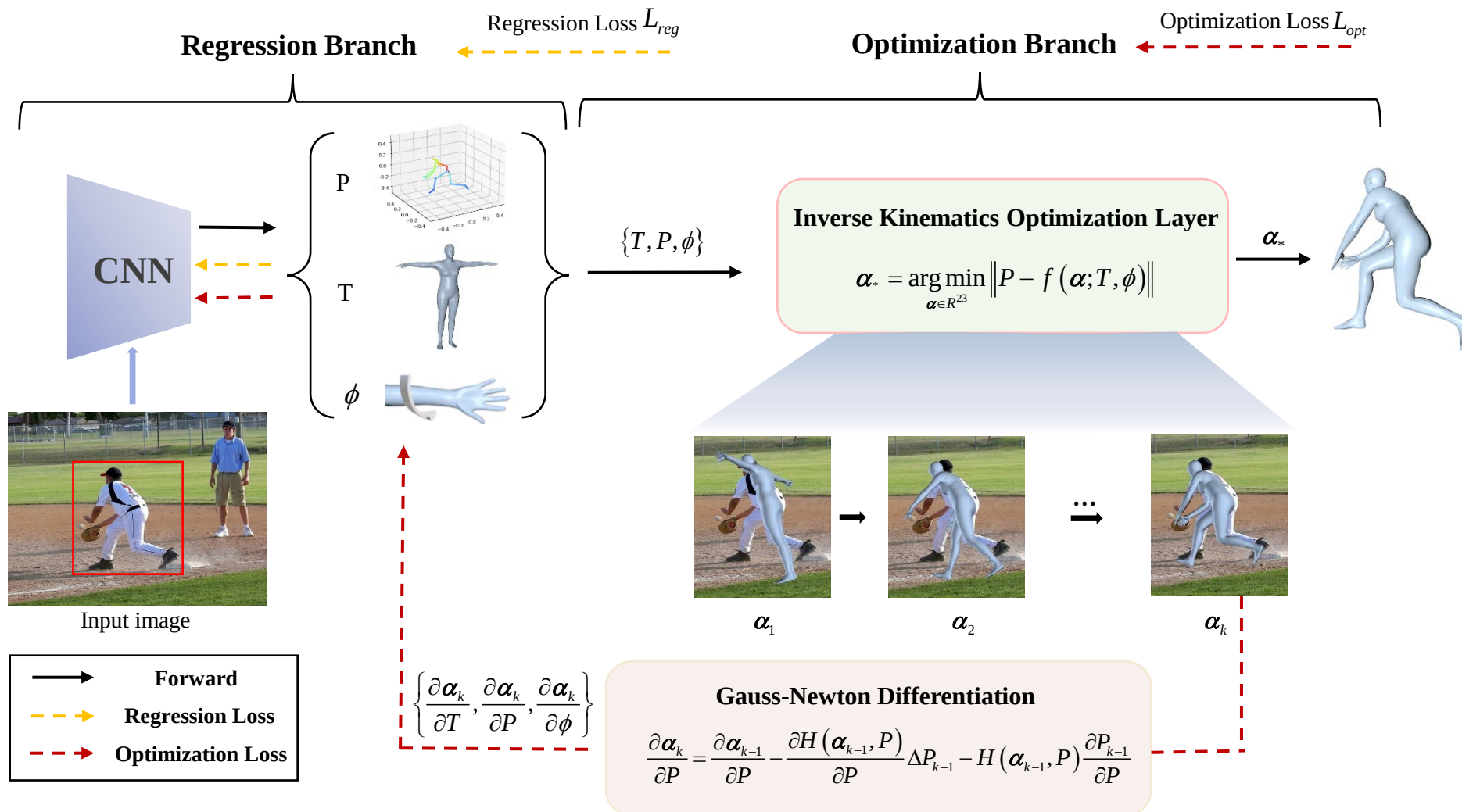
Challenge:

- Solve such minimal problem (IK) is **non-differentiable & non-convex & non-linear**.
- Existing **optimization layer** solver (such as Optnet¹ and CvxpyLayer²) can not handle this case.

[1] Amos et. al. Optnet: Differentiable optimization as a layer in neural networks. *ICML*, 2017.

[2] Agrawal et. al. Differentiable convex optimization layers. *NIPS*, 2019



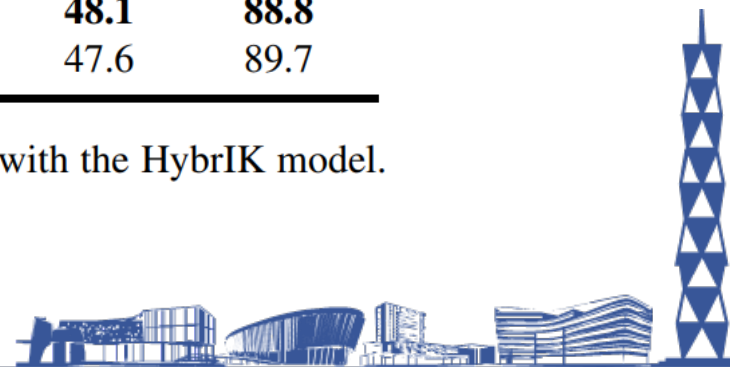


Results



Method	Backbone	3DPW			MPI-INF-3DHP		
		PVE ↓	MPJPE ↓	PA-MPJPE ↓	PCK ↑	AUC ↑	MPJPE ↓
HMR (Kanazawa et al. 2018)	ResNet-50	-	130.0	81.3	72.9	36.5	124.2
Kolotouros et al. 2019	ResNet-50	-	-	70.2	-	-	-
Arnab et al. 2019	-	-	-	72.2	-	-	-
SPIN (Kolotouros et al. 2019)	ResNet-50	116.4	96.9	59.2	76.4	37.1	105.2
I2L-MeshNet (Moon and Lee 2020)	ResNet-50	110.1	93.2	58.6	-	-	-
HybrIK (Li et al. 2021)	ResNet-34	94.5	80.0	48.8	86.2	42.2	91.0
SPEC (Kocabas et al. 2021b)	ResNet-50	118.5	96.5	53.2	-	-	-
HMR-EFT (Joo et al. 2021)	ResNet-50	98.7	85.1	52.2	-	-	-
PARE (Kocabas et al. 2021a)	HRNet-w32	94.2	79.1	46.4	-	-	-
ROMP (Sun et al. 2021)	HRNet-w32	93.4	76.7	47.3	-	-	95.11
BEV (Sun et al. 2022)	HRNet-w32	92.3	78.5	46.9	-	-	-
METRO (Lin et al. 2021a)	HRNet-w64	88.2	77.1	47.9	-	-	-
Mesh Graphormer (Lin et al. 2021b)	-	87.7	74.7	45.6	-	-	-
Ours(IKOL)	ResNet-34	86.4	73.3	45.5	87.9	48.1	88.8
Ours(IKOL*)	ResNet-34	84.1	71.1	44.5	87.0	47.6	89.7

Table 1: Performance comparison on 3DPW and MPI-INF-3DHP. ”*” means the results are finetune with the HybrIK model. ”-” means the results that are not available. Best in **bold**.



Outline



1. Controllable diffusion model for reliable generation

- Conditional Diffusion with Spherical Gaussian Constraint
- Gaze-guided Hand-object Interaction Synthesis
- Diffusion-based Reinforcement Learning

2. Fusing physical information into data-driven models

- Faster Implicit Differentiation
- More accurate 3D Motion Capture
- **Reliable Decision-making via Safe RL**

3. Federated Learning with Foundation Models

- Personalized Federated Learning with Transformer
- Prompt Tuning Boost Federated Learning
- Theoretical Analysis for prompt-based Federated Learning

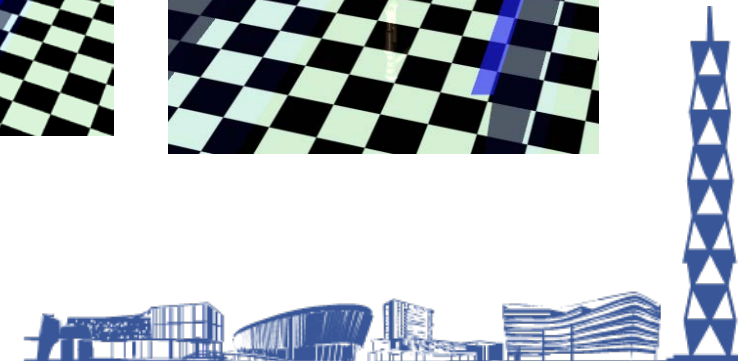
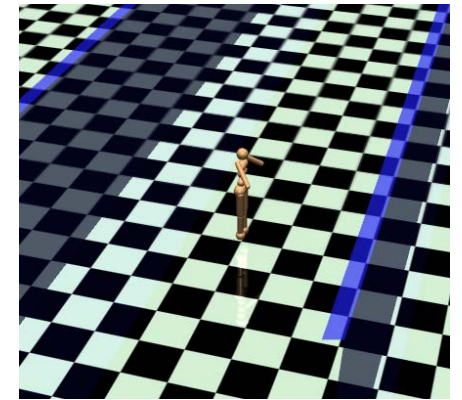
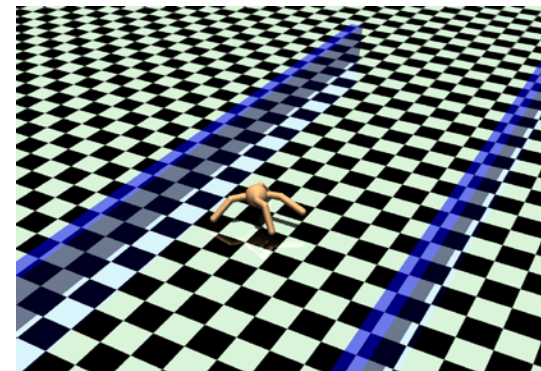
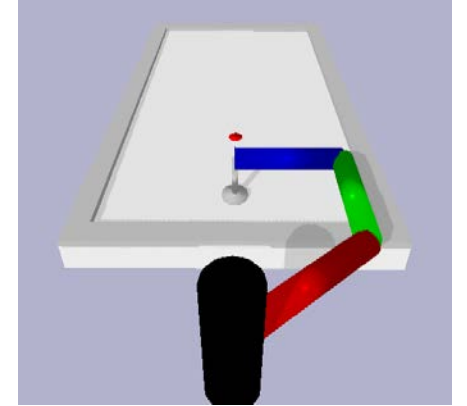
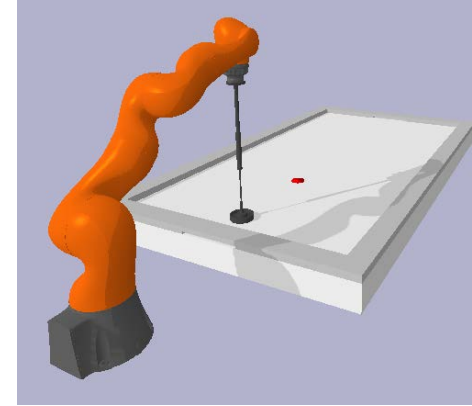


Reliable Decision-making via Safe RL



$$\begin{aligned} & \max_{\theta} J_R(\pi_{\theta}) \\ & \text{subject to} \quad f_i(\pi_{\theta}(s_t); s_t) = 0 \quad \forall i, t, \\ & \quad \quad \quad g_j(\pi_{\theta}(s_t); s_t) \leq 0 \quad \forall j, t, \end{aligned}$$

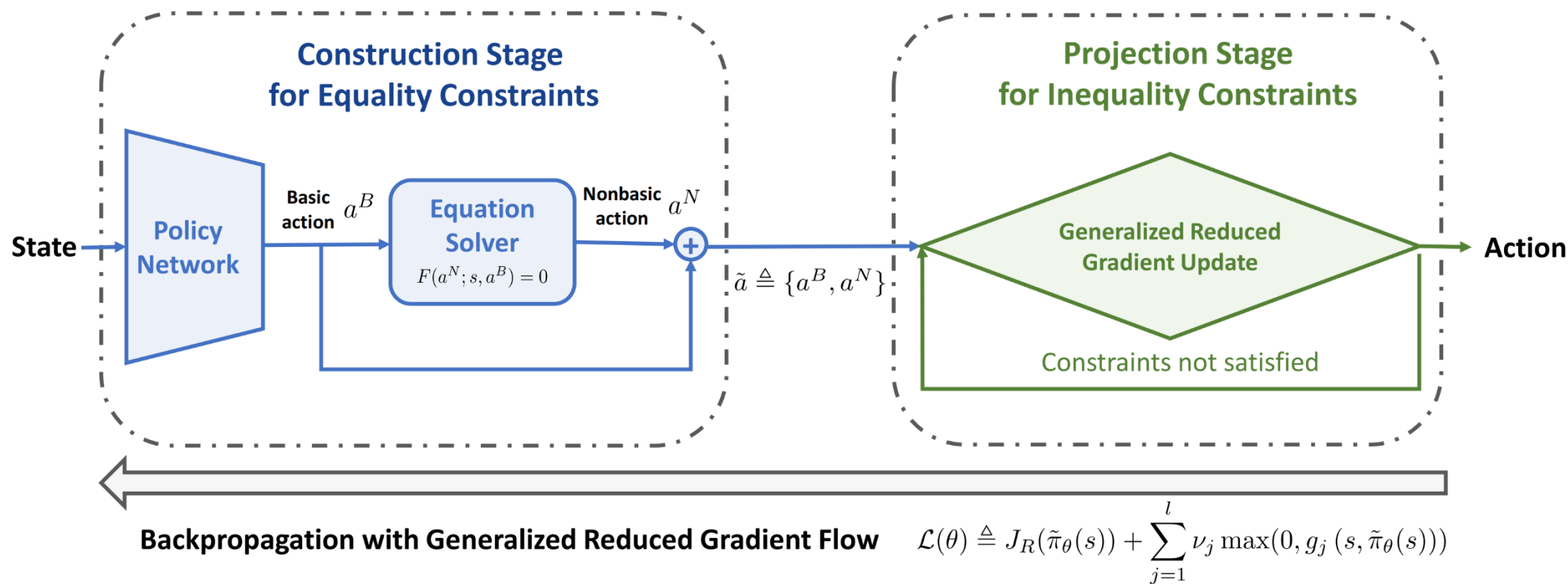
- **Hard & Instantaneous Constraints**
- **Equality & Inequality**



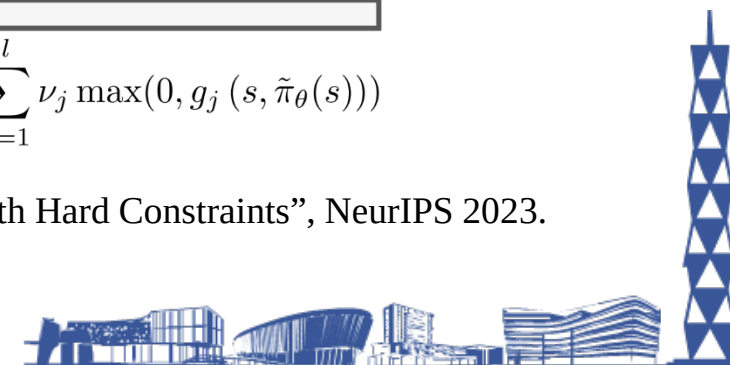
Reduced Policy Optimization



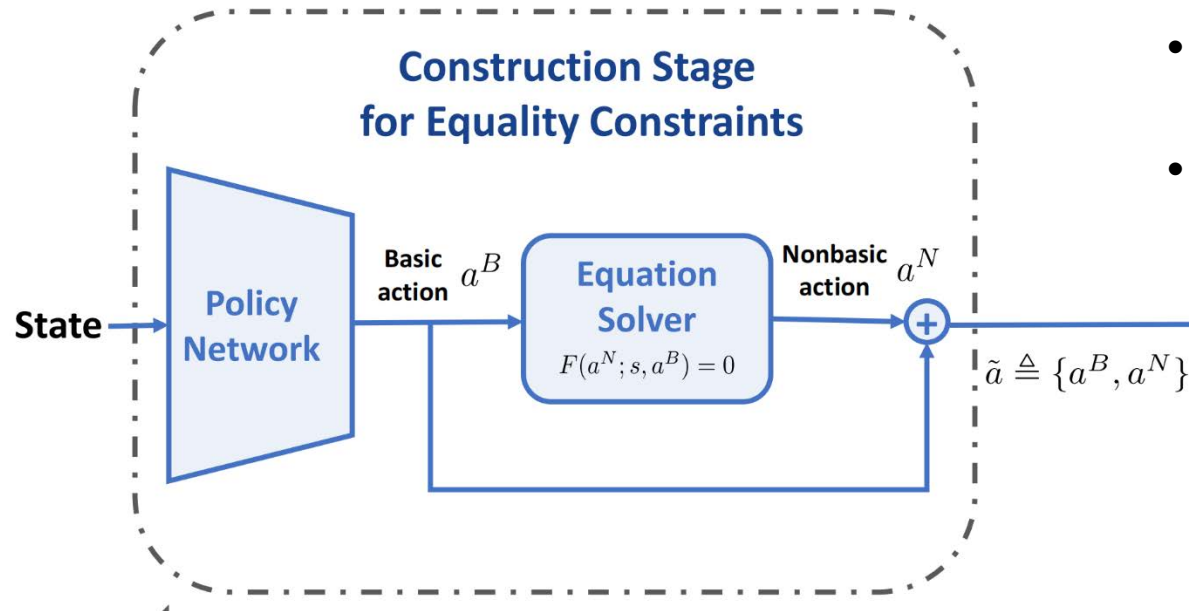
Incorporate the **GRG** into RL to handle **general hard constraints** 😊



Shutong Ding, Jingya Wang, Yali Du, **Ye Shi***, “Reduced Policy Optimization for Continuous Control with Hard Constraints”, NeurIPS 2023.



Construction Stage for Equality Constraints

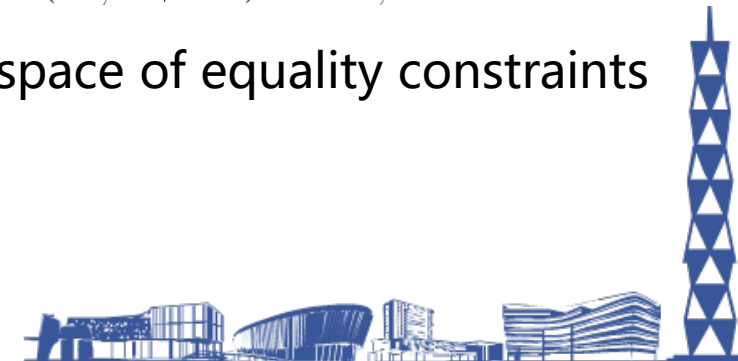


- Divide actions into **basic** & **nonbasic** actions
- Utilize policy network to predict basic actions
- Solve the nonbasic actions to ensure the equality constraints are satisfied

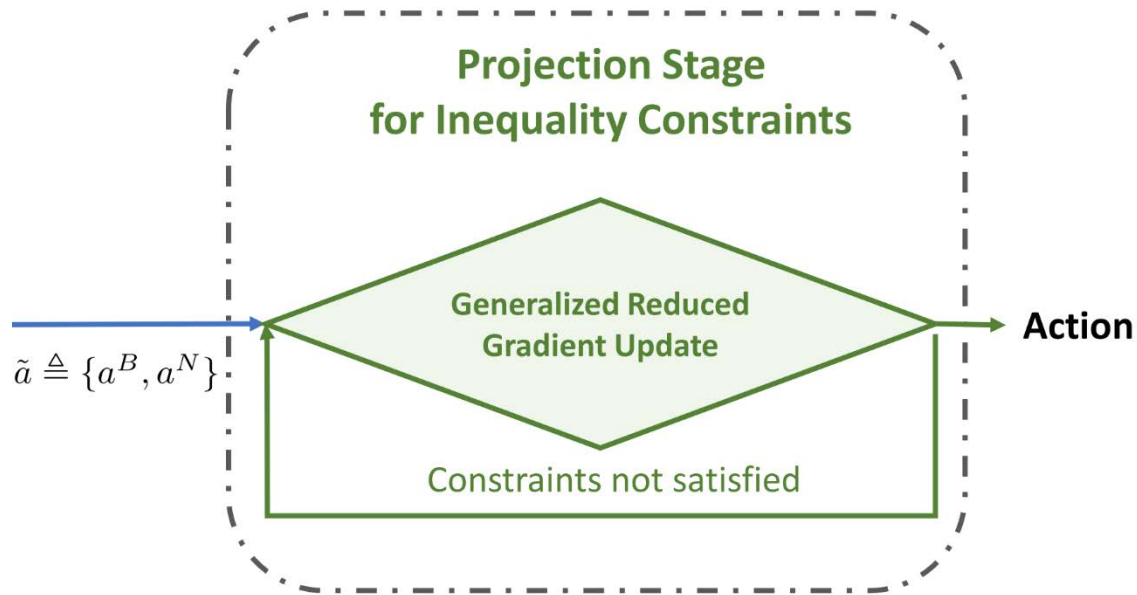
The gradient flow of equation-solving operation ϕ_N is defined as

$$\frac{\partial \phi_N(a^B)}{\partial a^B} = - \left(J_{:,m+1:n}^F \right)^{-1} J_{:,1:m}^F$$

J^F represent the tangent space of equality constraints



Projection Stage for Inequality Constraints



- Perform **GRG updates** until all the inequality constraints are satisfied

GRG update:

$$\nabla_{a^B} \mathcal{G}(a^B, a^N) \triangleq \frac{\partial \mathcal{G}(a^B, a^N)}{\partial a^B} + \frac{\partial \phi_N(a^B)}{\partial a^B} \frac{\partial \mathcal{G}(a^B, a^N)}{\partial a^N}$$
$$\Delta a^B = \nabla_{a^B} \mathcal{G}(a^B, a^N), \Delta a^N = \frac{\partial \phi_N(a^B)}{\partial a^B} \Delta a^B$$

$\mathcal{G}(a^B, a^N)$: the summation of the constraint violation



Theoretical Analysis



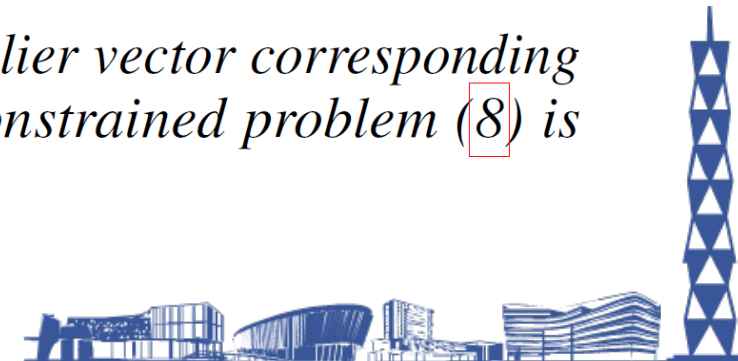
Proposition 1. (Gradient Flow in Construction Stage) Assume that we have $(n - m)$ equality constraints denoted as $F(a; s) = \mathbf{0}$ in each state s . Let $a^B \in \mathbb{R}^m$ and $a \in \mathbb{R}^n$ be the basic actions and integrated actions, respectively. Let $a^N = \phi_N(a^B) \in \mathbb{R}^{n-m}$ denote the nonbasic actions where ϕ_N is the implicit function that determined by the $n - m$ equality constraints. Then, we have

$$\frac{\partial \phi_N(a^B)}{\partial a^B} = - (J_{:,m+1:n}^F)^{-1} J_{:,1:m}^F \quad (4)$$

where $J^F = \frac{\partial F(a; s)}{\partial a} \in \mathbb{R}^{(n-m) \times n}$, $J_{:,1:m}^F$ is the first m columns of J^F and $J_{:,m+1:n}^F$ is the last $(n - m)$ columns of J^F .

Theorem 1. (GRG update in Tangent Space) If $\Delta a^N = \frac{\partial \phi_N(a^B)}{\partial a^B} \Delta a^B$, the GRG update for inequality constraints is in the tangent space of the manifold determined by linear equality constraints.

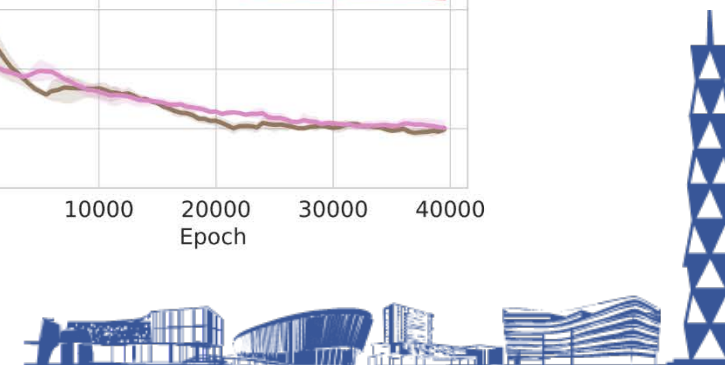
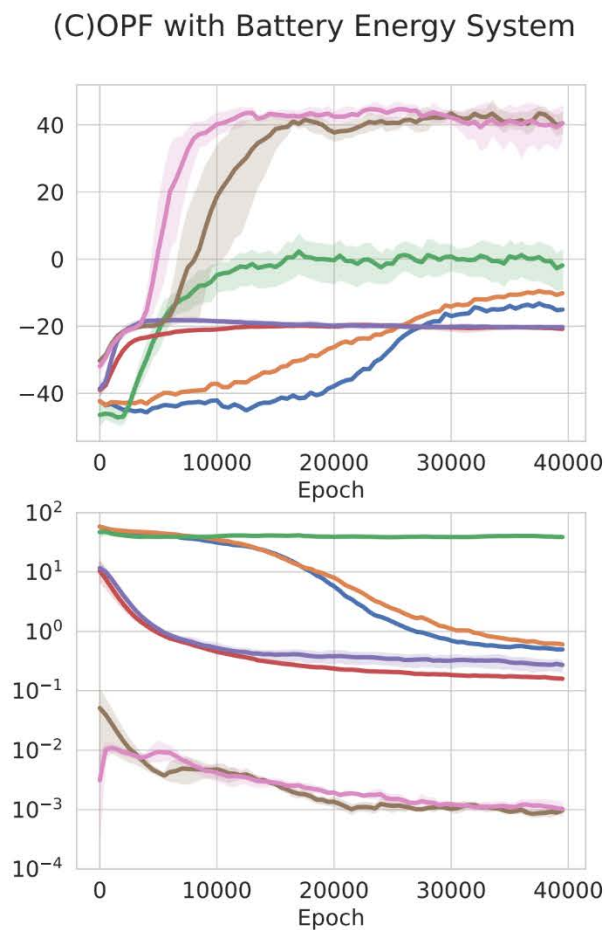
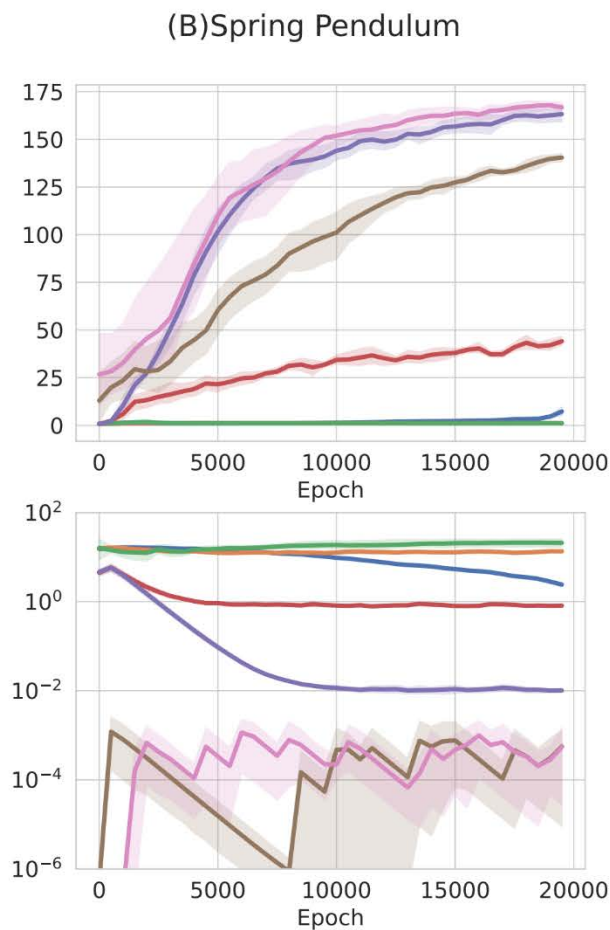
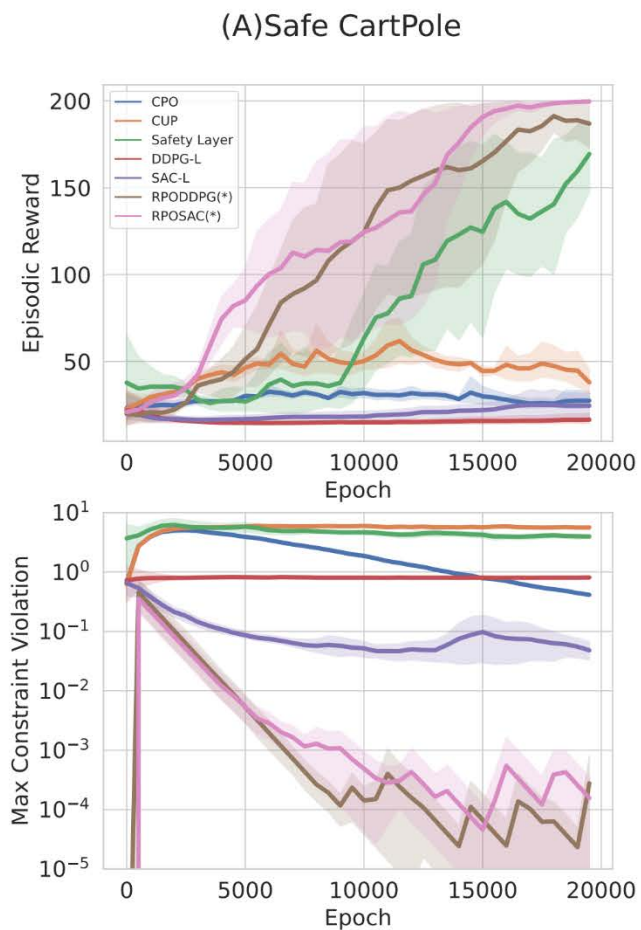
Theorem 2. (Exact Penalty Theorem) Assume ν_s^j is the Lagrangian multiplier vector corresponding to j th constraints in state s . If the penalty factor $\nu^j \geq \|\nu_s^j\|_\infty$, the unconstrained problem (8) is equivalent to the constrained problem (7).



Comparison with Safe RL algorithms



Learning Curves of different algorithms on our three benchmarks



Robot Manipulation with Foundation Models

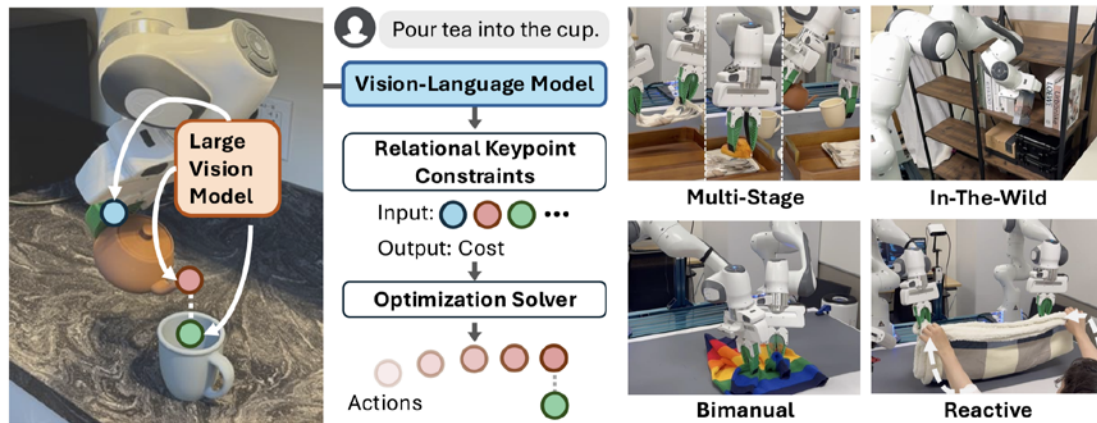
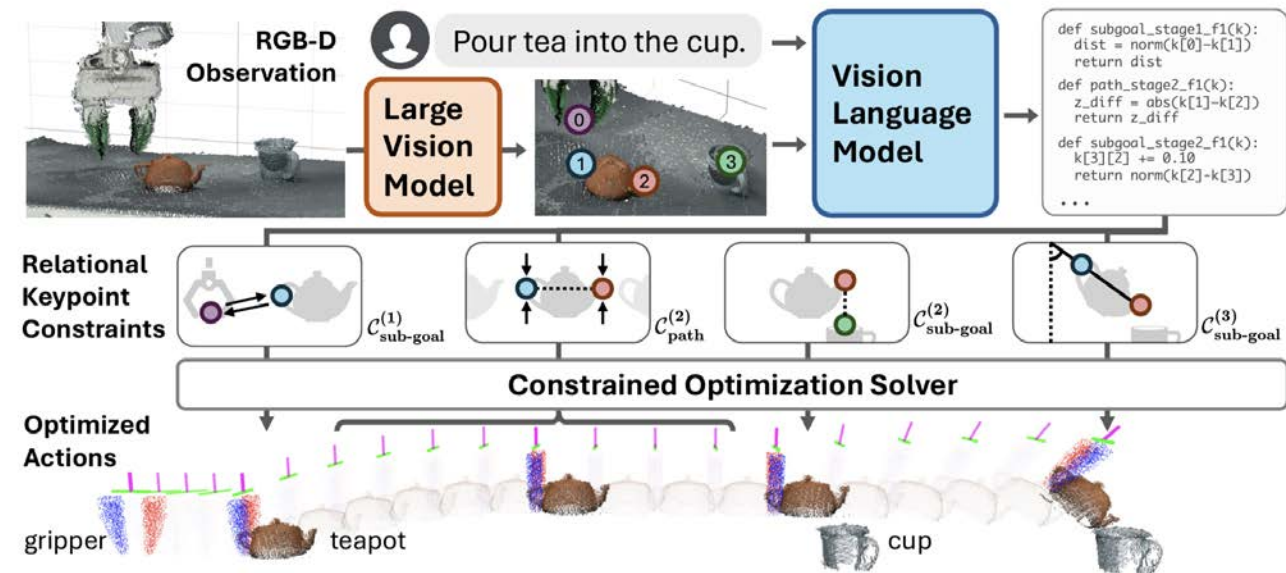


Figure 1: Relational Keypoint Constraints (ReKep) specify diverse manipulation behaviors as an optimizable spatio-temporal series of constraint functions operating on semantic keypoints. In the pouring task, one ReKep first constrains the grasping location at the handle of the teapot (*blue*). A subsequent ReKep pulls the teapot spout (*red*) towards the top of the cup opening (*green*) while another ReKep constrains the desired rotation of the teapot by associating the vector formed by the handle (*blue*) and the spout (*red*).

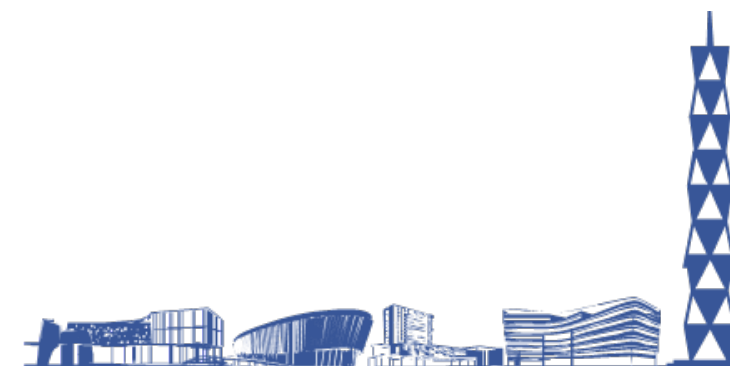


Wenlong Huang, Feifei Li, et al, ReKep: Spatio-Temporal Reasoning of Relational Keypoint Constraints for Robotic Manipulation, arXiv 2409.01652v1.





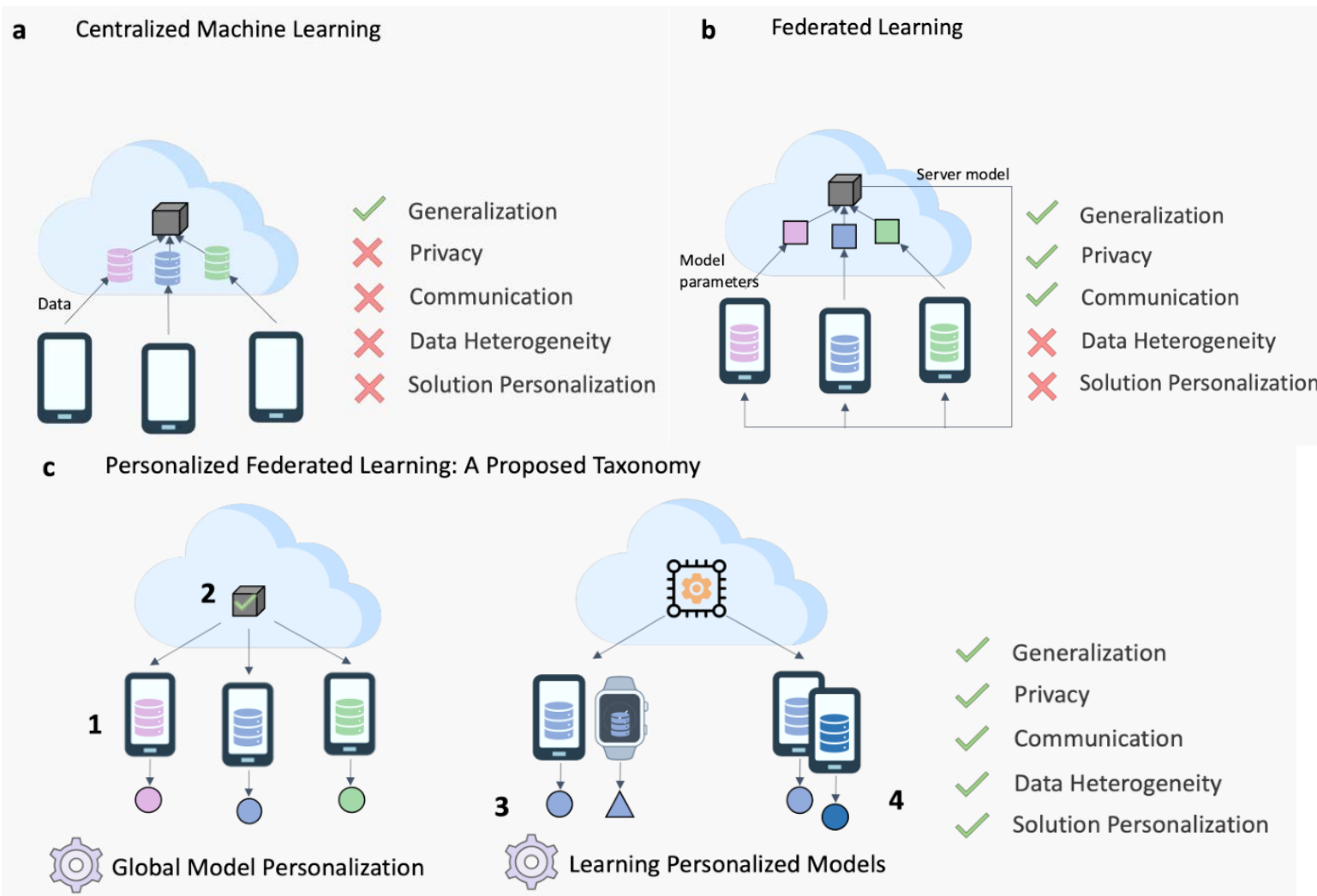
Federated Learning with Foundation Models



Federated Learning



上海科技大学
ShanghaiTech University



Server model



Trained server model



PFL algorithm



Local model



Personalized model



Local data

[1] TANAZ, YUH, CUIL, et al., 2022. Towards personalized federated learning [J]. IEEE Transactions on Neural Networks and Learning Systems.

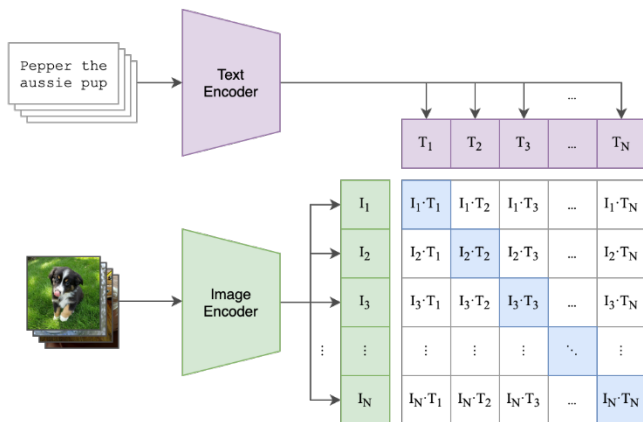
立志成才 报国裕民

Vision Language Foundation Models

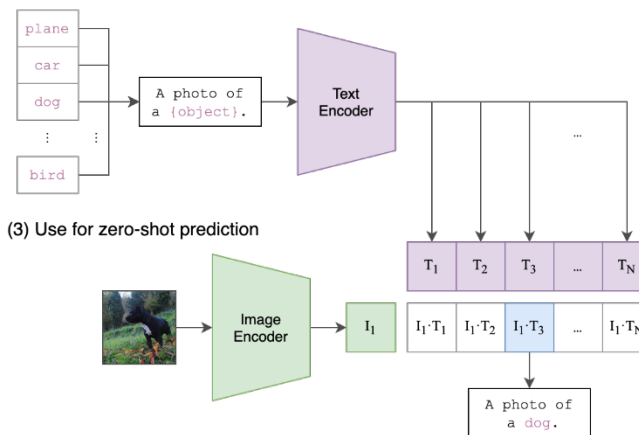


上海科技大学
ShanghaiTech University

(1) Contrastive pre-training

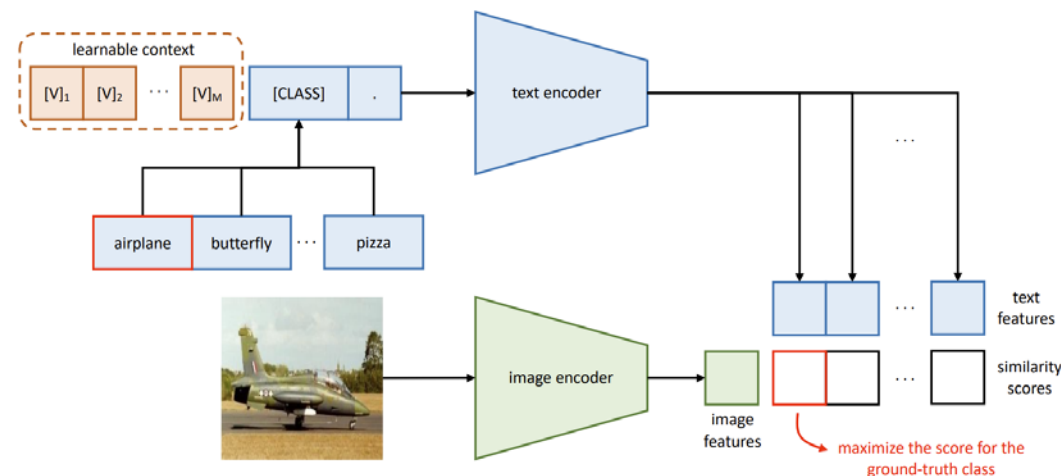


(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

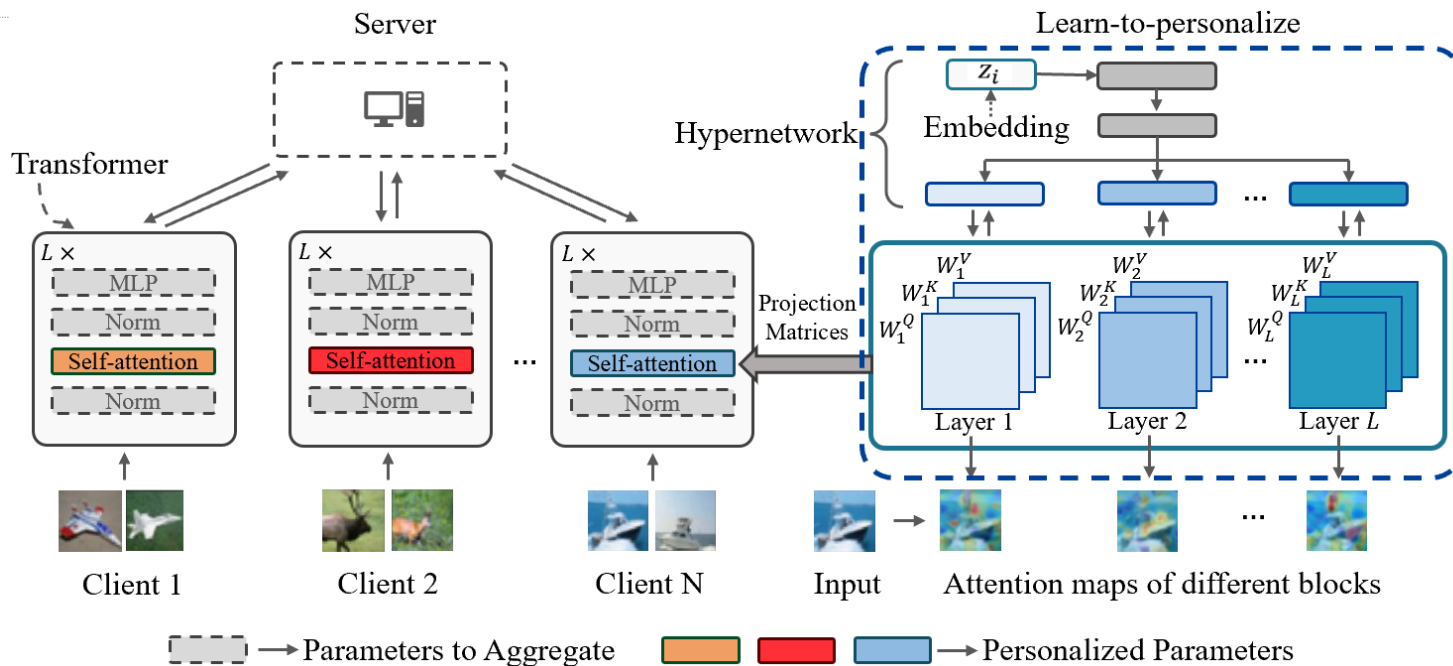
CLIP



Prompt Tuning for CLIP

立志成才 报国裕民

提出首个基于Transformer的异构联邦学习新框架，引入超网络生成各客户端的自注意力权重系数，并给出异构联邦学习泛化误差上界。



Theorem 1: Suppose $\hat{D}_1, \hat{D}_2, \dots, \hat{D}_N$ denote the empirical data distribution of N clients with $\hat{\phi}, \hat{z}_i, \hat{\xi}$ the parameters learned by the corresponding empirical distributions. Denote $\mathcal{H}$ as the personalized hypothesis and let d be the VC-dimension of $\mathcal{H}$. Suppose Assumptions 1 and 2 (described in paper) hold, with probability at least $1 - \delta$, we have

$$\left| \sum_{i=1}^N \frac{m_i}{M} \mathcal{L}_{\hat{D}_i}(h(\hat{\phi}; \hat{z}_i), \hat{\xi}) - \sum_{i=1}^N \frac{m_i}{M} \mathcal{L}_{D_i}(h(\phi^*; z_i^*), \xi^*) \right| \leq \sqrt{\frac{M}{2} \log \frac{N}{\delta}} + \sqrt{\frac{dN}{M} \log \frac{eM}{d}} + L_h R_h (L_\phi + L_z) + L_\xi R_t$$

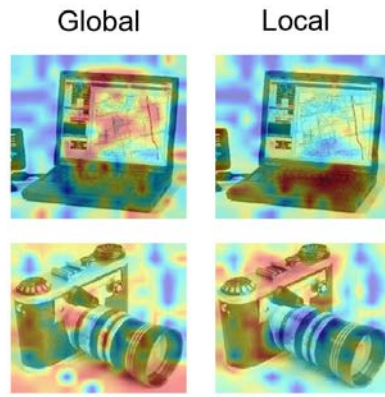
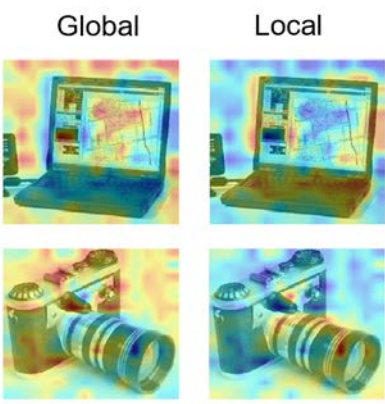
where ϕ^*, z_i^* , and ξ^* represent the optimal parameters corresponding to the real distribution of each client, respectively.

- 更具可扩展性和更强的泛化能力
- 更好的模型性能
- 对数据异构性更具鲁棒性

基于全局和本地提示合作的联邦预训练学习



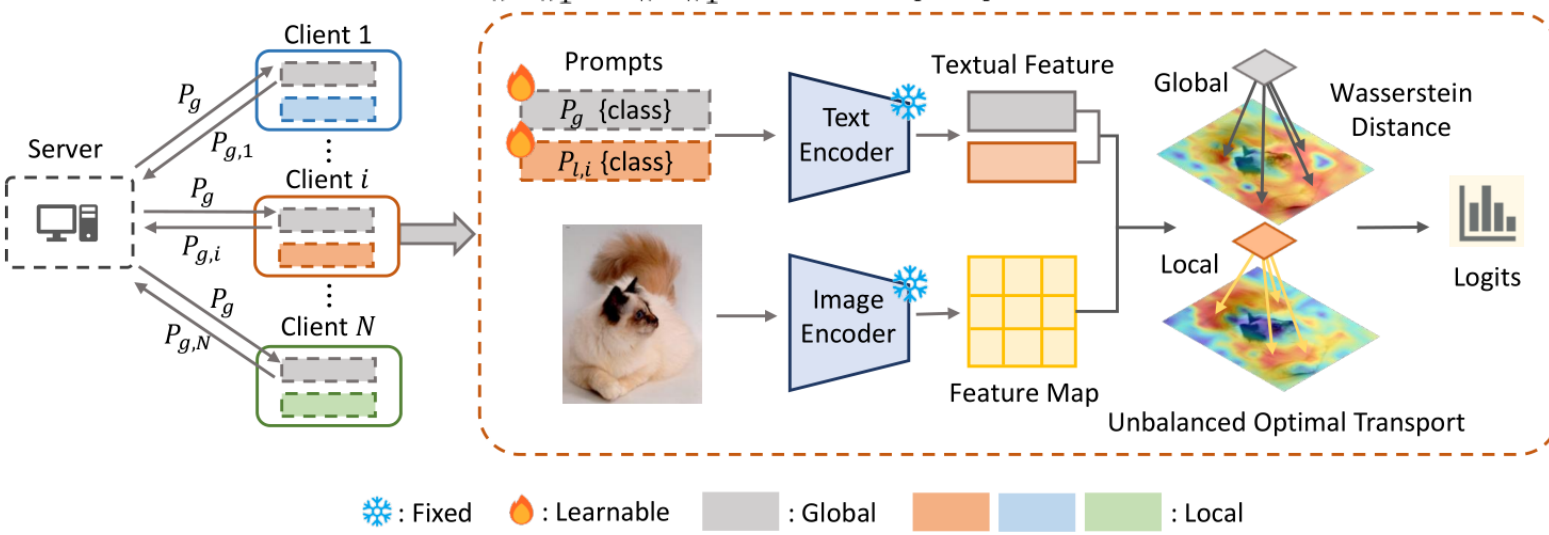
传统最优运输



不平衡最优运输

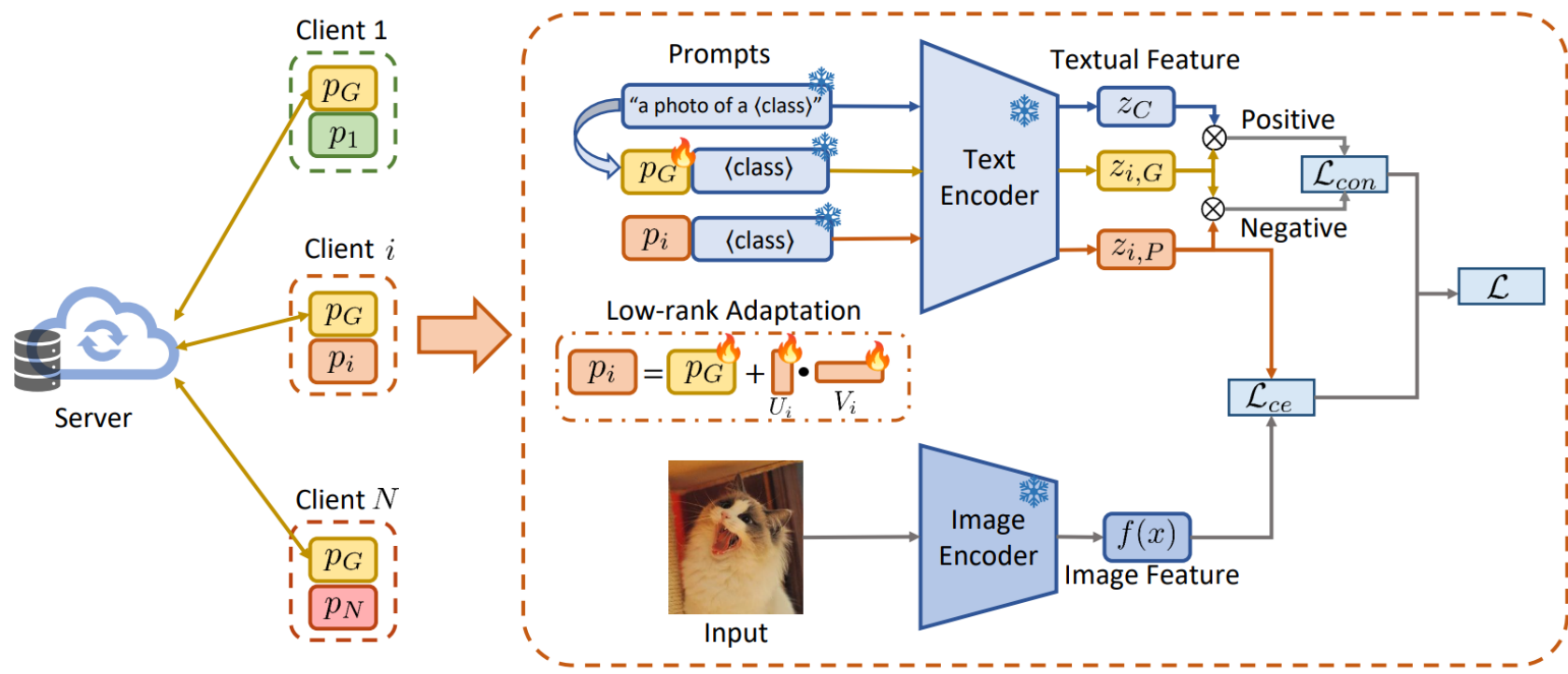
$$d_C(\alpha, \beta) = \min_{T \in U(\alpha, \beta)} \langle C, T \rangle,$$
$$U(\alpha, \beta) = \left\{ T \in \mathbb{R}_+^{|\alpha| \times |\beta|} \mid T \mathbf{1}_{|\beta|} = \alpha, T^\top \mathbf{1}_{|\alpha|} = \beta \right\}$$
$$d_{C,k}(\alpha, \beta) = \min_{T \in U(\alpha, \beta)} \langle C, T \rangle,$$
$$U(\alpha, \beta) = \left\{ T \in \mathbb{R}_+^{V \times 2} \mid \boxed{T \mathbf{1}_2 \leq \alpha}, T^\top \mathbf{1}_V = \beta \right\}$$
$$\|\alpha\|_1 \geq \|\beta\|_1 = \gamma \ (\gamma \in [0, 1])$$

- 在联邦学习框架中引入了预训练模型和提示学习，加速模型训练和降低通信成本；
- 提出了基于最优运输的联邦提示协作框架，为每个客户端学习全局提示和独特的本地提示；
- 利用不平衡最优运输将本地视觉特征与这些提示对齐，促进了全局提示和本地提示之间的协作。



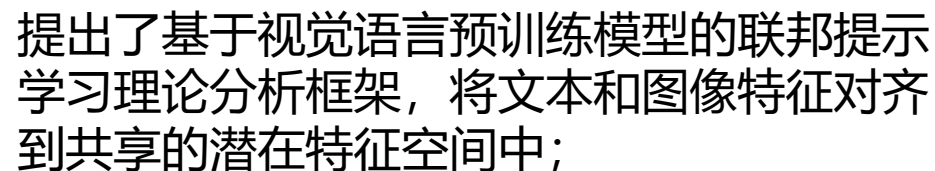
调和联邦提示学习的泛化和个性化

首次在联邦提示学习中同时考虑个性化和泛化，在异构联邦场景中为每个客户端学习个性化的提示，同时保留视觉语言模型中的泛化能力，实现泛化和个性化的平衡。



🔥 : Learnable ❄️ : Frozen ↪️ : Knowledge-guidance $\otimes$: Metric function p_G : Global Prompt p_1, p_i, p_N : Personalized Prompt

利用低秩分解自适应灵活地将全局提示调整为异构局部分布，防止在局部数据集上过度拟合。此外，整合了额外的对比损失，将全局和个性化提示的表示视为负对，将全局和人工提示的表示视为正对。



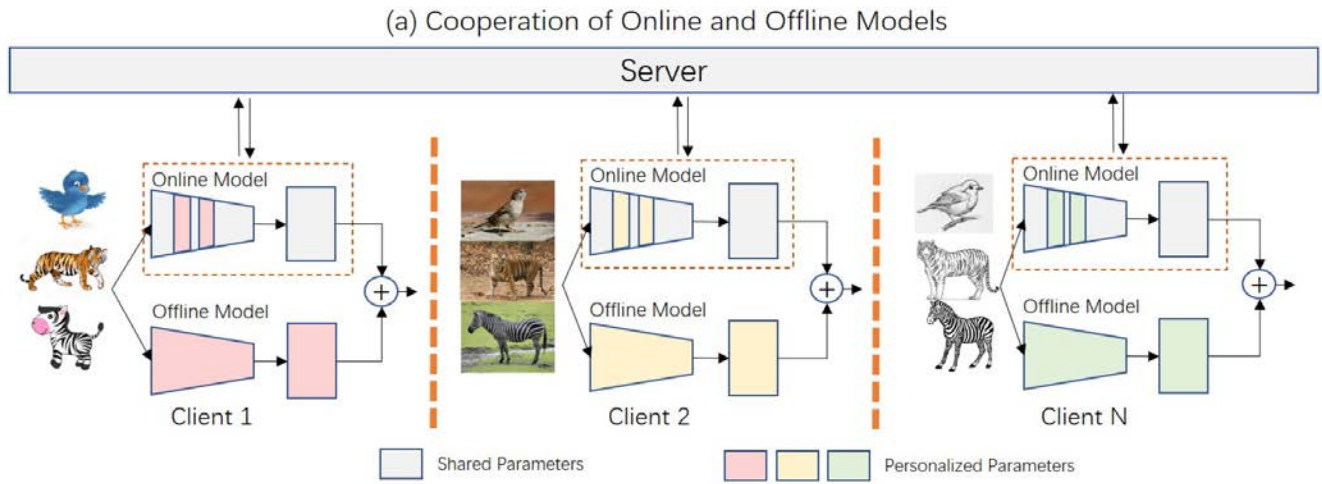
通过跟踪信号学习和噪声记忆，表明联邦提示学习的有效性可以通过任务相关系数与任务无关系数的比率来衡量；

引入Prompt组合来解决严重数据异质性问题，并平衡泛化和个性化；证明了Prompt组合比使用单个提示效果更好，并且给出了最佳混合比例。

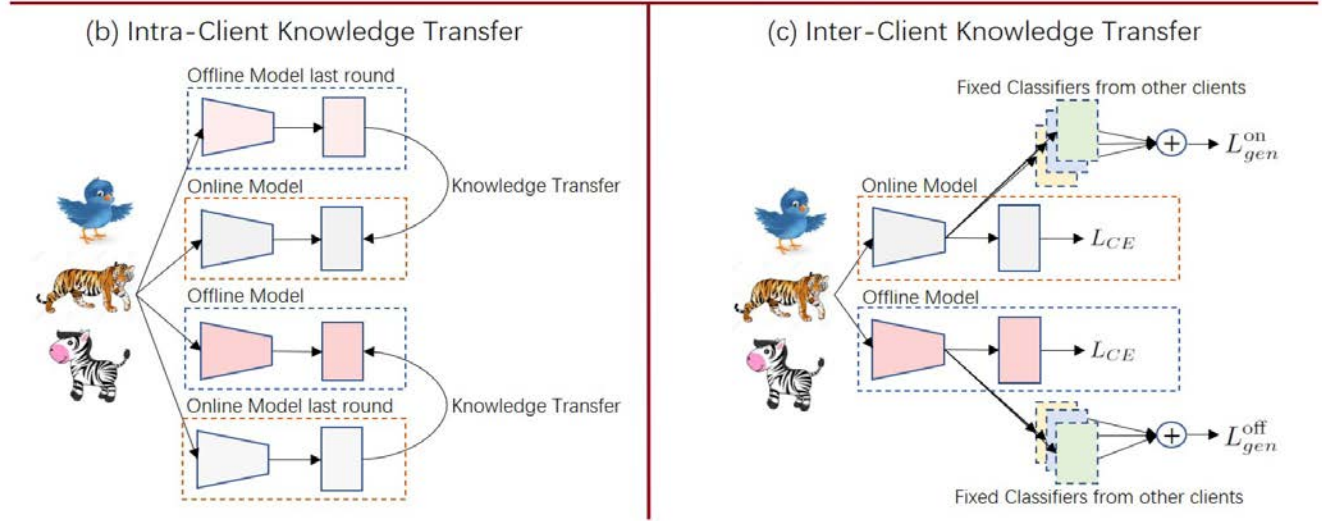
Theorem 5.1 (PromptFolio Advantage). *The mixed PromptFolio algorithm has a lower test loss than the mixing test loss of CoOp and PromptFL:*

$$L_{\mathcal{D}}((1-\theta)\mathbf{p}_G + \theta\mathbf{p}_L) \leq (1-\theta)L_{\mathcal{D}}(\mathbf{p}_G) + \theta L_{\mathcal{D}}(\mathbf{p}_L) \quad (10)$$

$$\forall \theta \in \left[0, \text{proj}_{[0,1]} \left(\frac{C_b - C_c}{2C_a} \right) \right], \quad \text{where} \quad \begin{cases} C_a = (b-a)(b^2 + 2\rho b + 1) \\ C_b = (a+b)(b^2 - 1) - 4b(\rho b - 1) \\ C_c = (b-1)\sqrt{(a+b)^2(b+1)^2 - 8ab^2(\rho+1)^2} \end{cases}.$$



提出一个同时处理标签分布偏差和特征偏差的联邦学习框架，通过在线和离线模型的合作机制解决数据异质性问题，在线模型学习共享知识，离线模型学习本地知识，并通过知识传递机制增强模型合作和领域泛化能力。



Theorem 4.3 For the V -dominated convergence, the convergence rate of Fed-CO₂ is faster than that of FedBN (the online model in Fed-CO₂).

Proof sketch The core is to show $\lambda_{\min}(\mathbf{V}_{\text{on}}^{\infty}) \leq \lambda_{\min}(\mathbf{V}^{\infty})$. To prove the theorem, we first show that $\lambda_{\min}(\mathbf{V}_{\text{on}}^{\infty}) \leq \lambda_{\min}(\mathbf{V}_{\text{off}}^{\infty})$. Comparing Eq. 14 and 15, $\mathbf{V}_{\text{off}}^{\infty}$ takes the $M \times M$ block matrices on the diagonal of $\mathbf{V}_{\text{on}}^{\infty}$. Let $\mathbf{V}_i^{\infty}$ be the i -th $M \times M$ block matrices on the diagonal of $\mathbf{V}_{\text{on}}^{\infty}$. According to linear algebra, we have $\lambda_{\min}(\mathbf{V}_i^{\infty}) \geq \lambda_{\min}(\mathbf{V}_{\text{on}}^{\infty})$, for $i \in [N]$. With $\mathbf{V}_{\text{off}}^{\infty} = \text{diag}(\mathbf{V}_1^{\infty}, \dots, \mathbf{V}_N^{\infty})$, it can be obtained that $\lambda_{\min}(\mathbf{V}_{\text{off}}^{\infty}) \geq \lambda_{\min}(\mathbf{V}_{\text{on}}^{\infty})$. Then, based on Eq. 10, 12, and 13, we can obtain $\lambda_{\min}(\mathbf{V}^{\infty}) = \lambda_{\min}(\frac{1}{2}(\mathbf{V}_{\text{on}}^{\infty} + \mathbf{V}_{\text{off}}^{\infty}))$. Therefore, we have $\lambda_{\min}(\mathbf{V}^{\infty}) \geq (\frac{1}{2}\lambda_{\min}(\mathbf{V}_{\text{on}}^{\infty}) + \frac{1}{2}\lambda_{\min}(\mathbf{V}_{\text{off}}^{\infty}))$. Since we have proven that $\lambda_{\min}(\mathbf{V}_{\text{off}}^{\infty}) \geq \lambda_{\min}(\mathbf{V}_{\text{on}}^{\infty})$, the result $\lambda_{\min}(\mathbf{V}^{\infty}) \geq \lambda_{\min}(\mathbf{V}_{\text{on}}^{\infty})$ can be achieved.

从神经正切核角度证明其更快的收敛性

从特征动力学角度证明了 FedAvg 算法收敛性，证明不依赖于传统权重空间中的线性化假设，表明 FedAvg 可以在多项式时间内实现近零的训练误差。

Theorem 4.3 (Convergence of FedAvg). *Suppose $\epsilon > 0$, and let $T = \tilde{\Theta}(\eta^{-1} K m \bar{n} \sigma_p^{-2} d^{-1} + \eta^{-1} \epsilon^{-1} m \bar{n} \sigma_p^{-2} K^2 d^{-1})$. Under Assumption 4.2, then with probability at least $1 - d^{-1}$, there exists $0 \leq t \leq T$ such that the training loss of FedAvg converges to ϵ , i.e., $L_S(\bar{\mathbf{W}}^{(t)}) \leq \epsilon$.*

首次通过特征动力学中的信噪比建立FedAvg和局部算法之间的泛化差距。此外，我们的理论结果揭示了基于特征学习理论中客户端输入数据之间的相似性在 FL 中合作的重要性。

Theorem 4.4 (Generalization of FedAvg). *Under the same assumption as Theorem 4.3, there exists $0 \leq t \leq T$ such that if $\frac{\bar{n} \chi_k^2 \|\boldsymbol{\mu}\|_2^4}{K \sigma_p^4 d} = \Omega(1)$, the trained model by FedAvg achieves a small test loss: $L_{\mathcal{D}_k}^{0-1}(\bar{\mathbf{W}}^{(t)}) \leq \exp\left(-c \frac{\bar{n} \chi_k^2 \|\boldsymbol{\mu}\|_2^4}{K \sigma_p^4 d}\right)$. Here, $\chi_k = \sum_{k'=1}^K \frac{\langle \boldsymbol{\mu}_k, \boldsymbol{\mu}_{k'} \rangle}{\|\boldsymbol{\mu}_k\|_2^2}$. Otherwise, if $\frac{\bar{n} \chi_k^2 \|\boldsymbol{\mu}\|_2^4}{K \sigma_p^4 d} = O(1)$, the trained model by FedAvg achieves a large constant test error $L_{\mathcal{D}_k}^{0-1}(\bar{\mathbf{W}}^{(t)}) = \Omega(1)$.*

Conclusions

Integrating Models into Deep Learning for

(accurate) Recognition/ (reliable) Generation/ (Safe) Decision

Integrating Loss Guidance to diffusion models for Controllable Generation.

Integrating Optimization/Constraints to DL/RL for end to end learning (Robustness/Safety).

Integrating/Applying Foundation Models to ...

Challenges: Not differentiable, Manifold deviation, Computation Increase,
Foundation Model Personalization with limited resource.



Thank you
Any Question?

A black and silver pen is positioned diagonally, pointing towards the text.